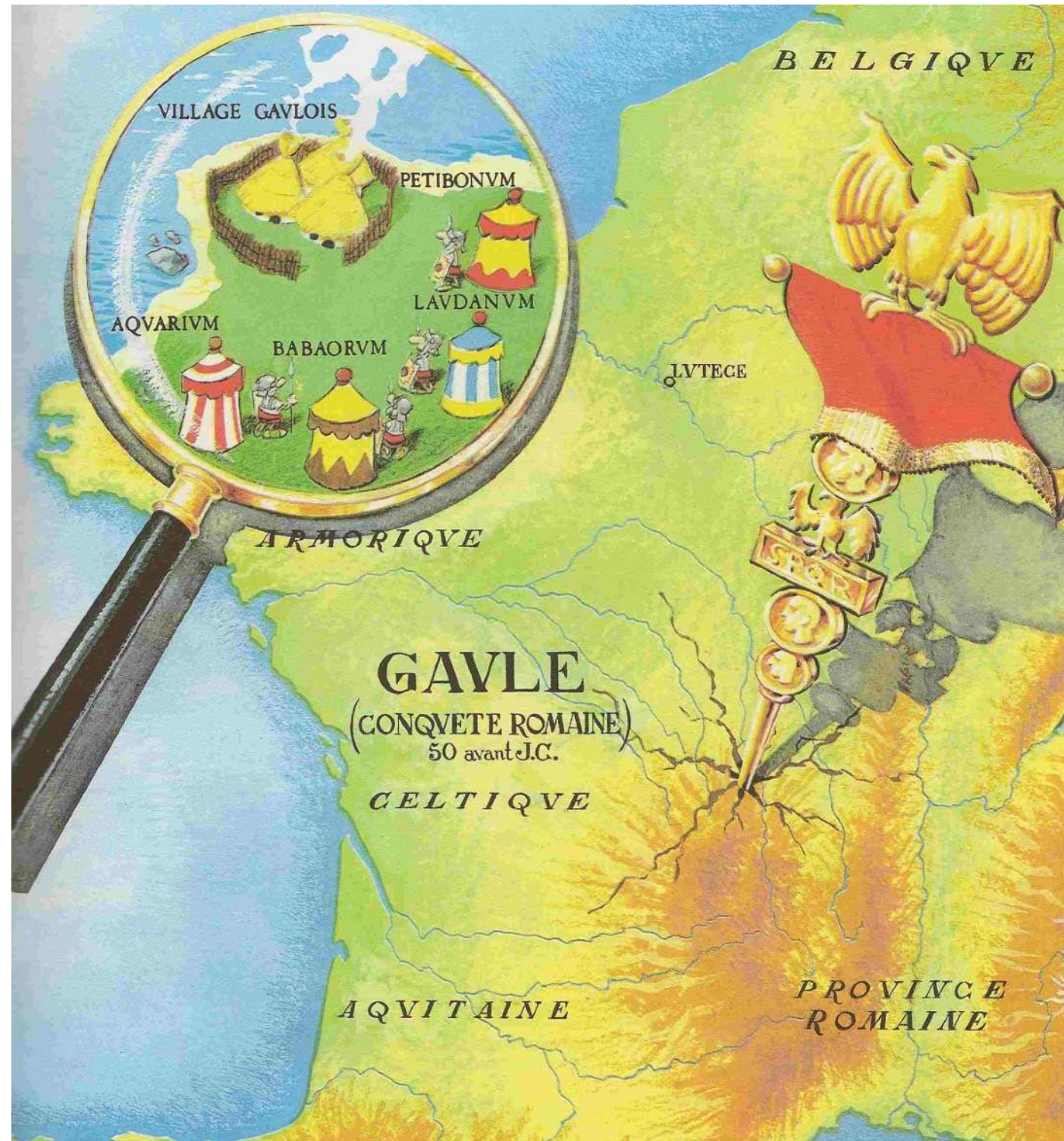

A few applications of GSP to machine learning and medicine

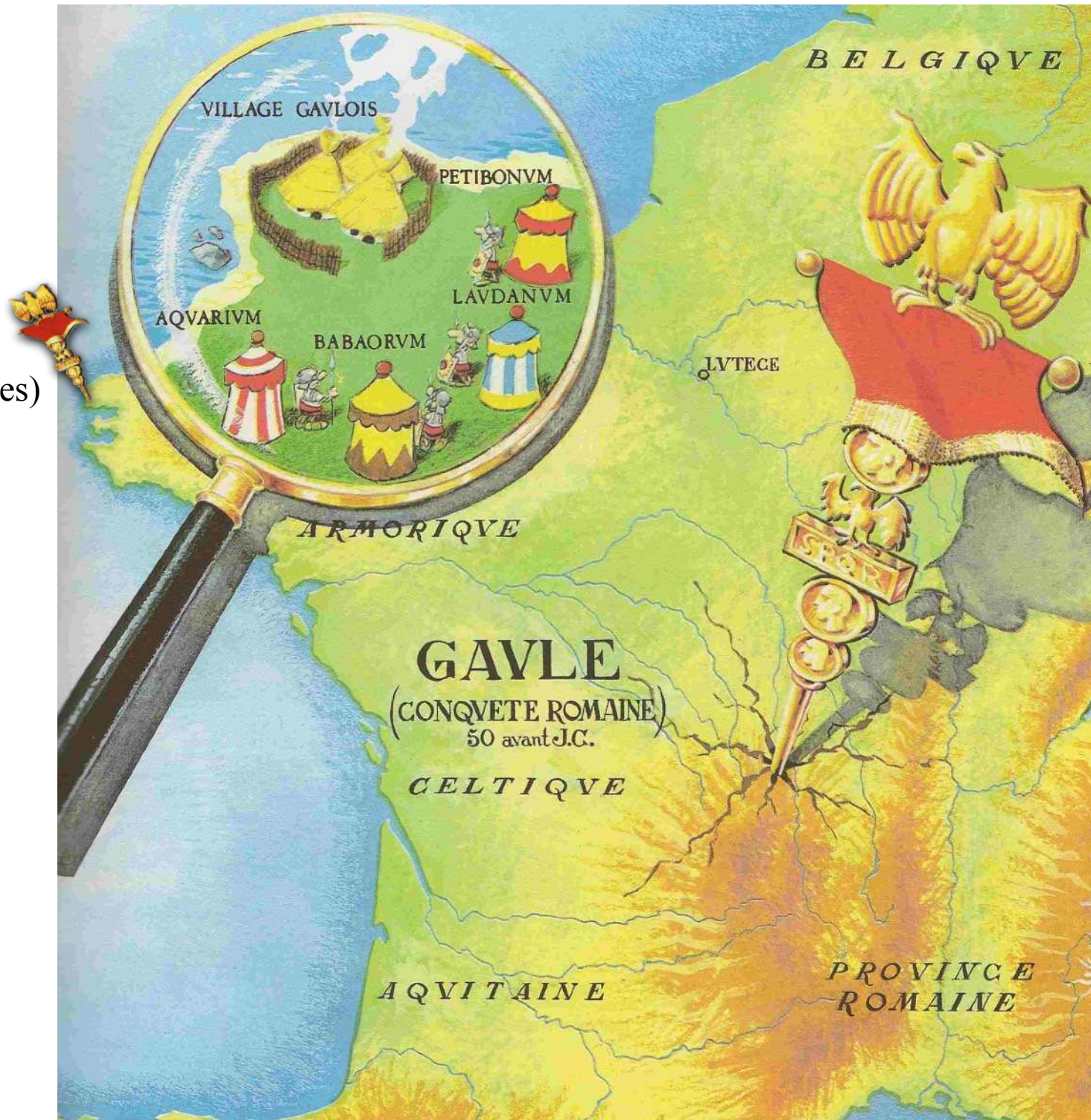
Bastien Pasdeloup
LTS4 - EPFL - Lausanne - Switzerland
`bastien.pasdeloup@epfl.ch`

Hello :)

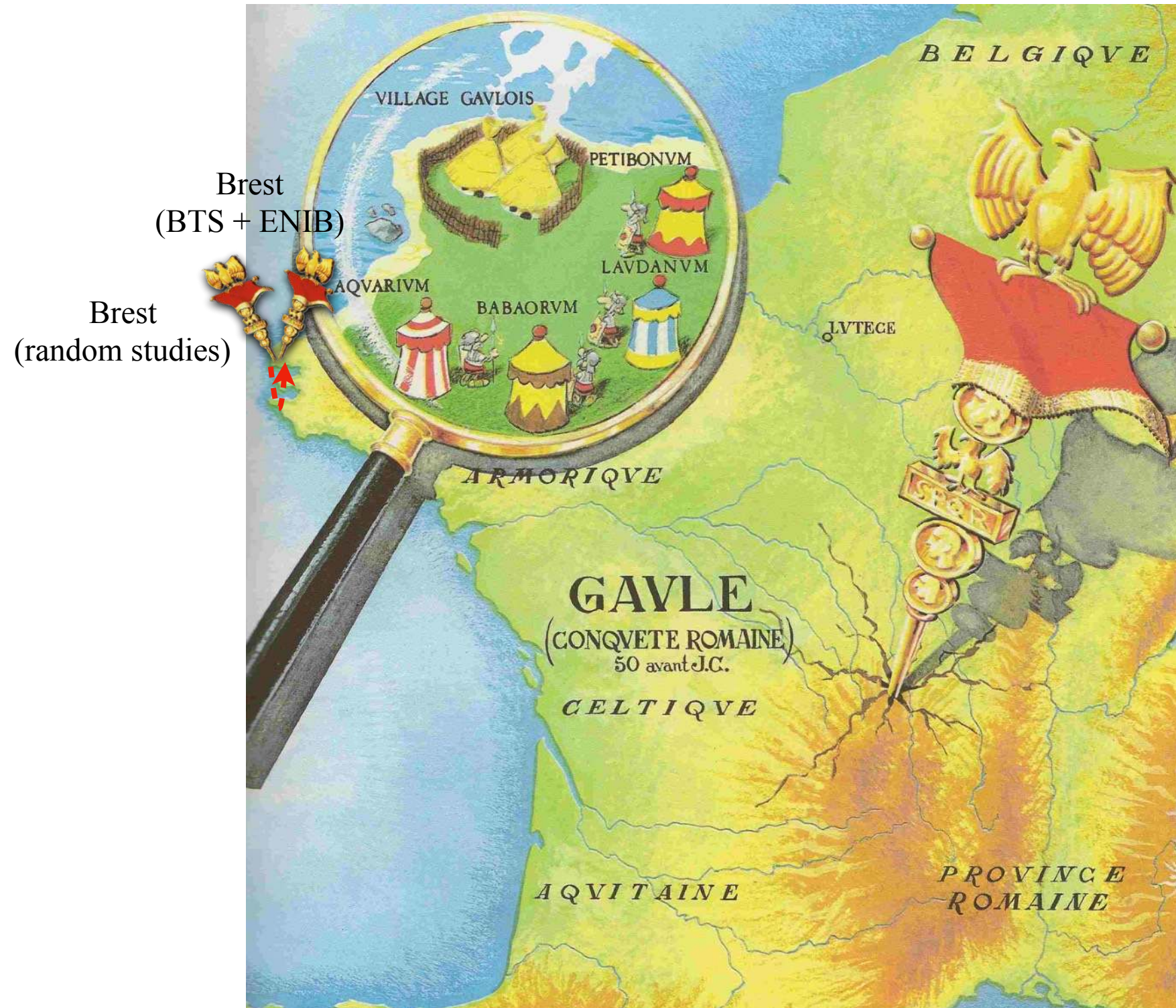


Hello :)

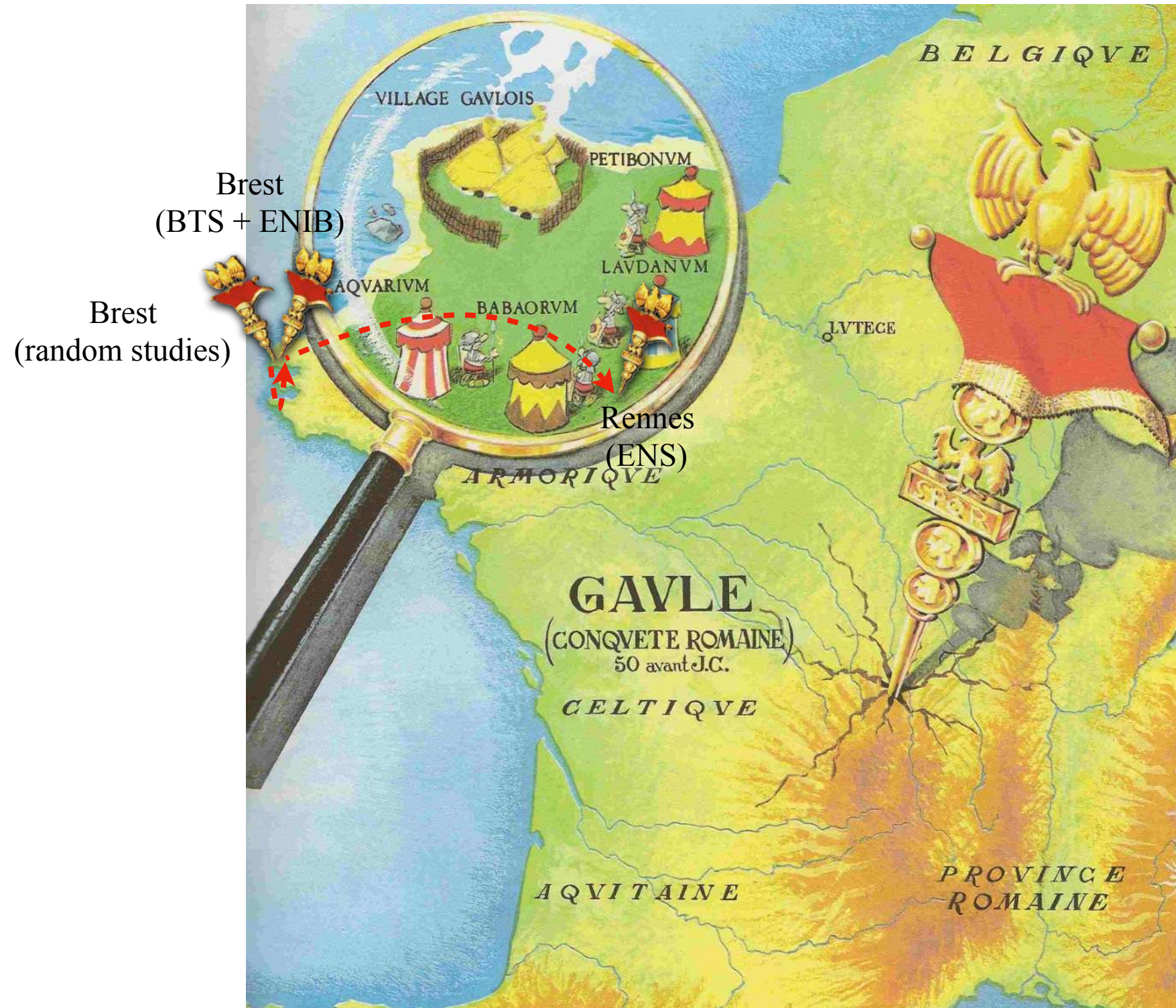
Brest
(random studies)



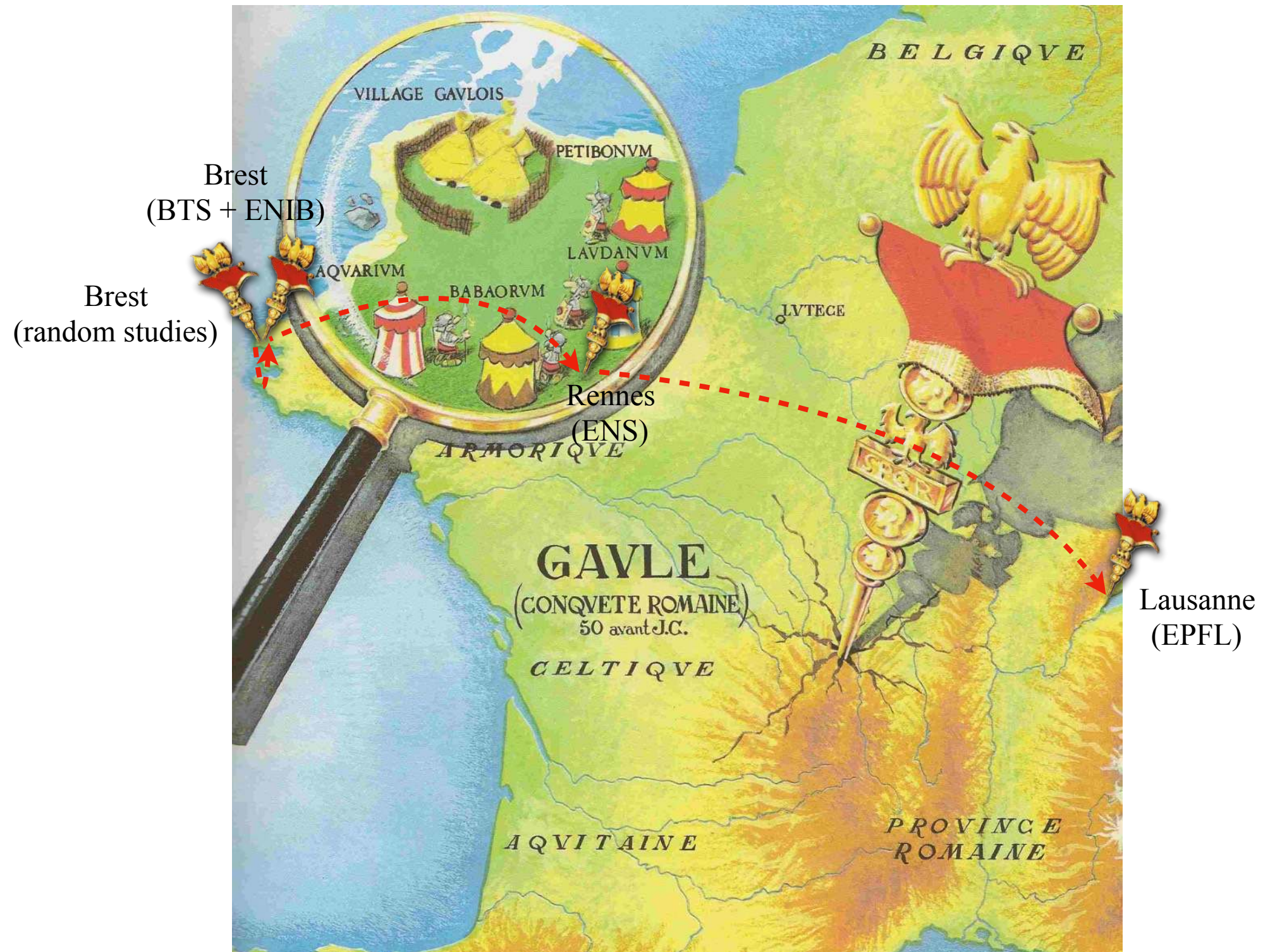
Hello :)



Hello :)



Hello :)



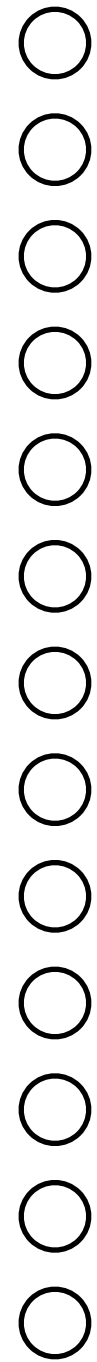
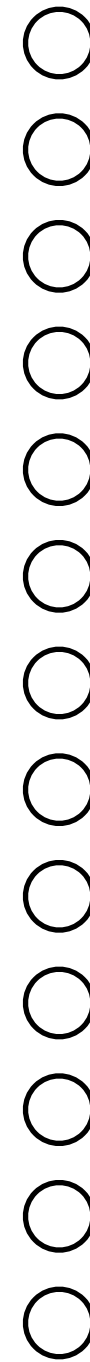
Outline

- GSP for AI, *a.k.a.* "What I did during my Ph.D. and a bit after"
- GSP and personalized healthcare
- GSP and brain signals

Outline

- **GSP for AI, *a.k.a.* "What I did during my Ph.D. and a bit after"**
- GSP and personalized healthcare
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How to design a convolutional layer?

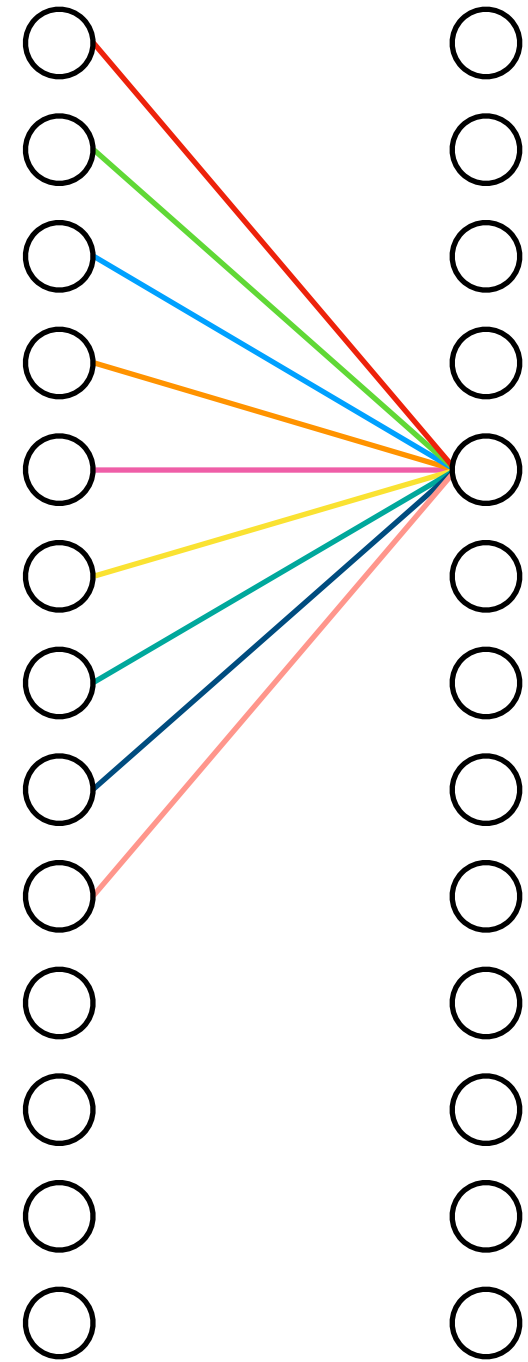


How to design a convolutional layer?

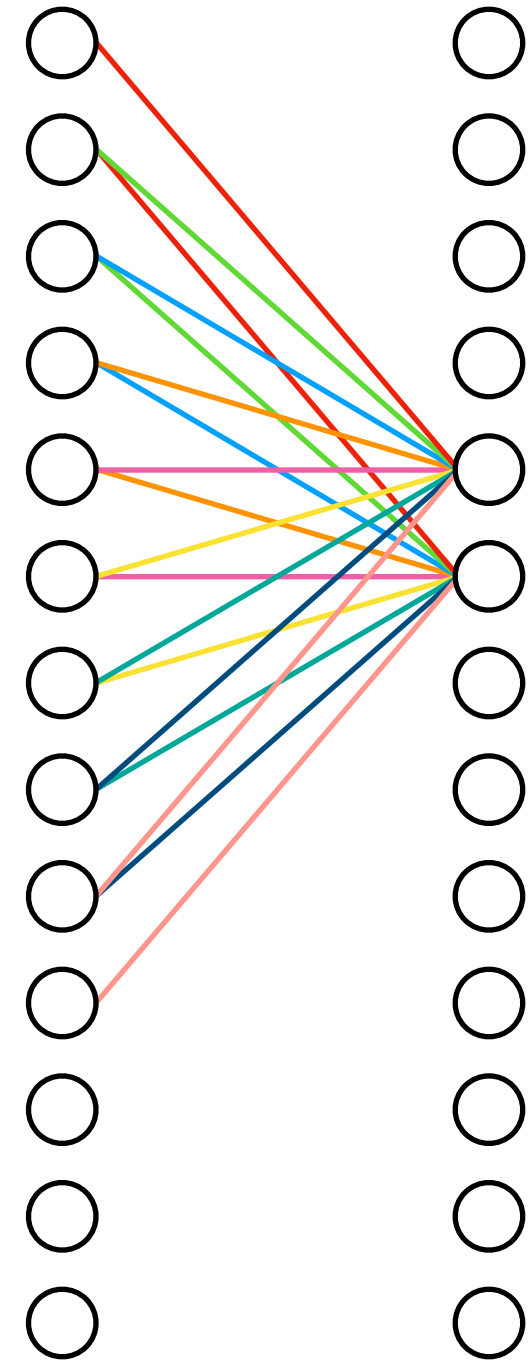


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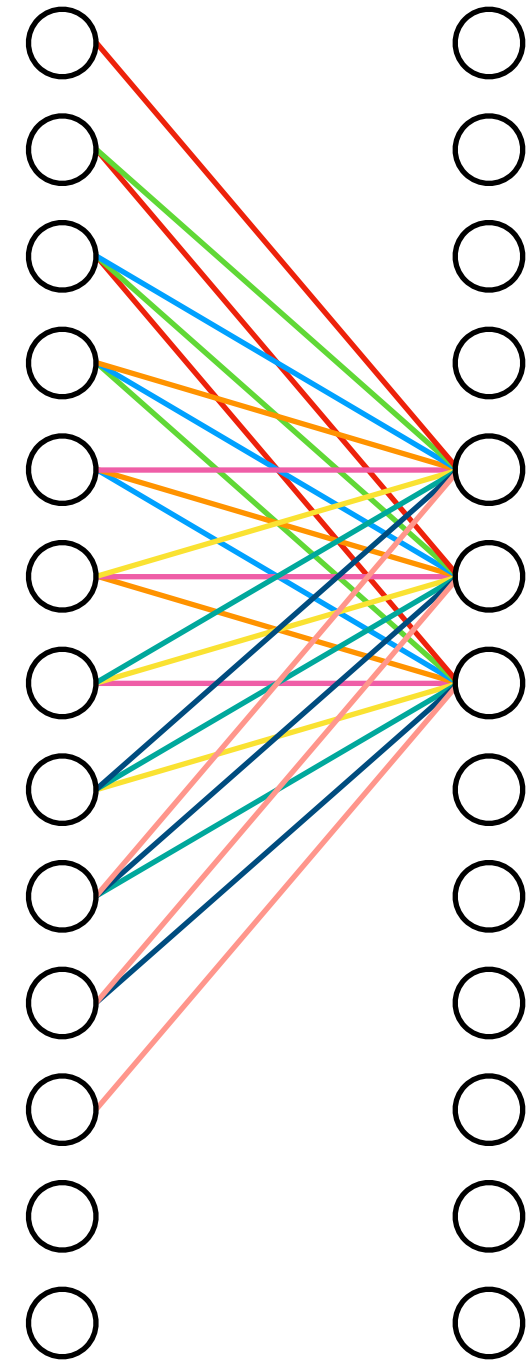
How to design a convolutional layer?



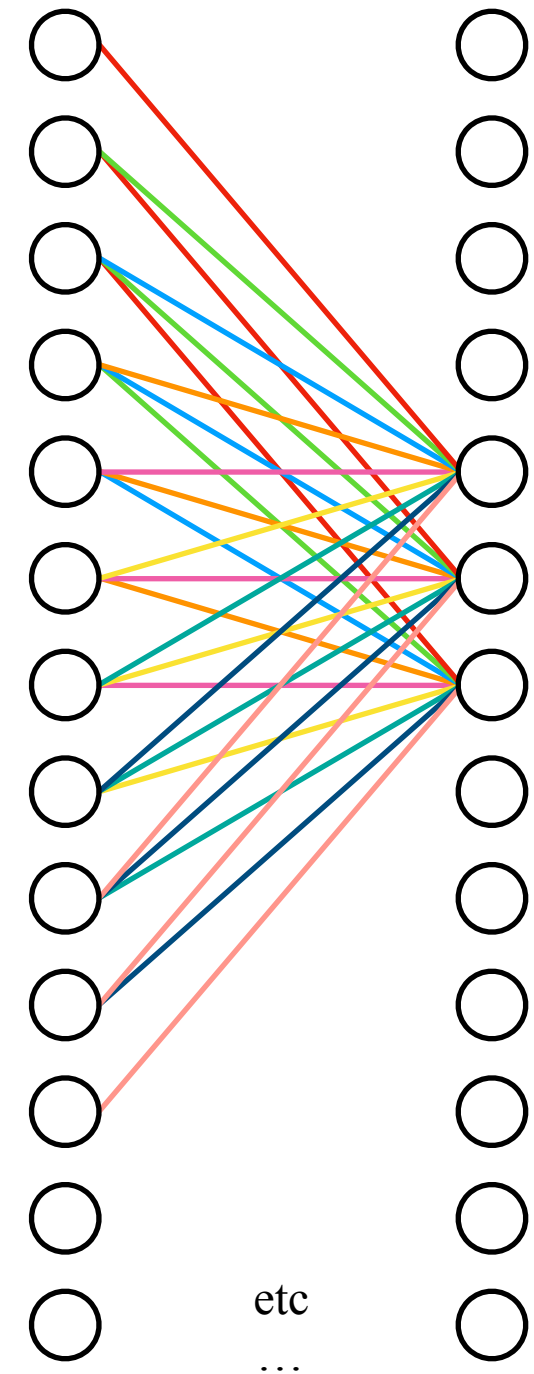
How to design a convolutional layer?



How to design a convolutional layer?



How to design a convolutional layer?



Why does it work?

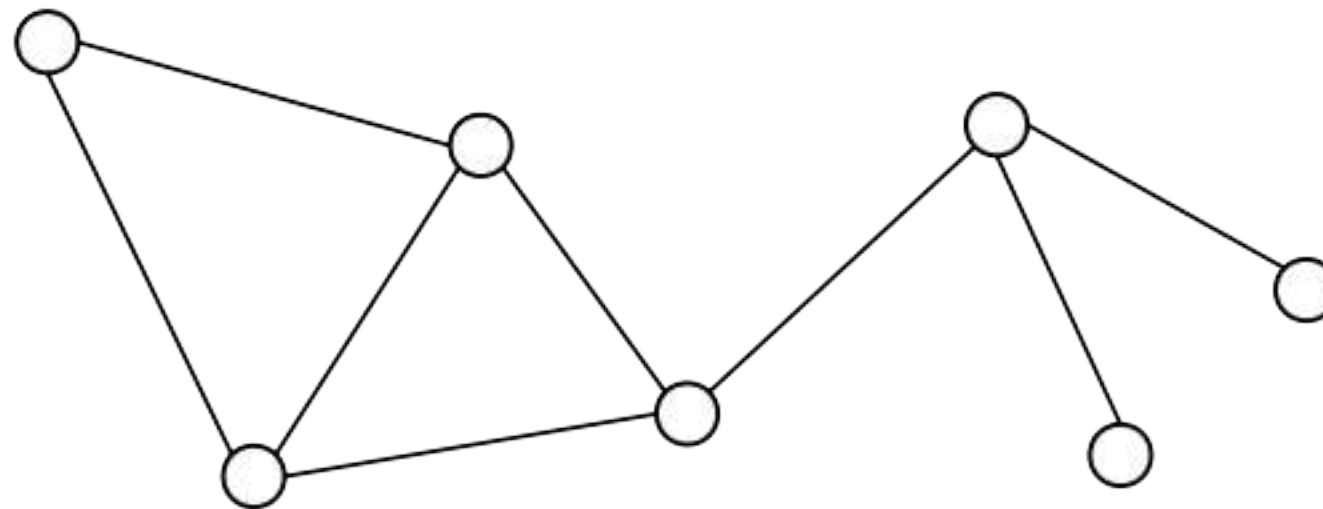
Discriminating features are localized in the image (= locality)

Statistical properties of discriminating features are invariant by translation (= stationarity)

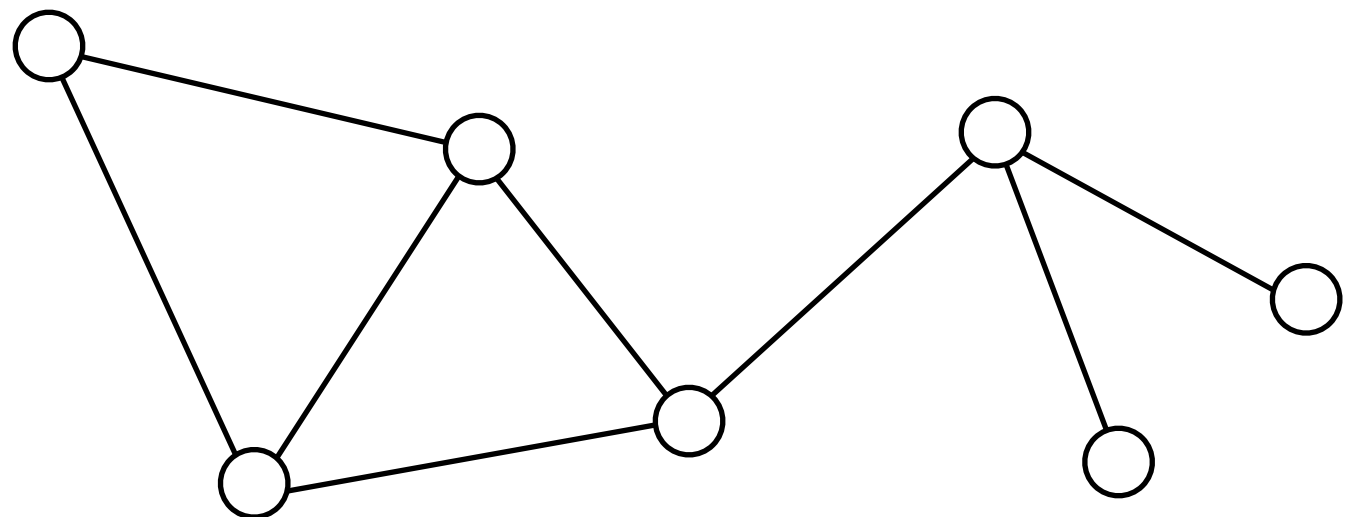
A natural scene has an important low-frequency component (= smoothness)



Beyond images



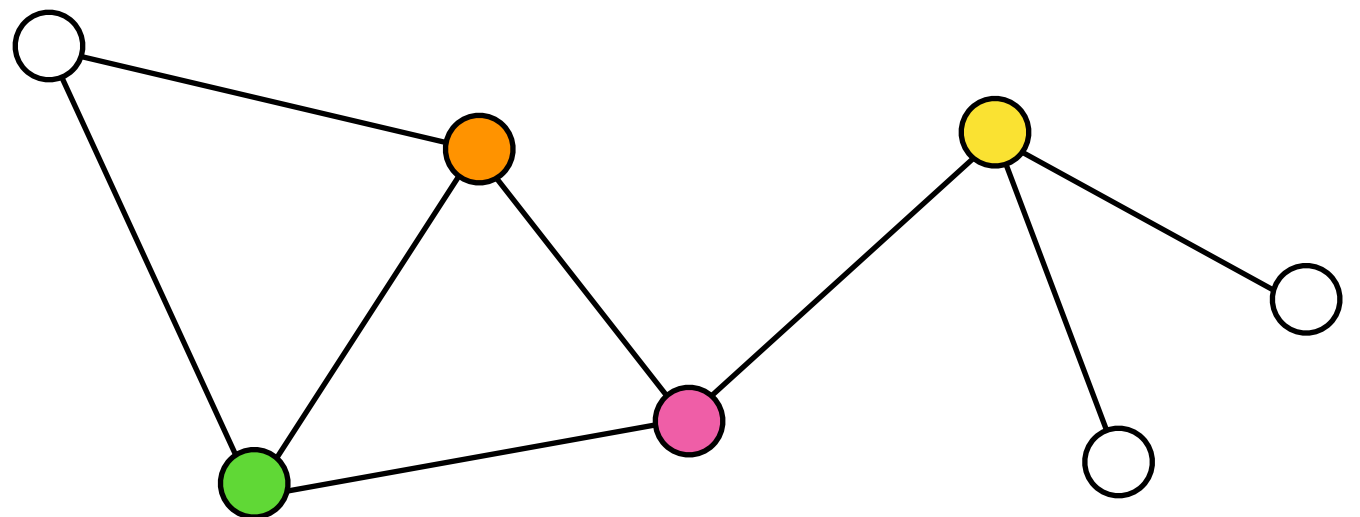
What do we need?



What do we need?



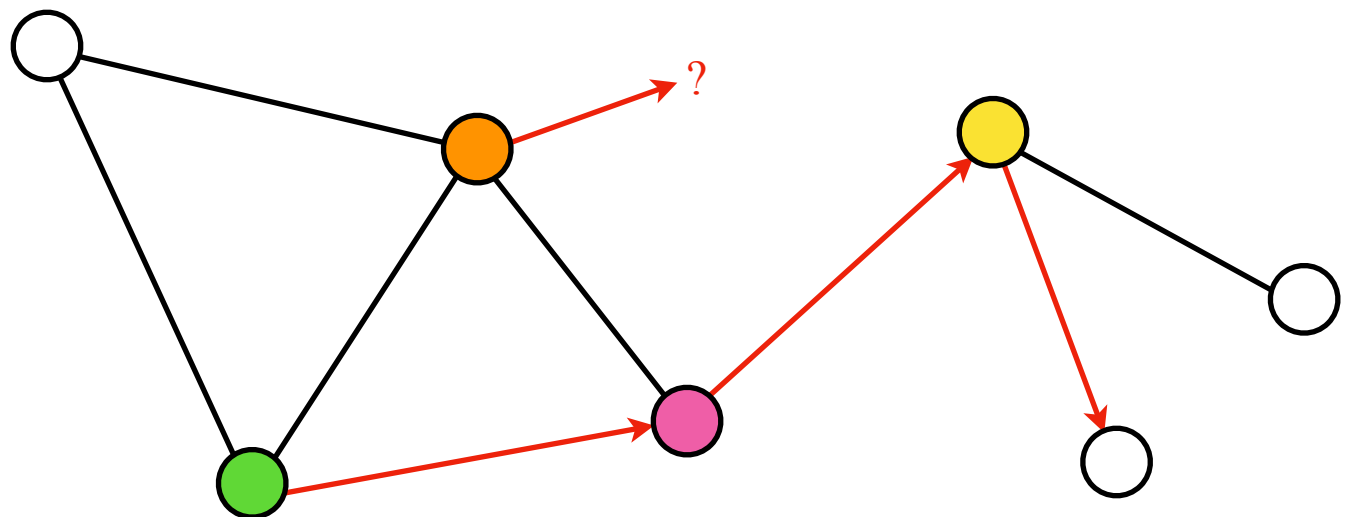
Assumption that discriminating features are localized



What do we need?



What is a translation on a graph domain?



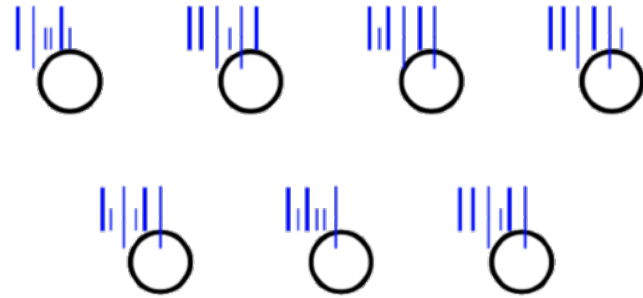
What do we need?



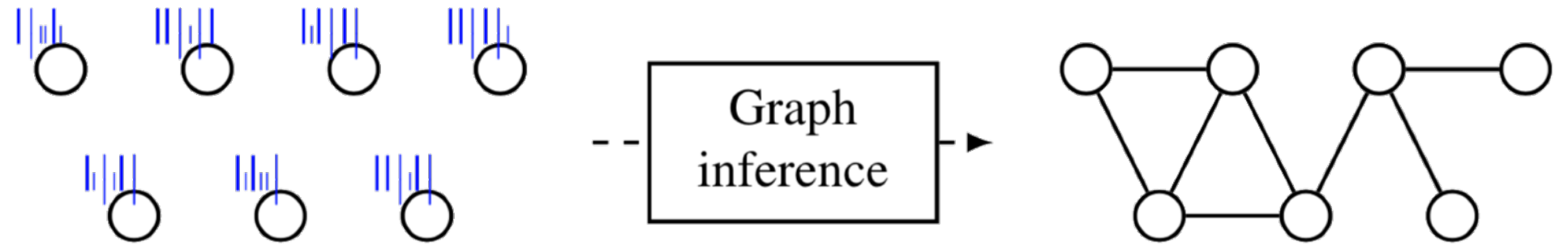
Which graph does data live on?



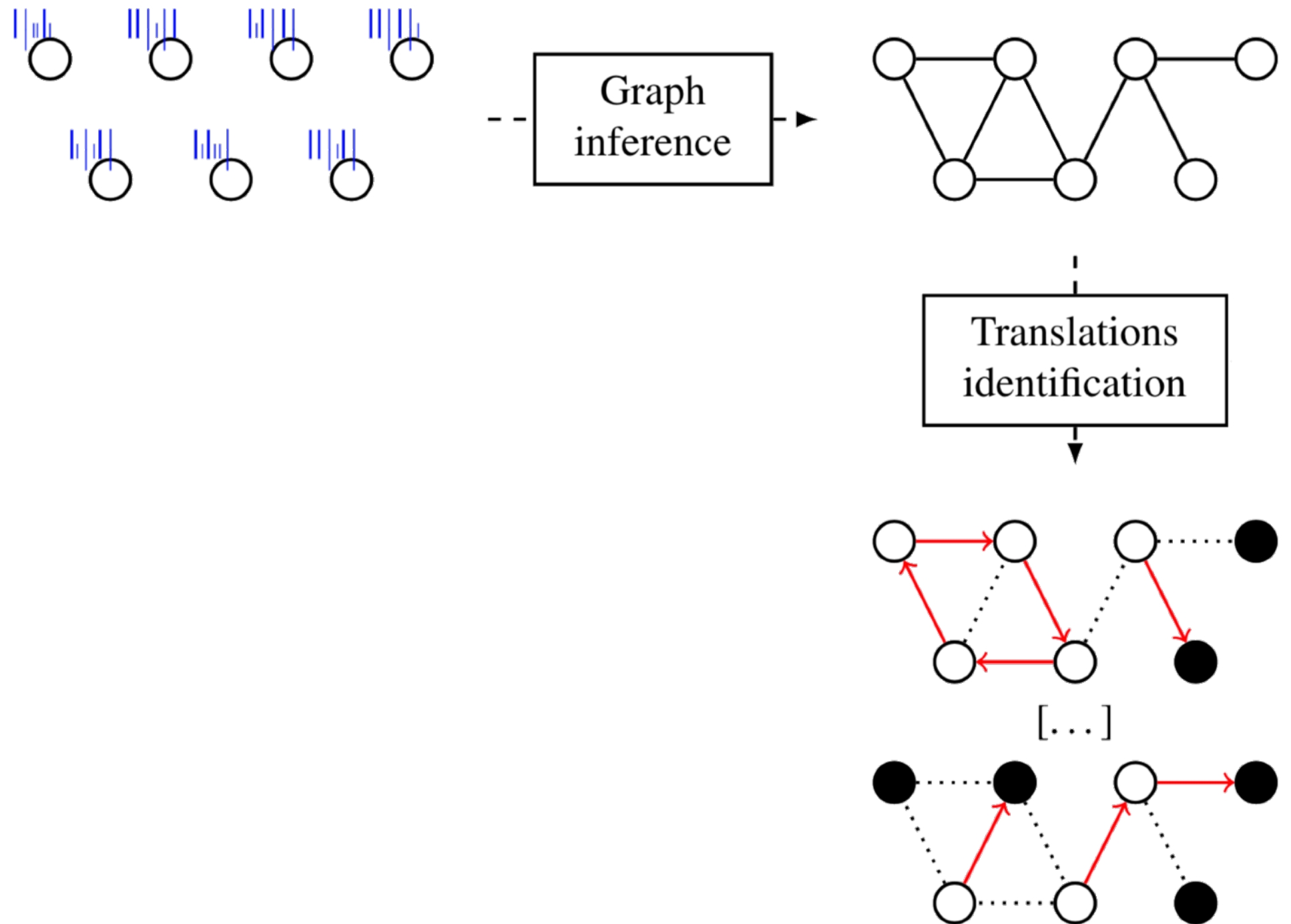
Extending CNNs to graph signals



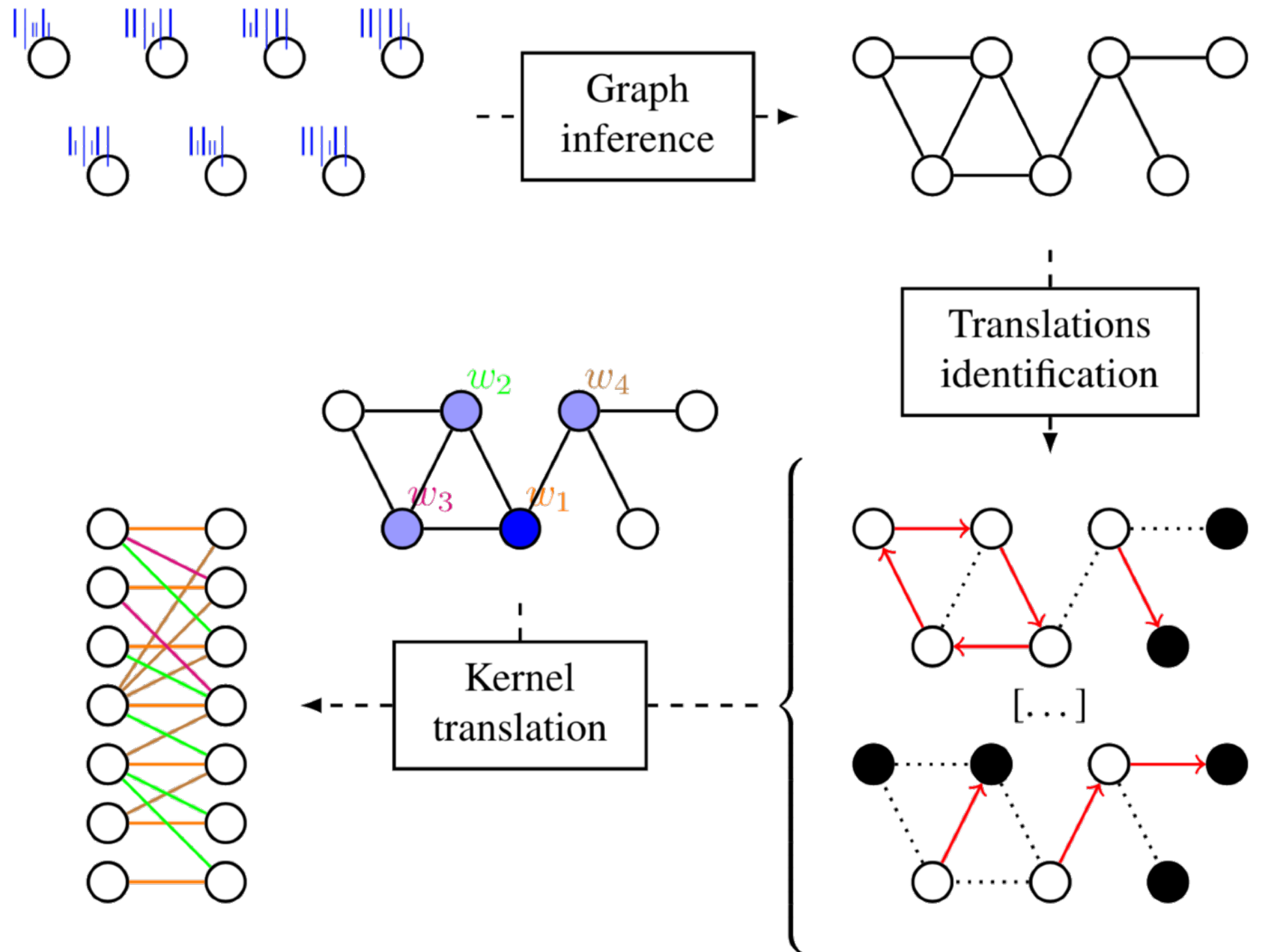
Extending CNNs to graph signals



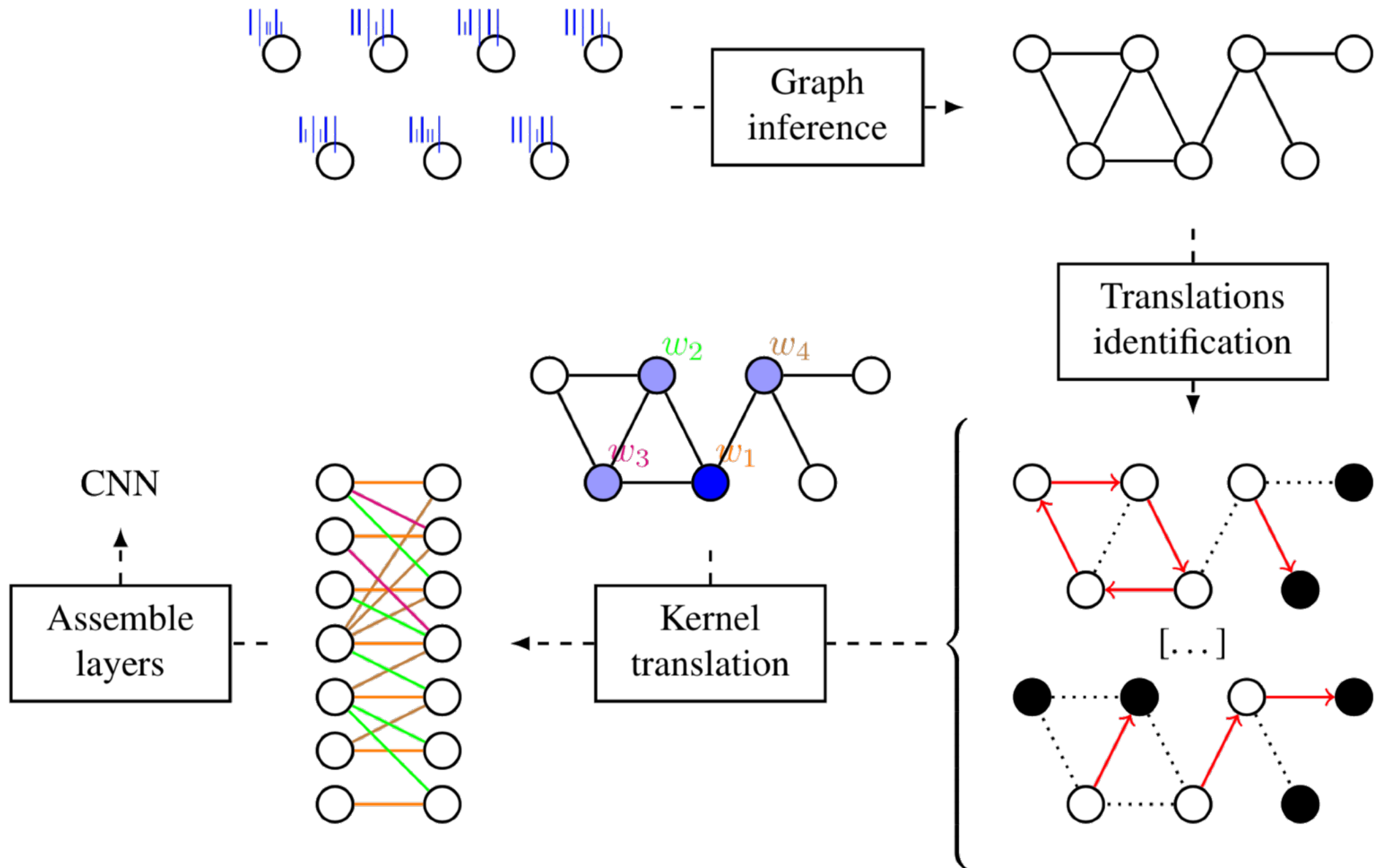
Extending CNNs to graph signals



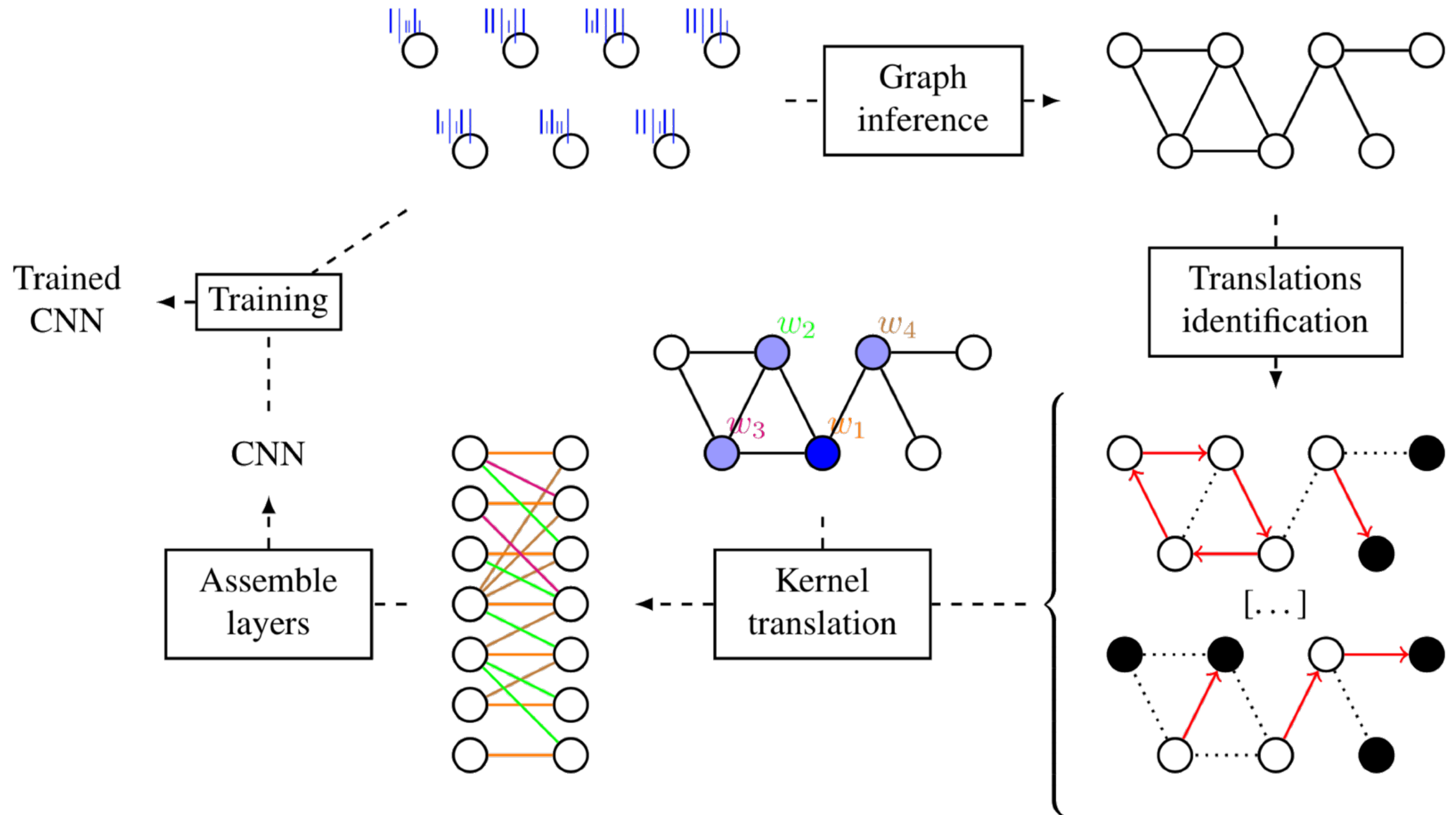
Extending CNNs to graph signals



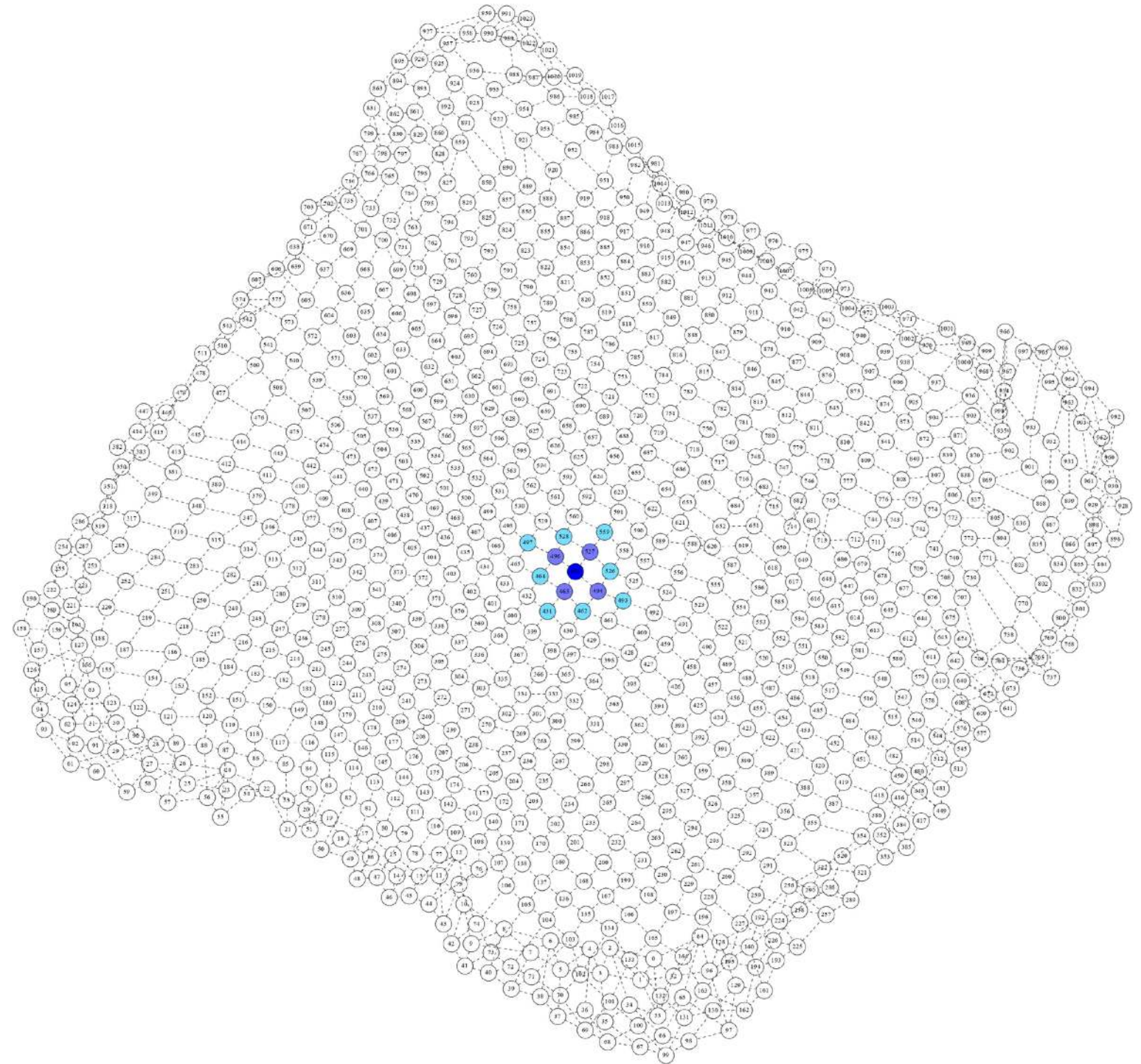
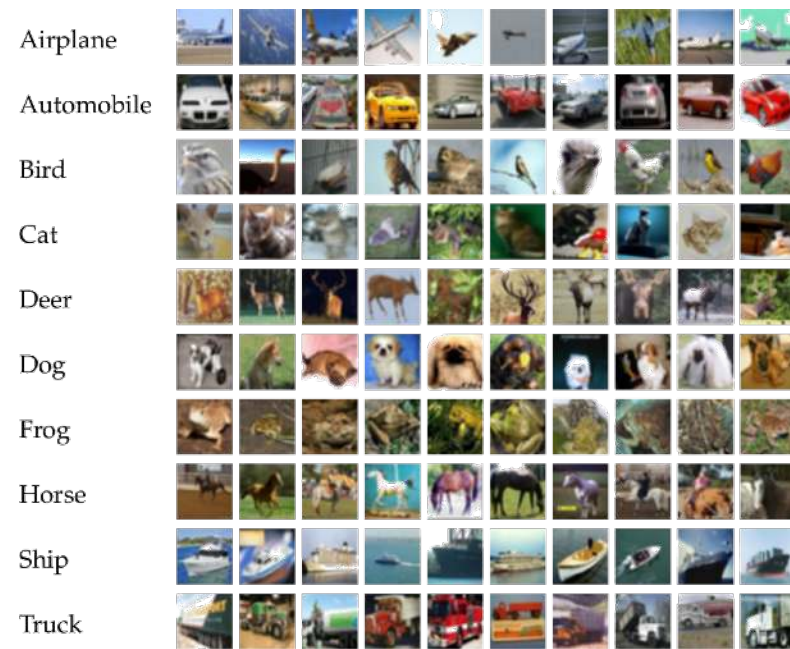
Extending CNNs to graph signals



Extending CNNs to graph signals



Results

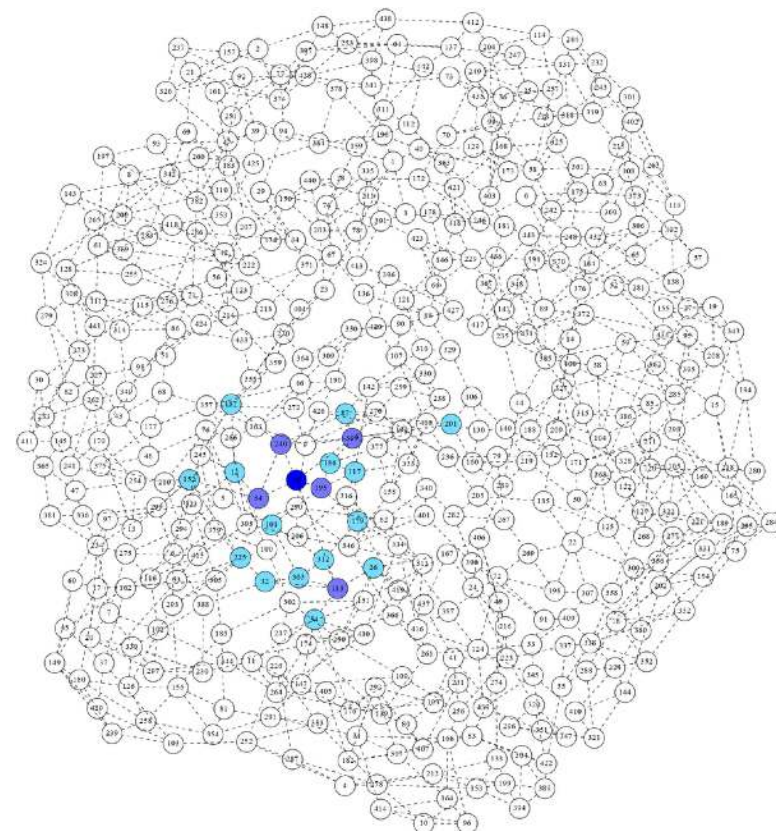
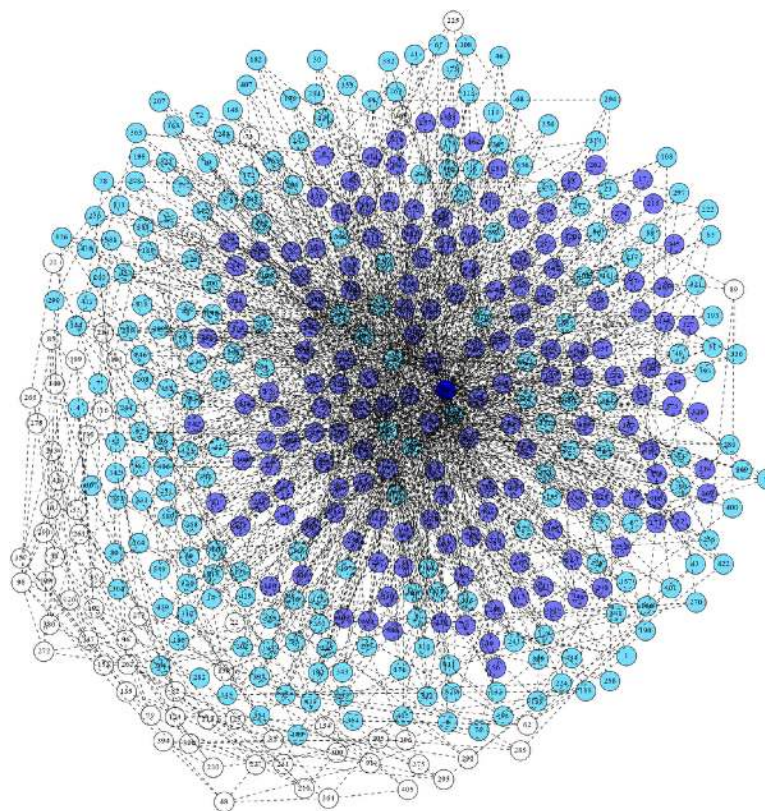
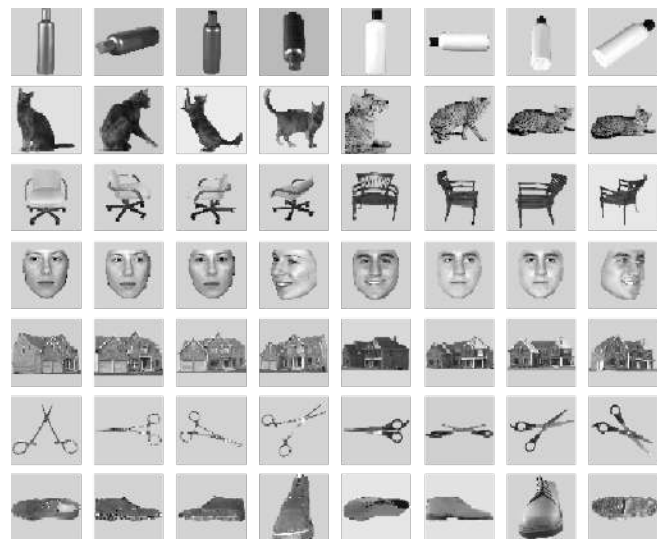


No knowledge on topology [1]: 72.7%

Introduced method: 85.52%

Same architecture with 3x3 kernels: 85.98%

Results



	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Subject 6	Average
Our method with $\mathcal{G}_{geo}$	56%	52%	47%	53%	42%	49%	49.9%
Def ₅ ($\mathcal{G}_{geo}$)	60%	56%	51%	54%	45%	51%	52.8%
Def ₁₀ ($\mathcal{G}_{geo}$)	61%	55%	50%	55%	42%	51%	52.3%
Def ₂₀ ($\mathcal{G}_{geo}$)	62%	54%	51%	56%	43%	53%	53.2%
Def ₅ ($\mathcal{G}_K^*$)	61%	53%	50%	53%	43%	49%	51.5%
Def ₁₀ ($\mathcal{G}_K^*$)	59%	54%	50%	54%	45%	51%	52.2%
Def ₂₀ ($\mathcal{G}_K^*$)	61%	51%	49%	54%	42%	49%	51%
MLP	57%	51%	51%	51%	42%	49%	50.2%
SVM	50%	42%	35%	32%	31%	40%	38.2%
LR ($C = 1, l_1$)	51%	48%	44%	34%	42%	42%	43.6%
LR ($C = 50, l_1$)	47%	47%	38%	28%	34%	40%	38.9%
LR ($C = 1, l_2$)	49%	47%	41%	34%	31%	38%	39.7%
Ridge	35%	34%	28%	22%	23%	31%	28.8%

Ongoing work on this

$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

Too strict
NP-complete

$$\begin{aligned} \forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \\ \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp) \end{aligned}$$

Ongoing work on this

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$$\begin{aligned} \forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \\ \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp) \end{aligned}$$

$$\begin{aligned} s(\tilde{\psi}) &= \alpha \text{loss}(\tilde{\psi}) \\ &+ \beta |\{v \in \mathcal{V}_1 : \{v, \tilde{\psi}(v)\} \notin \mathcal{E} \wedge \tilde{\psi}(v) \neq \perp\}| \\ &+ \gamma \sum_{\substack{\{v_1, v_2\} \in \binom{\mathcal{V}_1}{2} \\ \tilde{\psi}(v_1) \neq \perp \\ \tilde{\psi}(v_2) \neq \perp}} \left| d(v_1, v_2) - d(\tilde{\psi}(v_1), \tilde{\psi}(v_2)) \right| \end{aligned}$$

Ongoing work on this

Algorithm 1: $v_{src_to_v_{tgt}}$ ($\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle, \mathcal{V}_1, v_{src}, v_{tgt}$)

```

1 foreach  $v \in \mathcal{V}$  do
2    $visited[v] := \mathbf{false};$ 
3    $predecessors[v] := \mathbf{NULL};$ 
4  $Q := \{\langle 0, v_{src}, \mathbf{NULL}, \mathbf{NULL} \rangle\};$ 
5 while  $Q \neq \emptyset$  do
6    $total\_score_{v_1}, v_1, pred, \tilde{\psi}_{pred \rightarrow v_1} :=$ 
7      $\underline{extract\_min}(Q);$ 
8   if not  $visited[v_1]$  then
9      $visited[v_1] := \mathbf{true};$ 
10     $predecessors[v_1] := \langle pred, \tilde{\psi}_{pred \rightarrow v_1} \rangle;$ 
11    if  $v_1 = v_{tgt}$  then
12      break;
13    if  $\tilde{\psi}_{pred \rightarrow v_1} \neq \mathbf{NULL}$  then
14       $\mathcal{V}_1 := \{v \in \mathcal{V} \mid \tilde{\psi}_{pred \rightarrow v_1}(v) \neq \perp\};$ 
15       $\mathcal{V}_2 := \underline{choose\_subset}(\mathcal{V});$ 
16      foreach  $v_2 \in \mathcal{V}_2 \setminus \{v_1\}$  do
17        if not  $visited[v_2]$  then
18           $\tilde{\psi}_{v_1 \rightarrow v_2}, score_{\tilde{\psi}_{v_1 \rightarrow v_2}} :=$ 
19             $\underline{minimize\_s}(v_1, v_2, \mathcal{G}, \mathcal{V}_1, \mathcal{V}_2);$ 
20           $total\_score_{v_2} :=$ 
21             $total\_score_{v_1} + score_{\tilde{\psi}_{v_1 \rightarrow v_2}};$ 
22           $Q :=$ 
23             $Q \cup \{\langle total\_score_{v_2}, v_2, v_1, \tilde{\psi}_{v_1 \rightarrow v_2} \rangle\};$ 
24 return  $predecessors;$ 

```

From here

To here

Target of
approx. trans.

NP-complete
Heuristics!

Ongoing work on this

Algorithm 1: $v_{src_to_v_{tgt}}$ ($\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle, \mathcal{V}_1, v_{src}, v_{tgt}$)

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18           $\tilde{\psi}_{v_1 \rightarrow v_2}, score_{\tilde{\psi}_{v_1 \rightarrow v_2}} :=$ 
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20             $total\_score_{v_2} :=$ 
21               $total\_score_{v_1} + score_{\tilde{\psi}_{v_1 \rightarrow v_2}};$ 
22             $Q :=$ 
23               $Q \cup \{\langle total\_score_{v_2}, v_2, v_1, \tilde{\psi}_{v_1 \rightarrow v_2} \rangle\};$ 
24 return  $predecessors;$ 

```

Algorithm 2: minimize_s ($v_1, v_2, \mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle, \mathcal{V}_1, \mathcal{V}_2$)

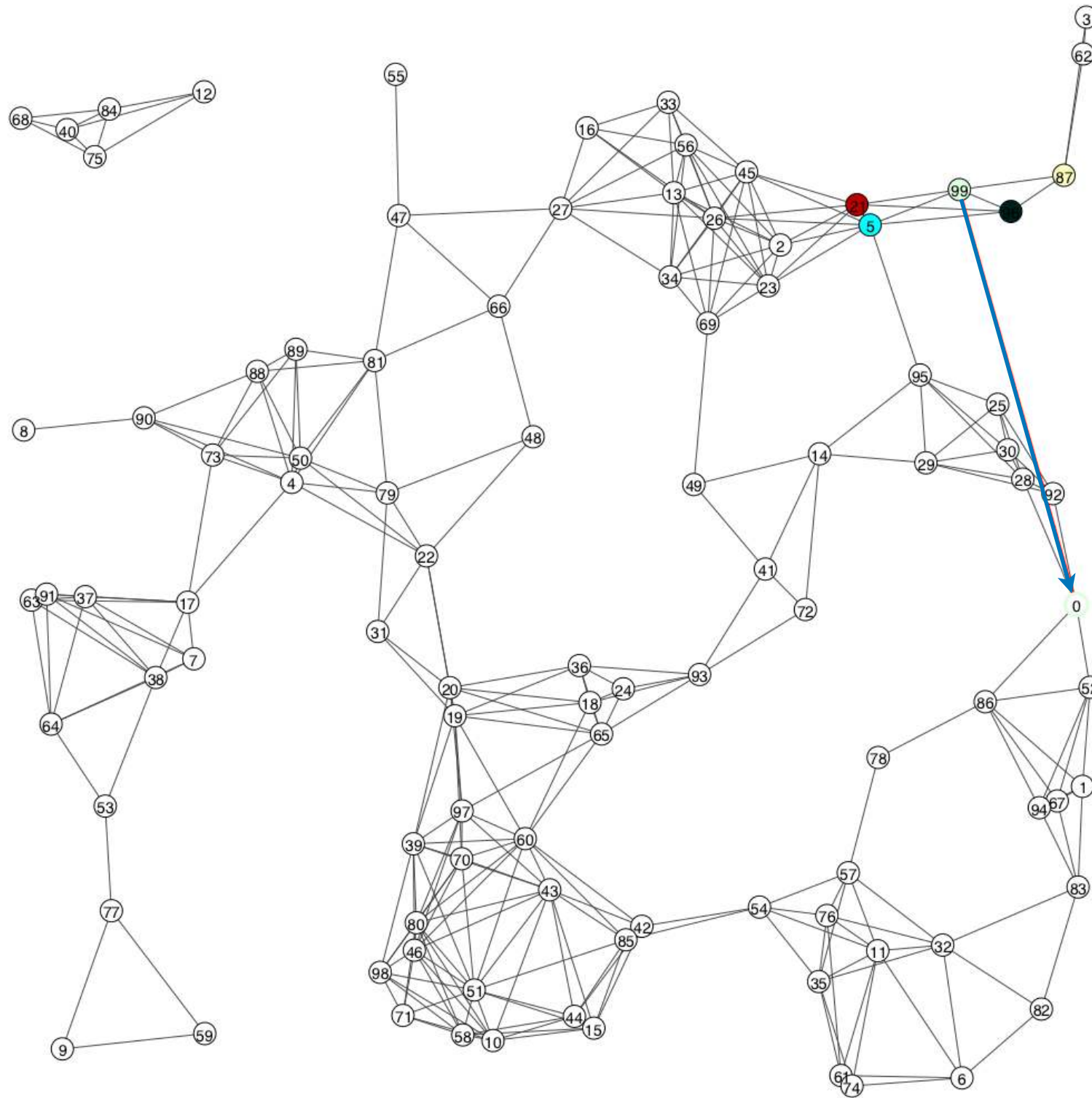
```

1 foreach  $v \in \mathcal{V}_1$  do
2    $\tilde{\psi}_{v_1 \rightarrow v_2}(v) := \text{NULL};$ 
3    $\tilde{\psi}_{v_1 \rightarrow v_2}(v_1) := v_2;$ 
4    $score_{\tilde{\psi}_{v_1 \rightarrow v_2}} := 0;$ 
5 for  $i = 1$  to  $\lceil \frac{|\mathcal{V}_1|}{K} \rceil$  do
6    $current\_best_{\tilde{\psi}} := \text{NULL};$ 
7    $current\_best\_score := \infty;$ 
8    $\mathcal{V}_1^- := \mathcal{V}_1 \setminus \{v \in \mathcal{V}_1 \mid best_{\tilde{\psi}_{v_1 \rightarrow v_2}}(v) \neq \text{NULL}\};$ 
9    $\mathcal{V}_2^- := \mathcal{V}_2 \setminus \{best_{\tilde{\psi}_{v_1 \rightarrow v_2}}(v), \forall v \in \mathcal{V}_1\} \cup \underbrace{\{\perp, \dots, \perp\}}_{K \text{ times}};$ 
10  foreach  $(v_{11}, \dots, v_{1K}) \in \text{permutations}(\mathcal{V}_1^-, K)$  do
11    foreach  $(v_{21}, \dots, v_{2K}) \in$ 
12       $\text{unique\_permutations}(\mathcal{V}_2^-, K)$  do
13       $\tilde{\psi} := \tilde{\psi}_{v_1 \rightarrow v_2};$ 
14      for  $j = 1$  to  $K$  do
15         $\tilde{\psi}[v_{1j}] := v_{2j};$ 
16       $score := s(\tilde{\psi});$ 
17      if  $score < current\_best\_score$  then
18         $current\_best\_score := score;$ 
19         $current\_best_{\tilde{\psi}} := \tilde{\psi};$ 
20   $\tilde{\psi}_{v_1 \rightarrow v_2} := current\_best_{\tilde{\psi}};$ 
21   $score_{\tilde{\psi}_{v_1 \rightarrow v_2}} := current\_best\_score;$ 
22 return  $\tilde{\psi}_{v_1 \rightarrow v_2}, score_{\tilde{\psi}_{v_1 \rightarrow v_2}};$ 

```

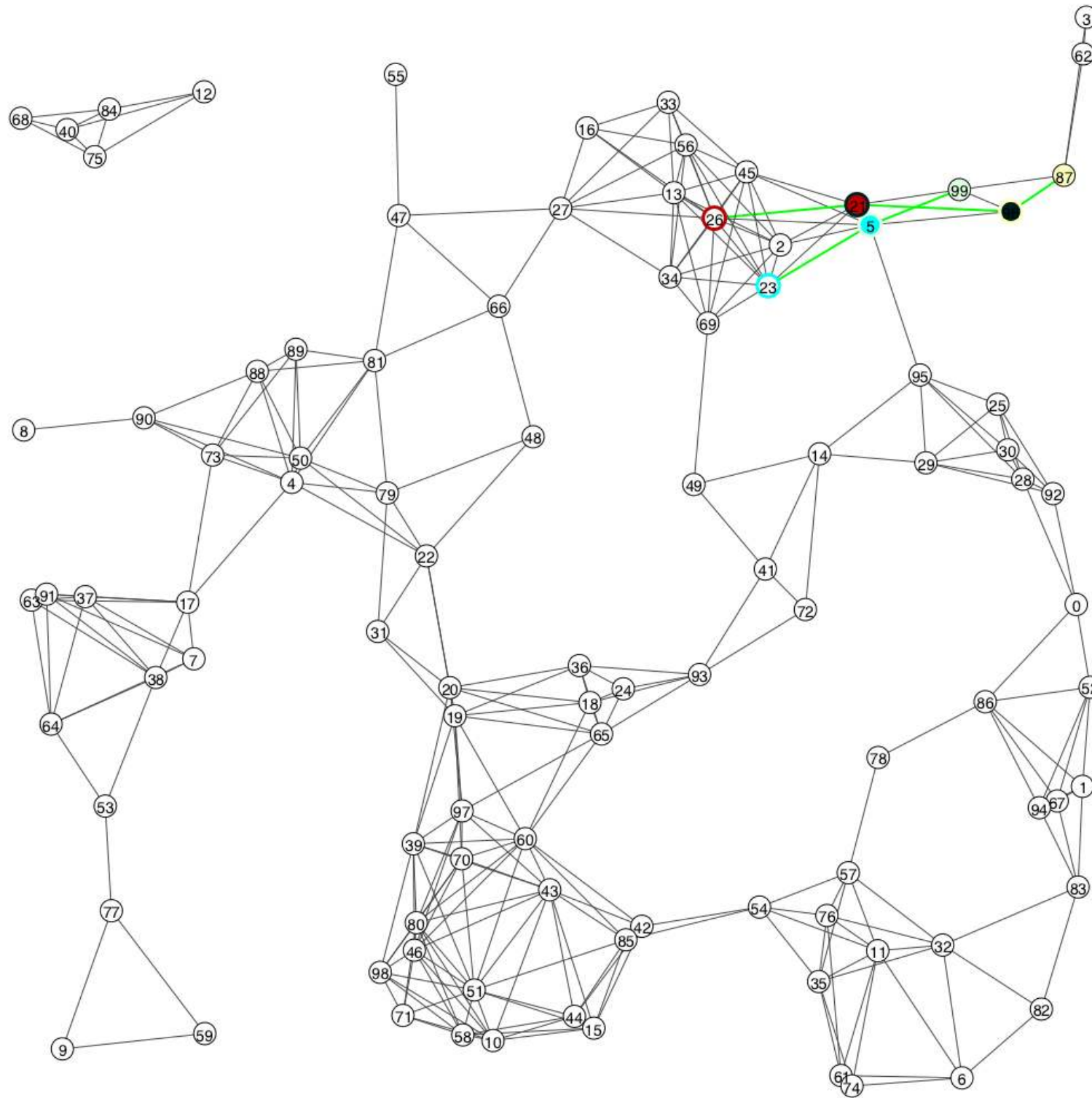
Controls
complexity

Results — Random geometric graph (1)



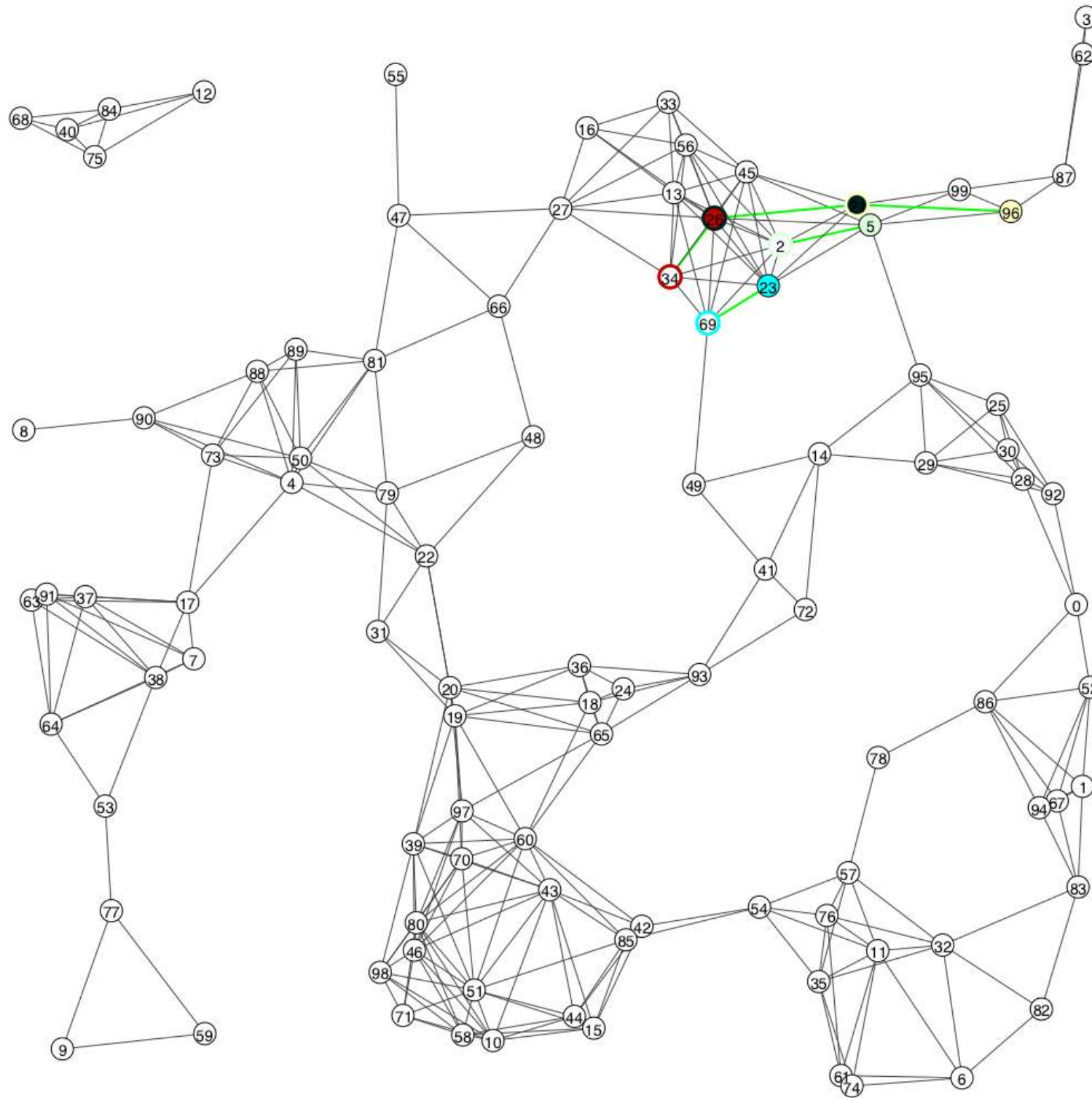
Objective

Results — Random geometric graph (1)



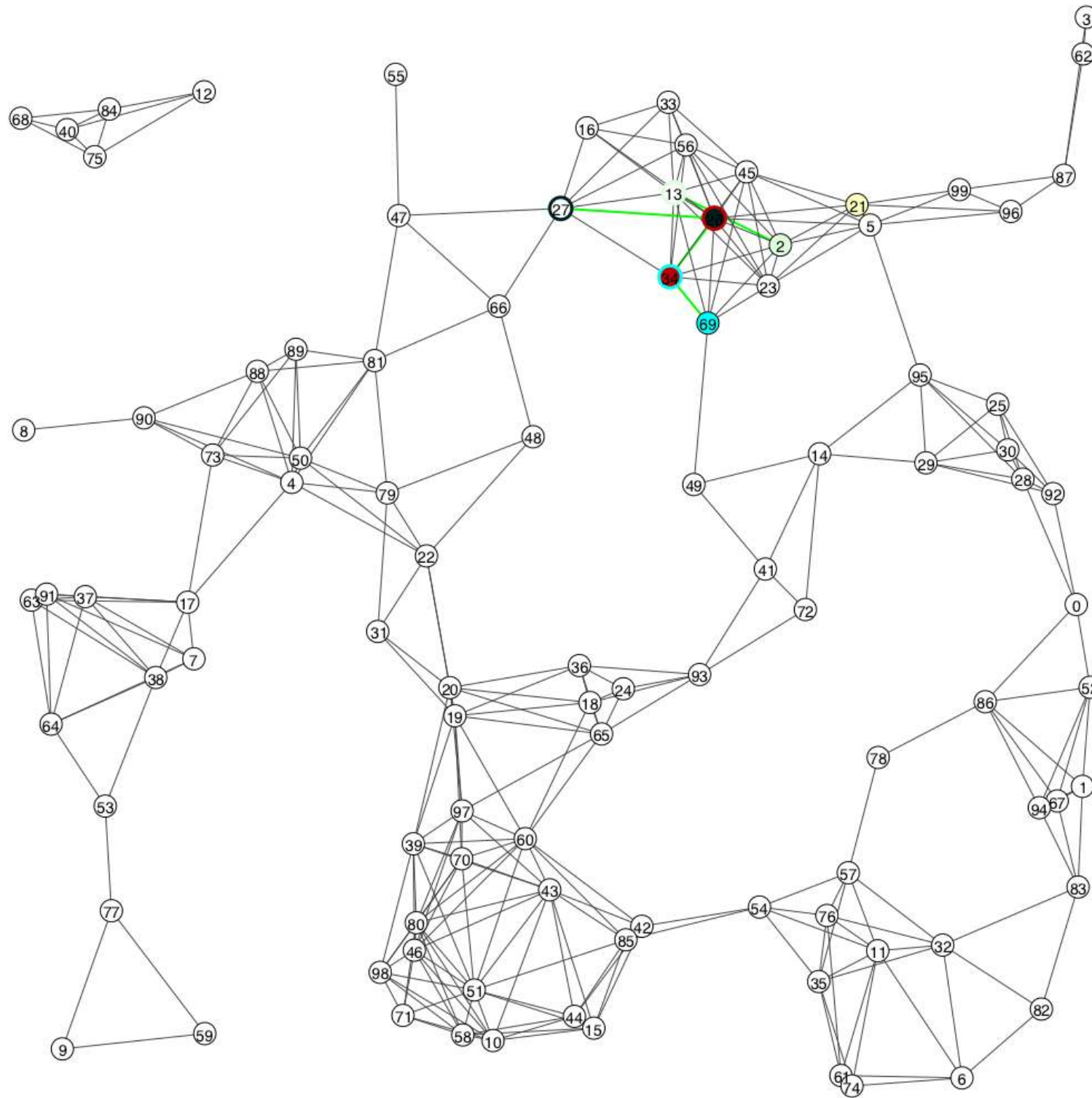
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



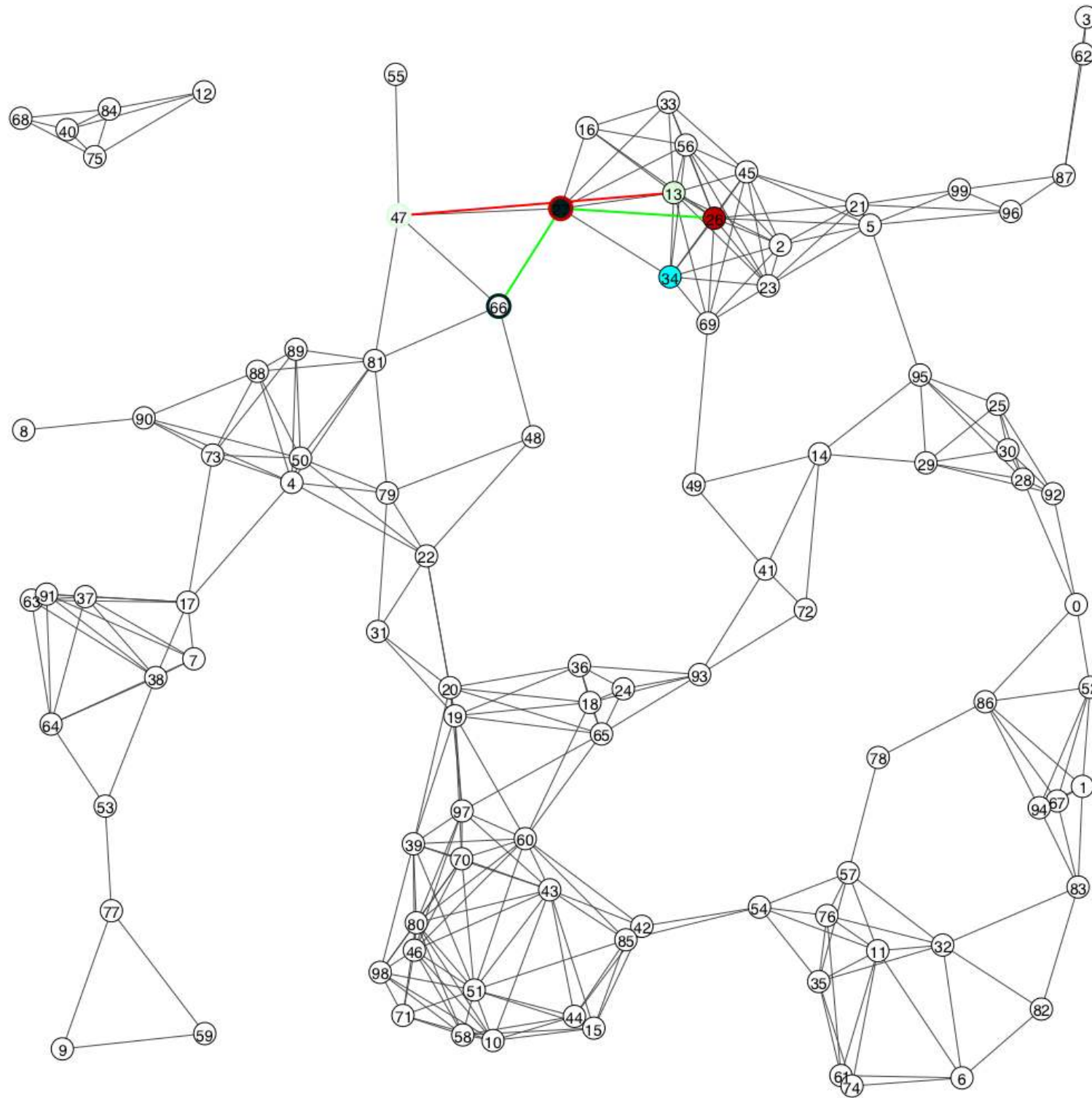
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Results — Random geometric graph (1)



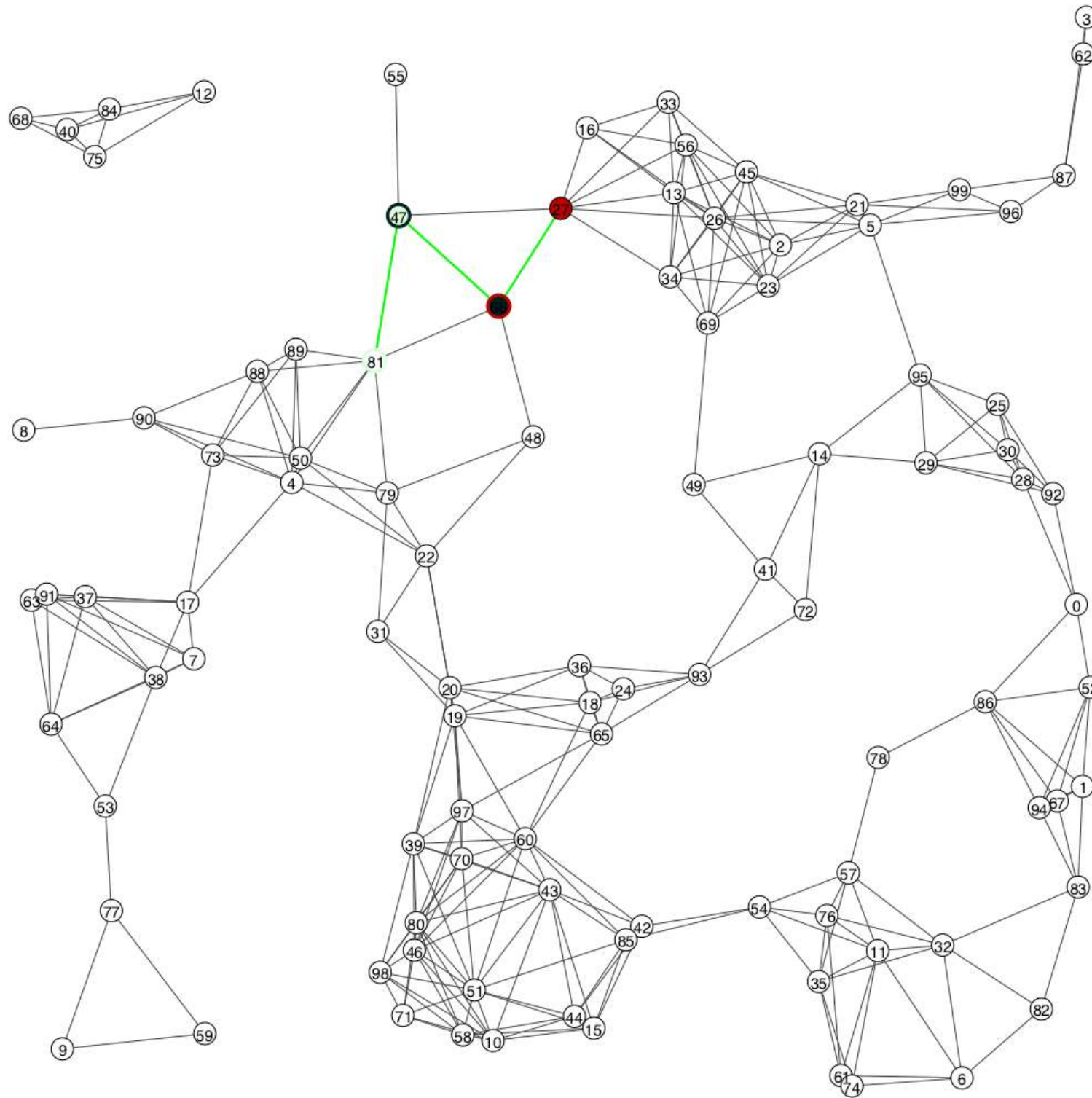
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Results — Random geometric graph (1)



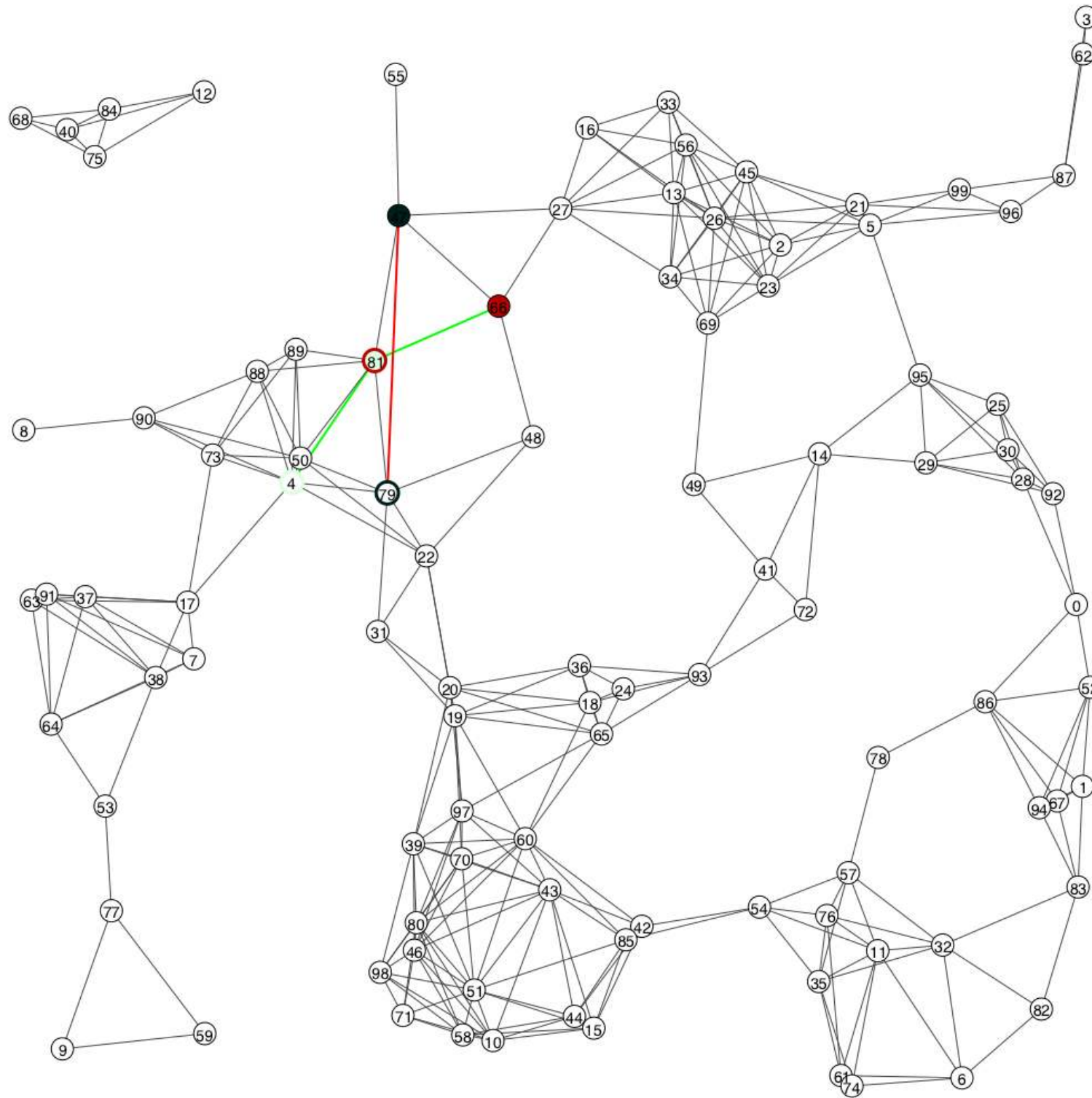
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Results — Random geometric graph (1)



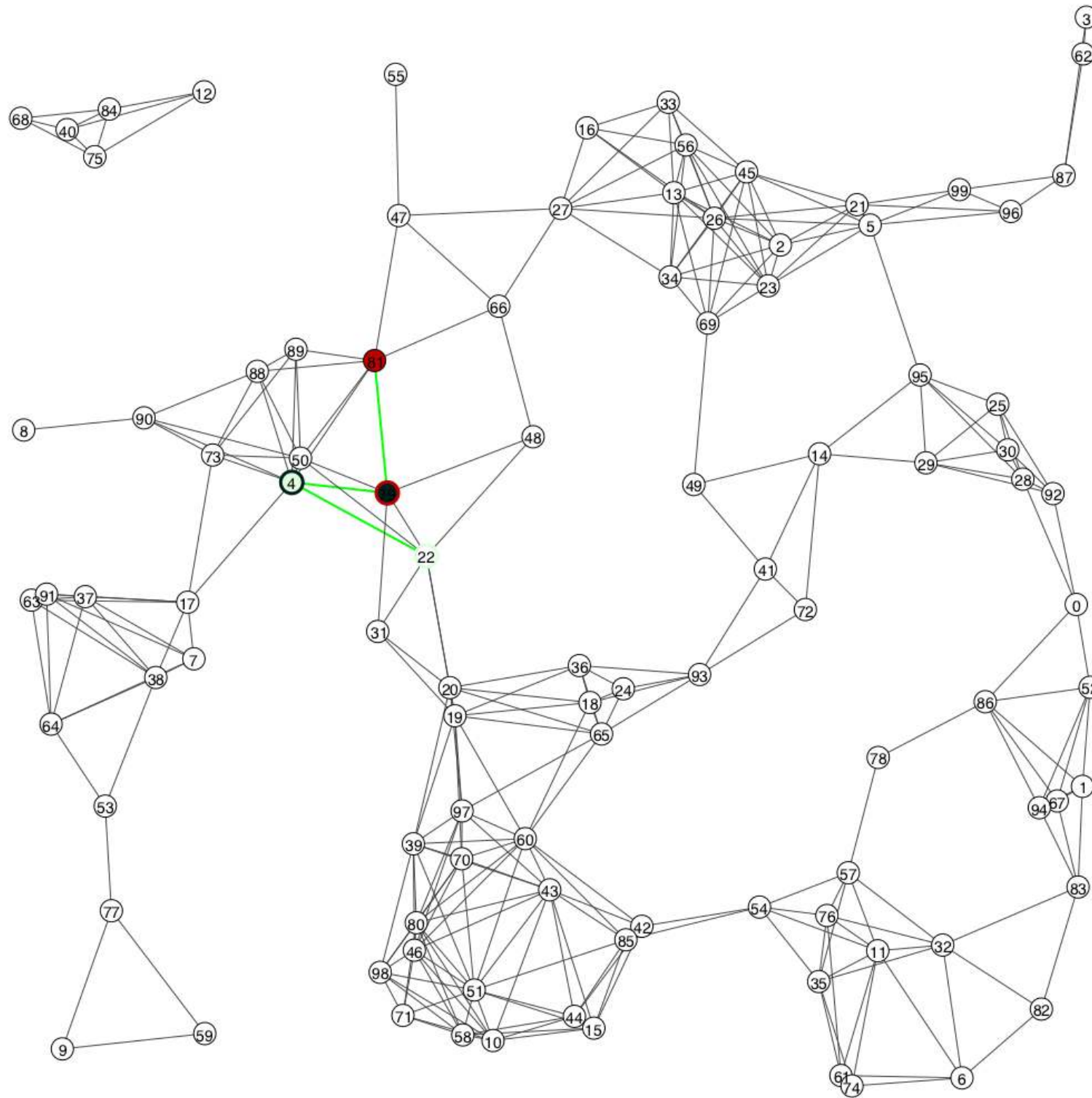
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Results — Random geometric graph (1)



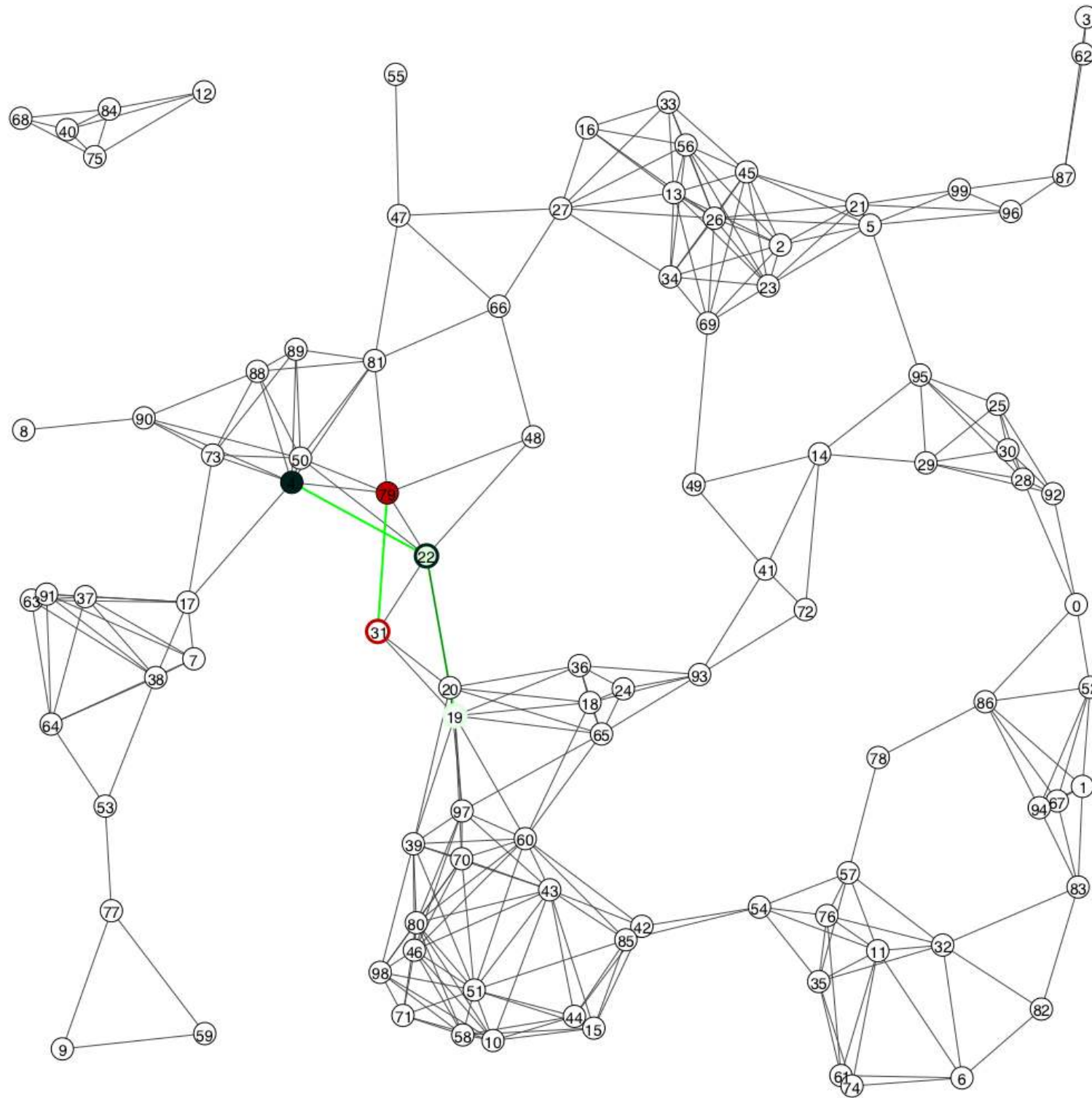
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Results — Random geometric graph (1)



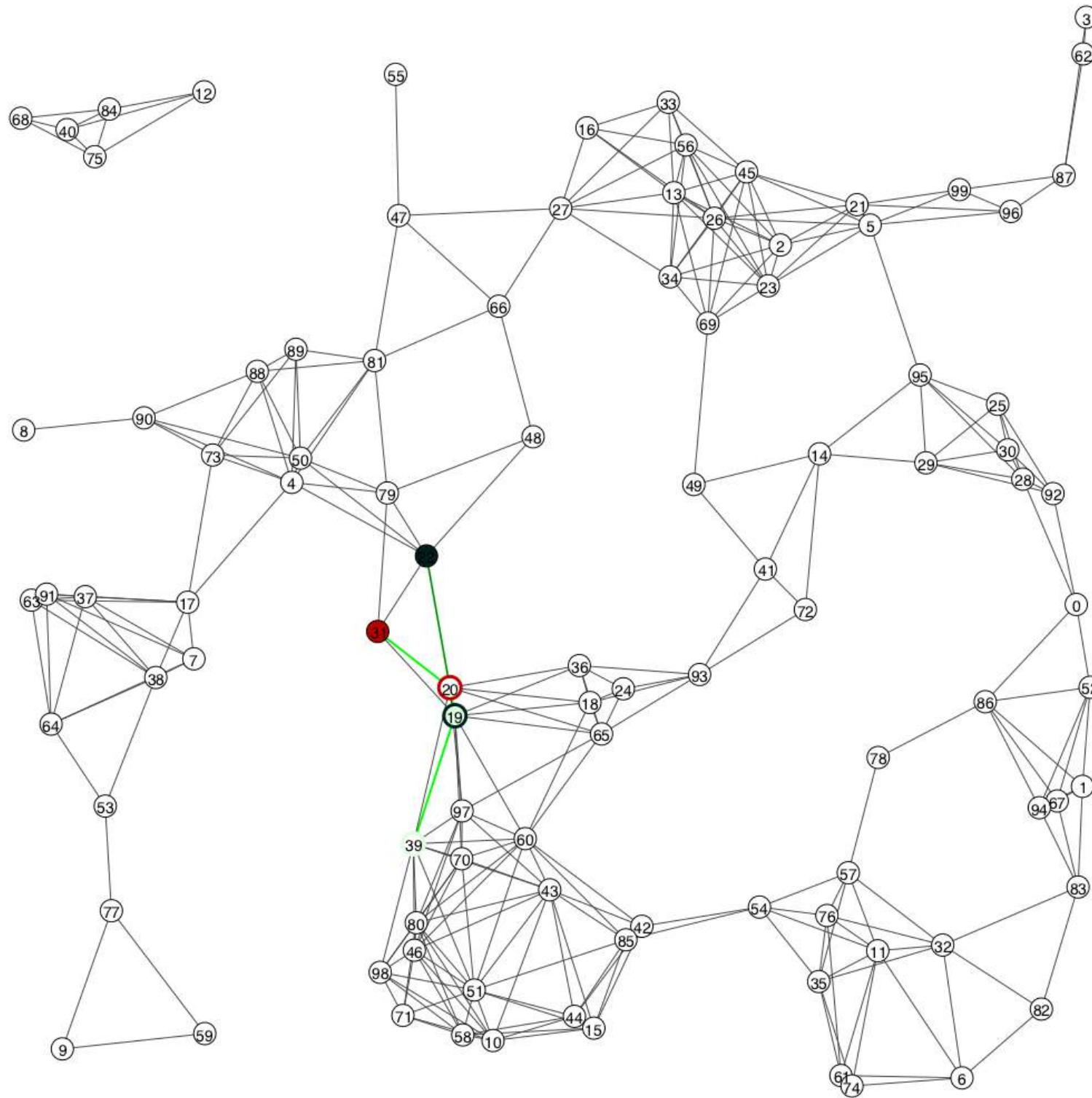
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



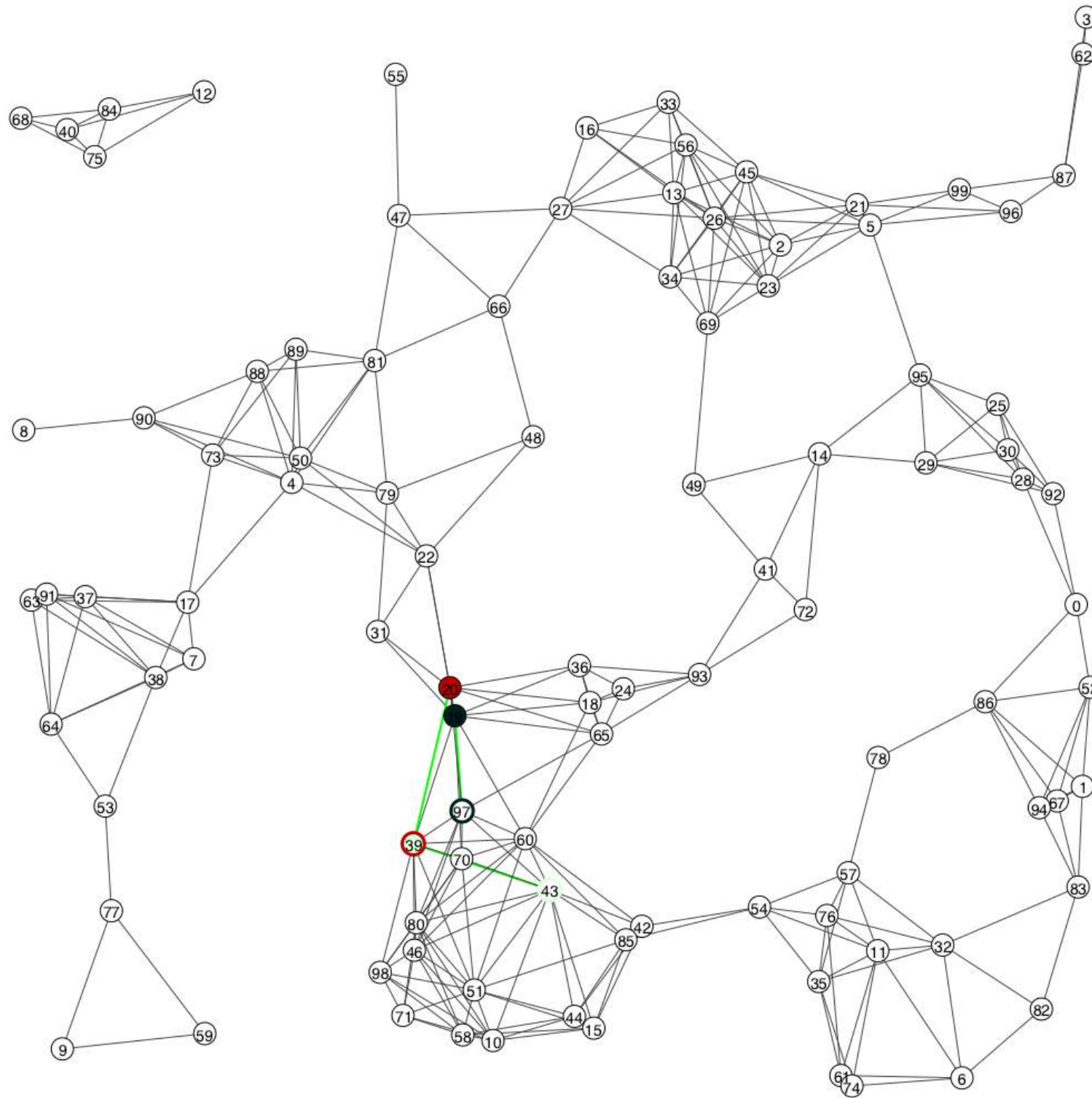
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



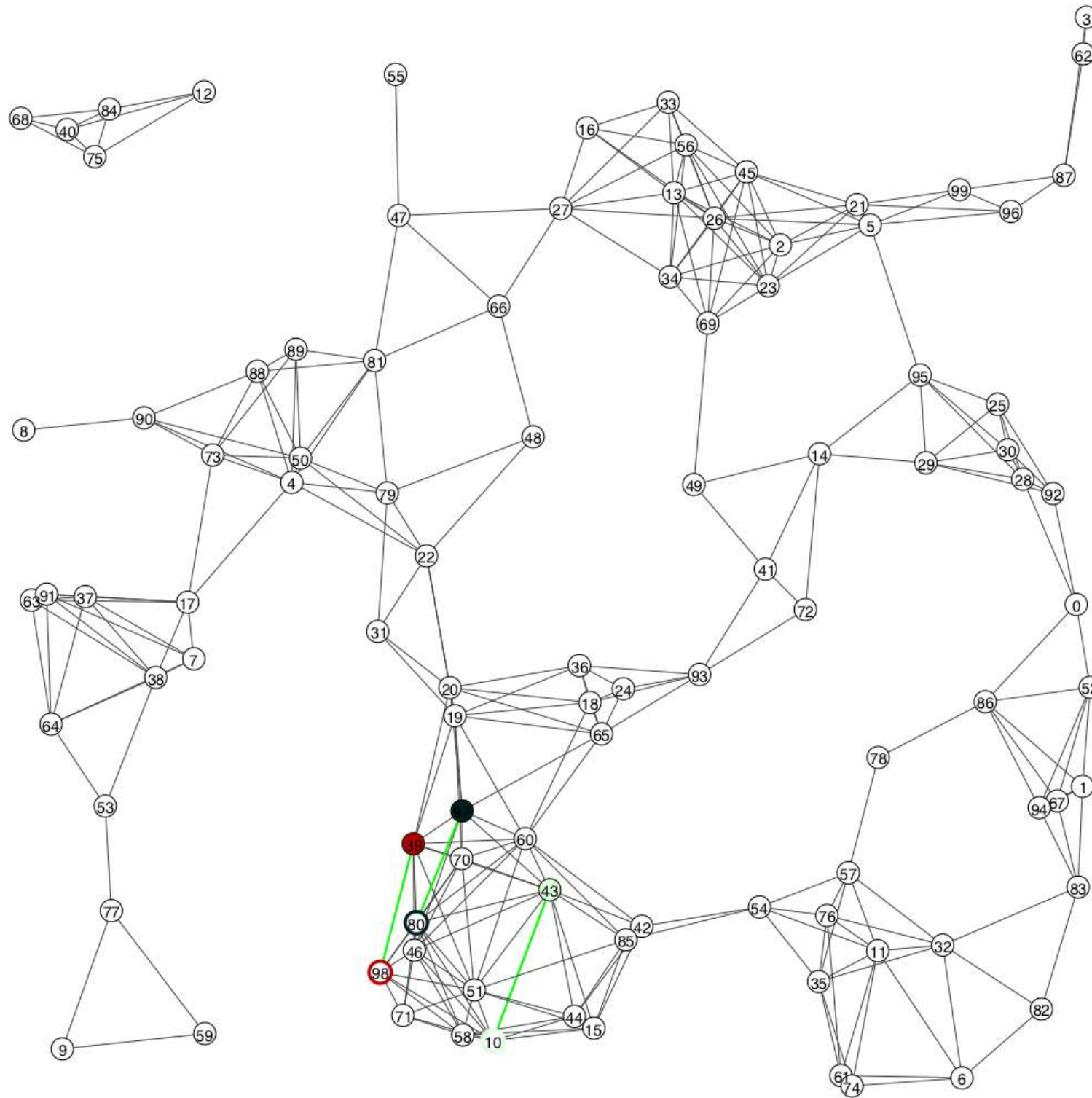
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Results — Random geometric graph (1)



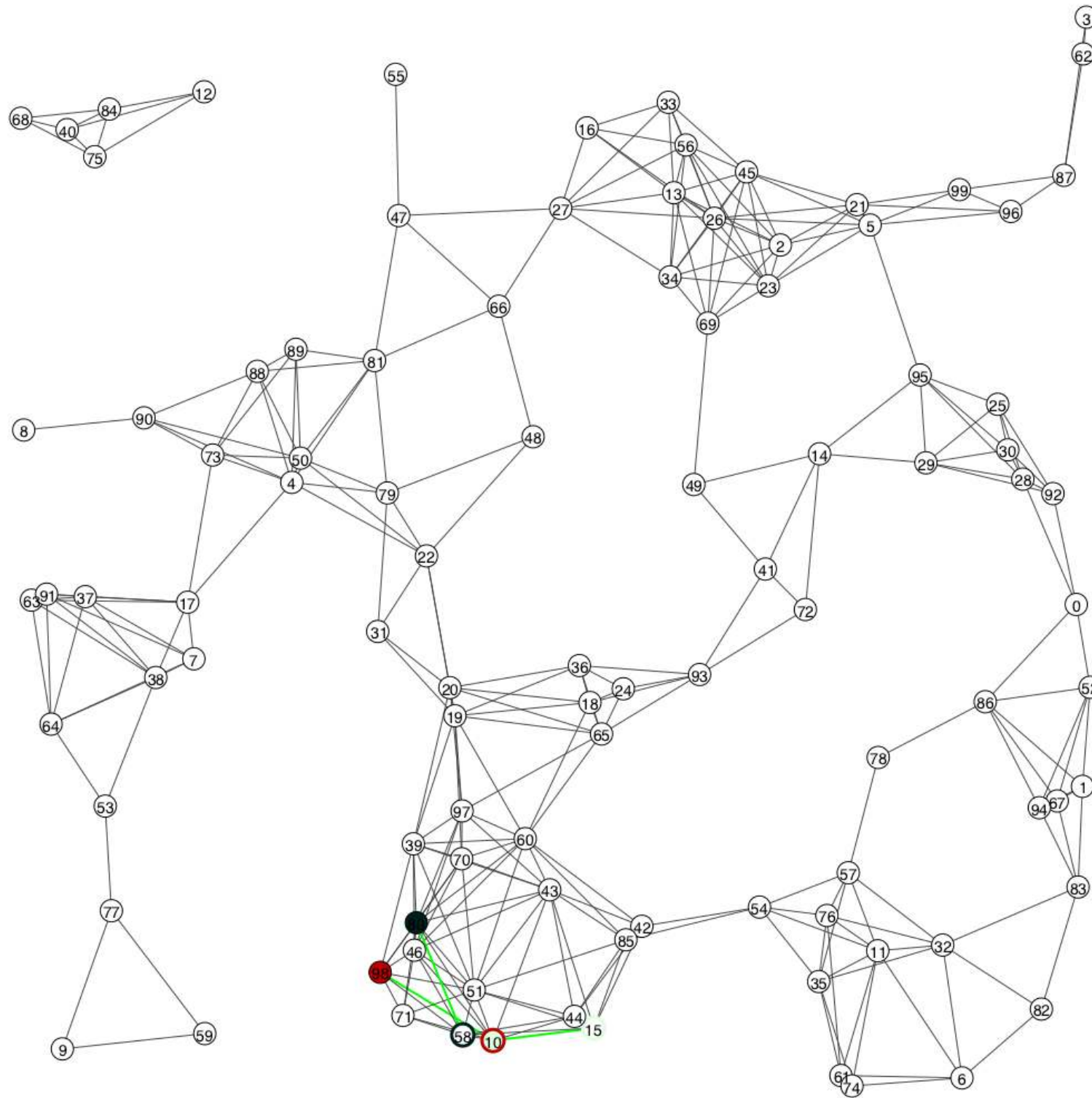
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Results — Random geometric graph (1)



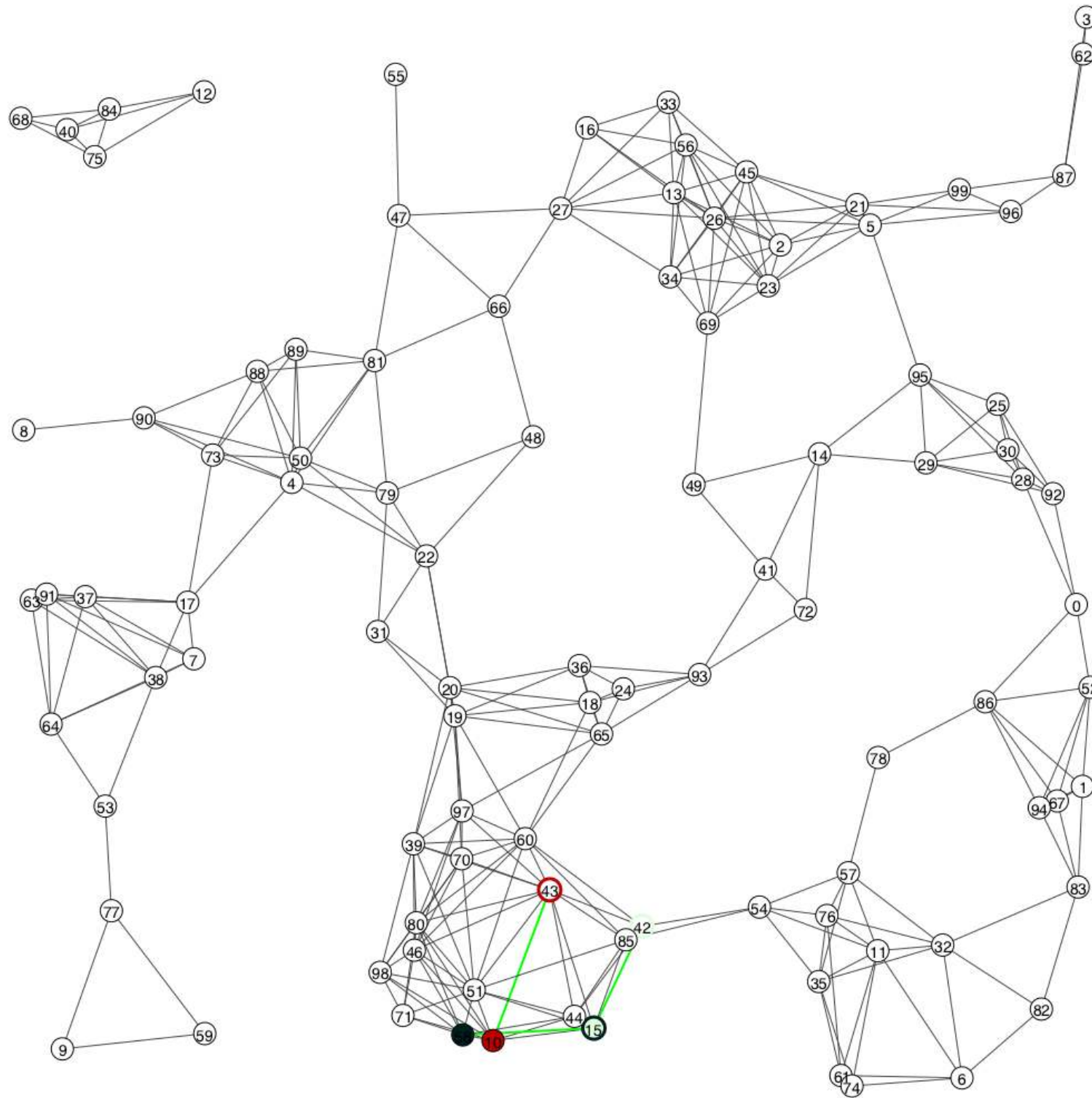
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



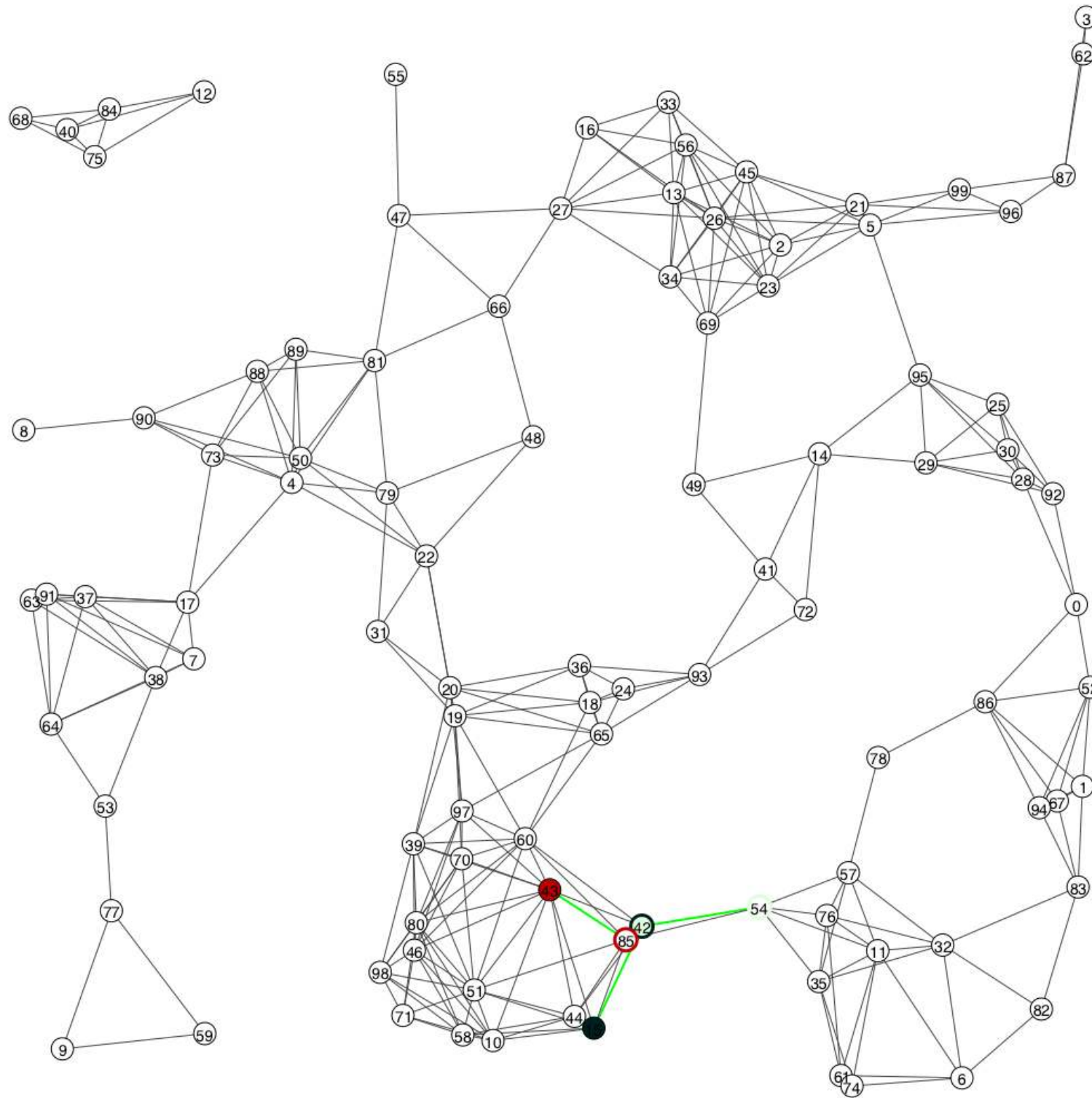
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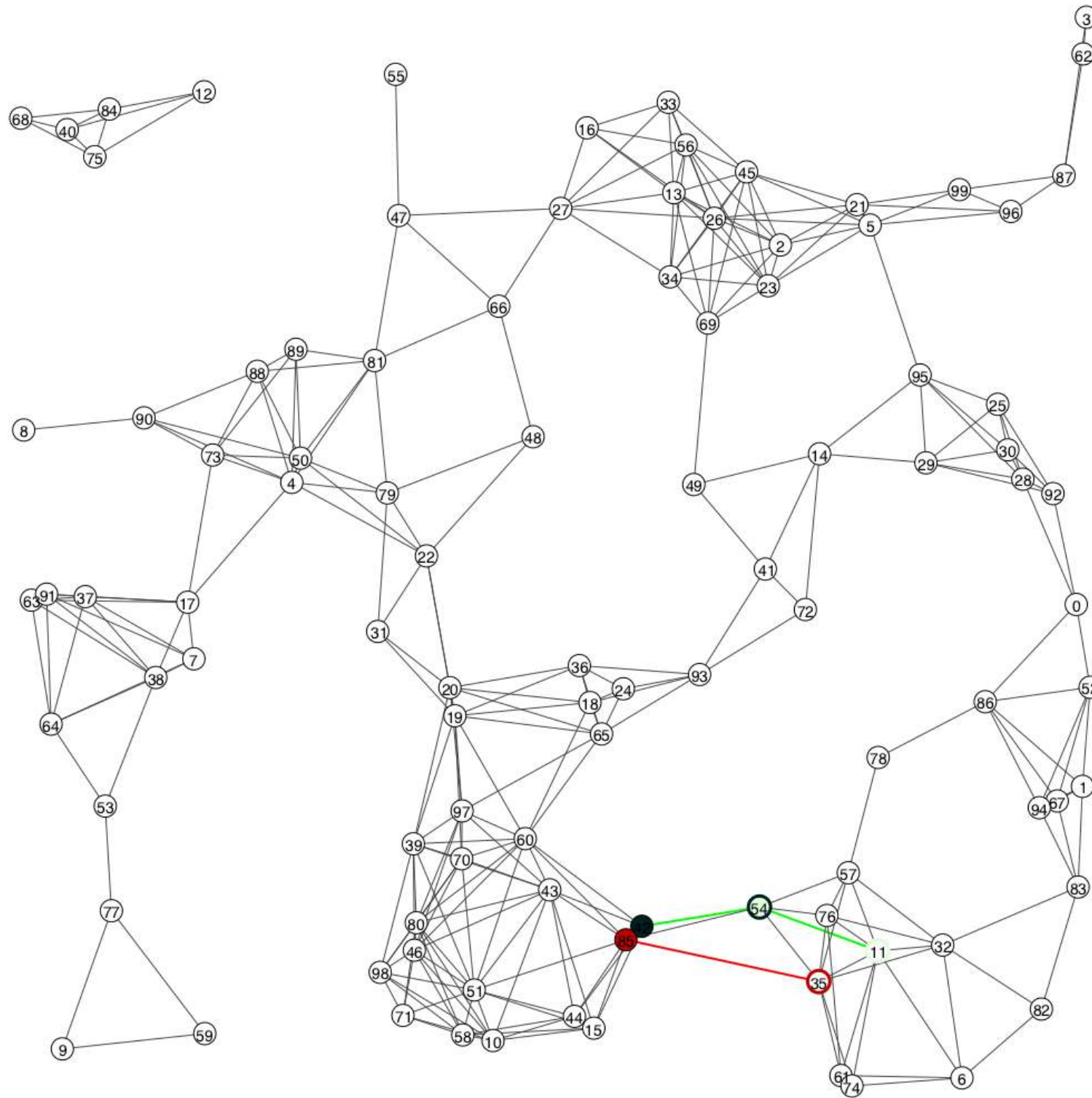
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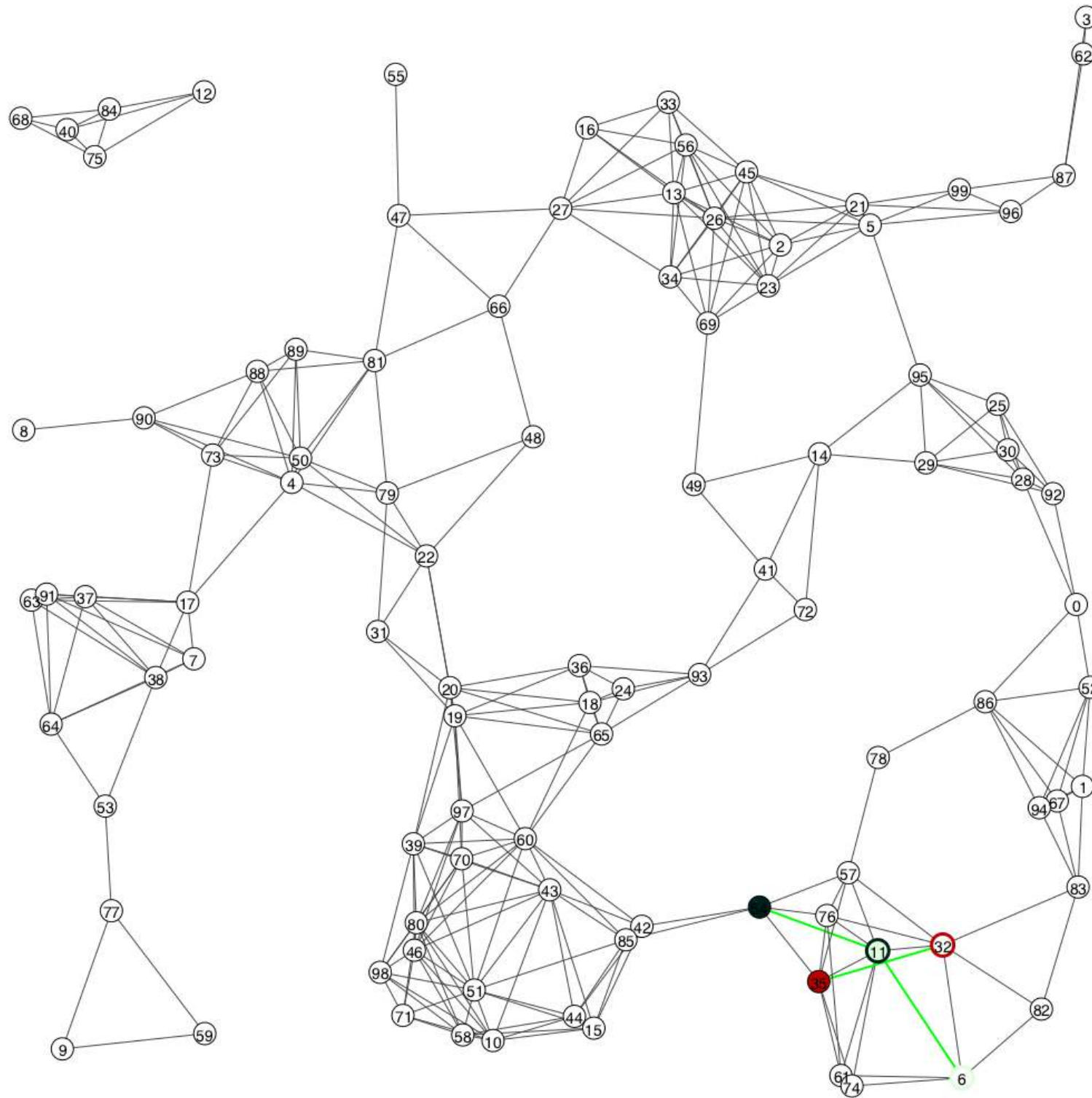
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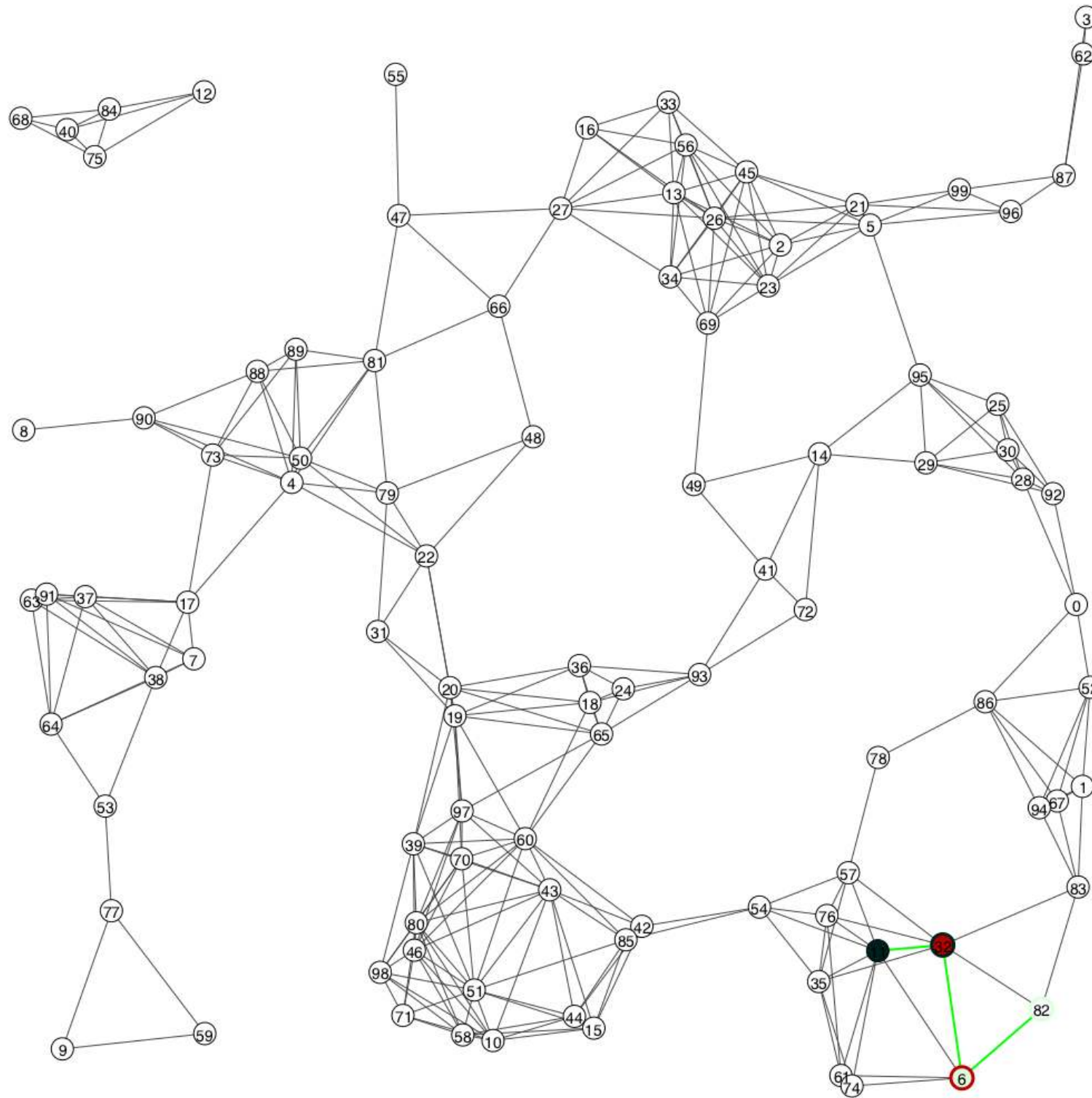
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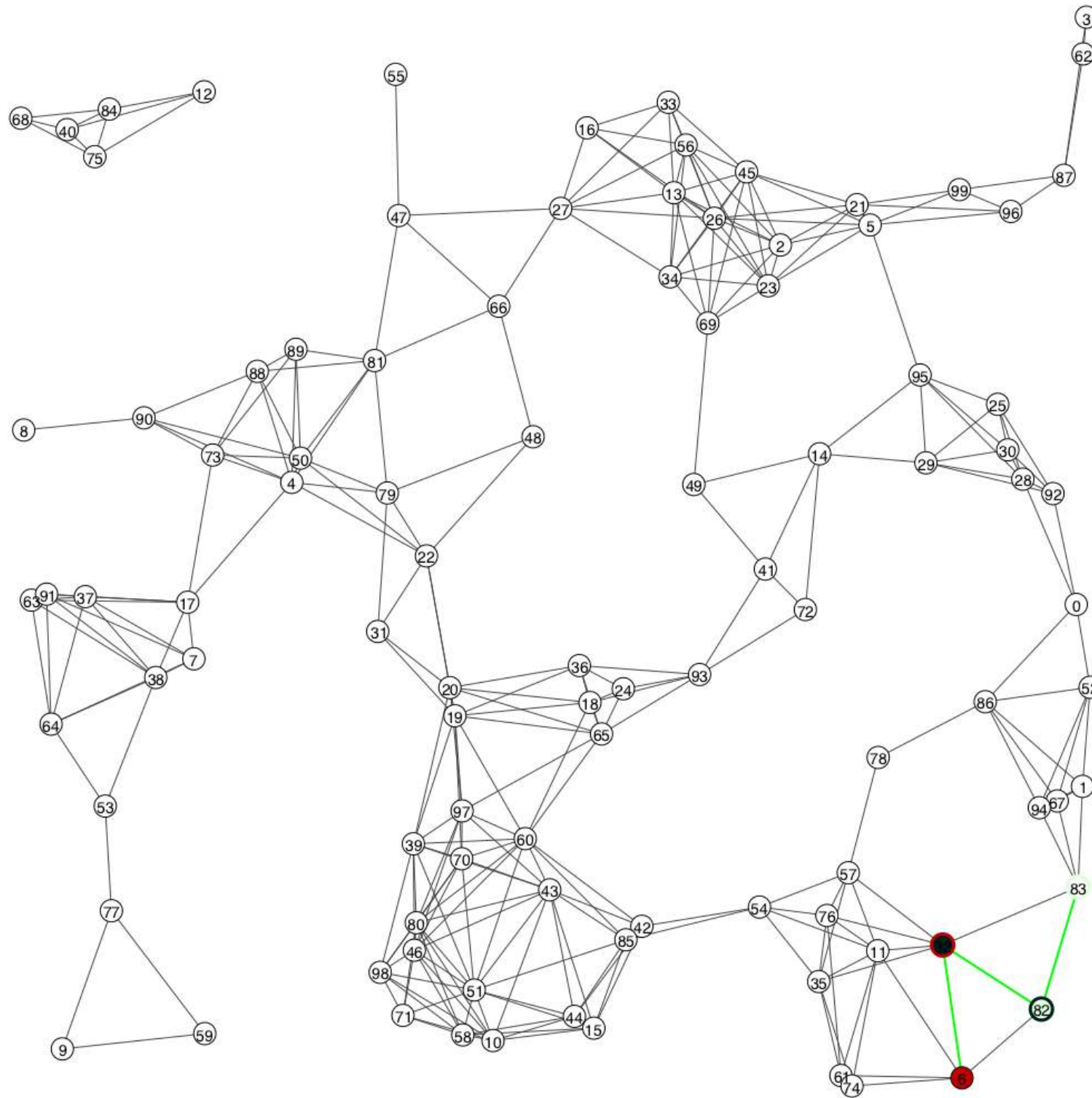
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



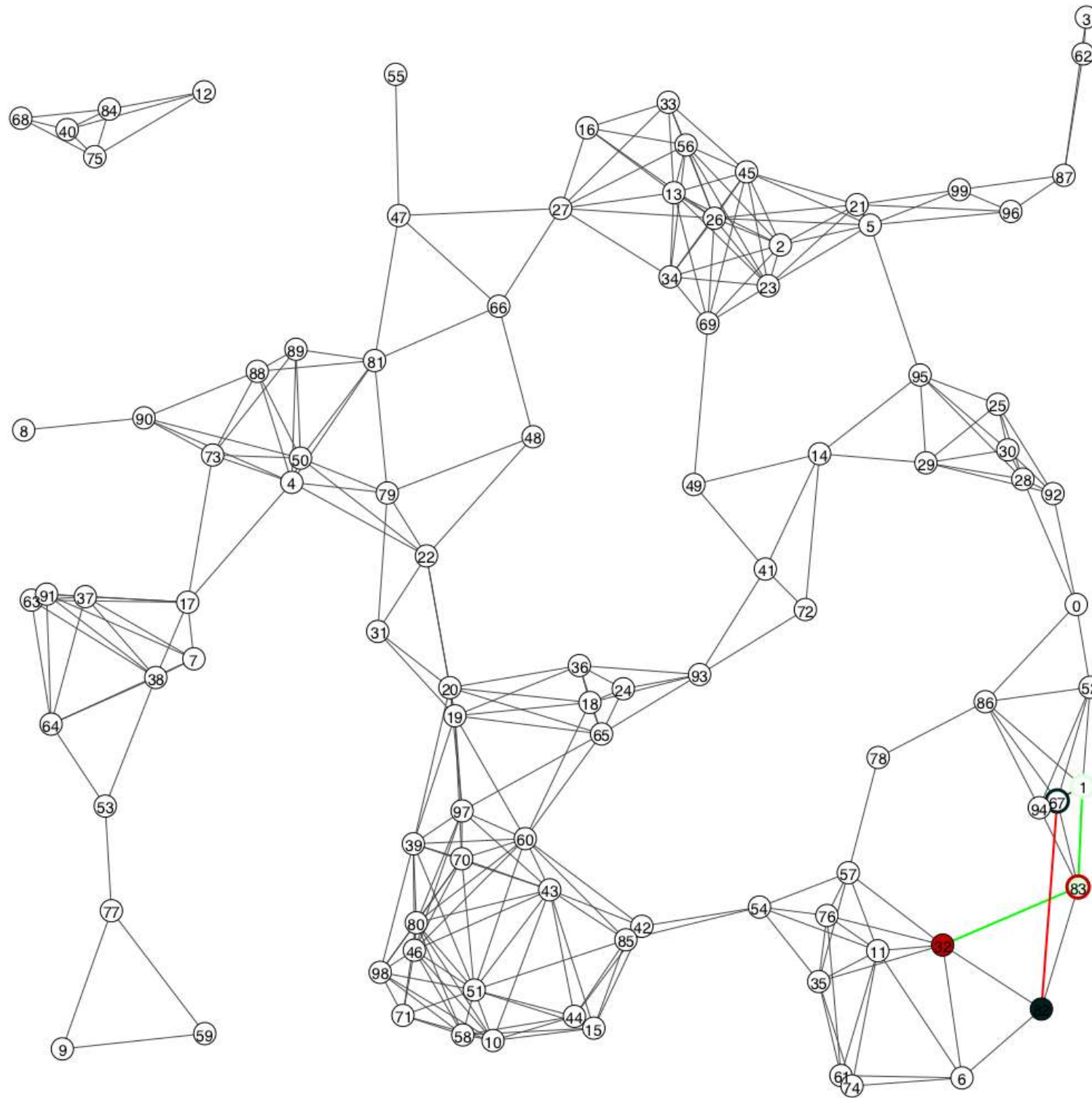
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



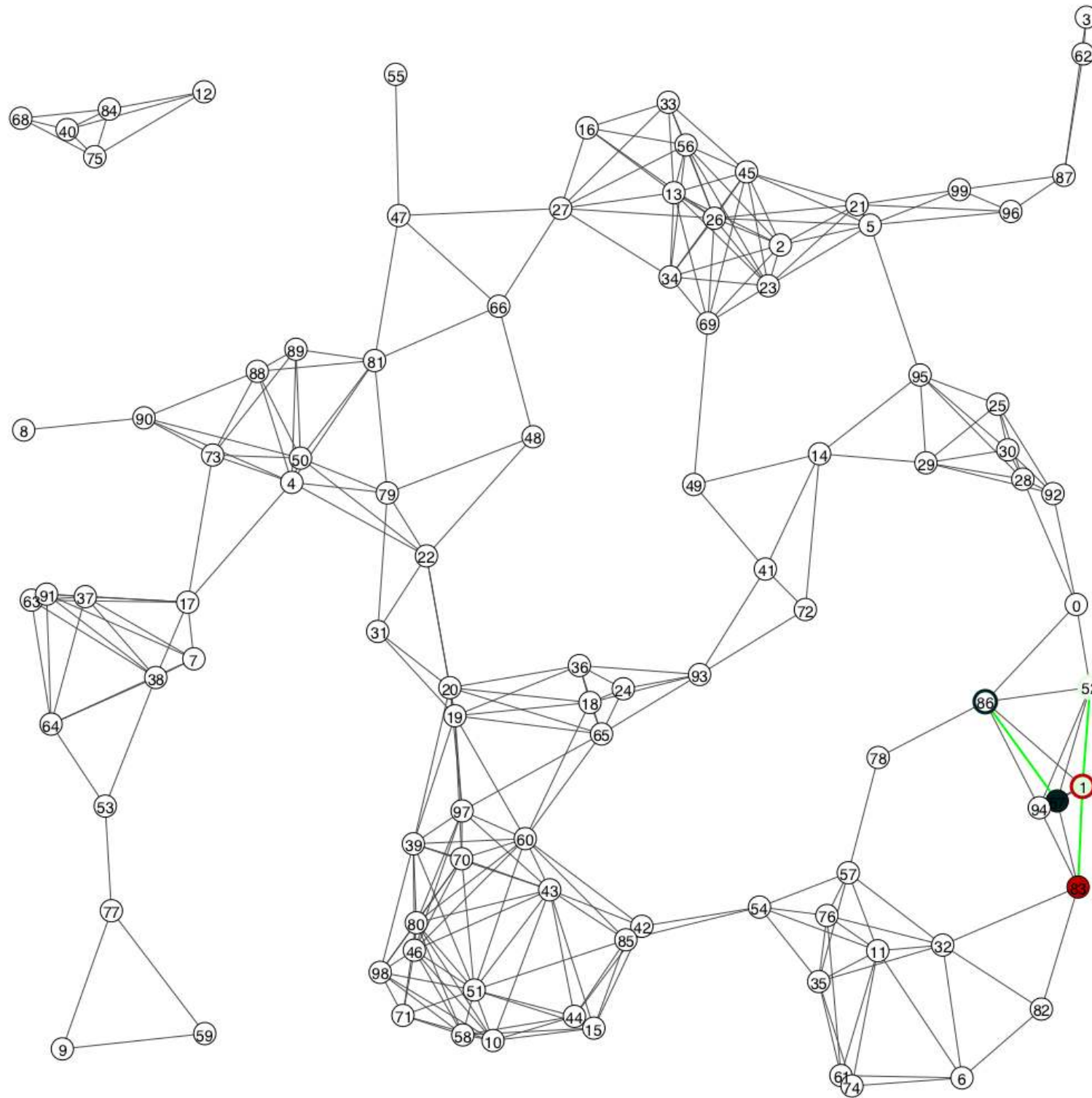
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



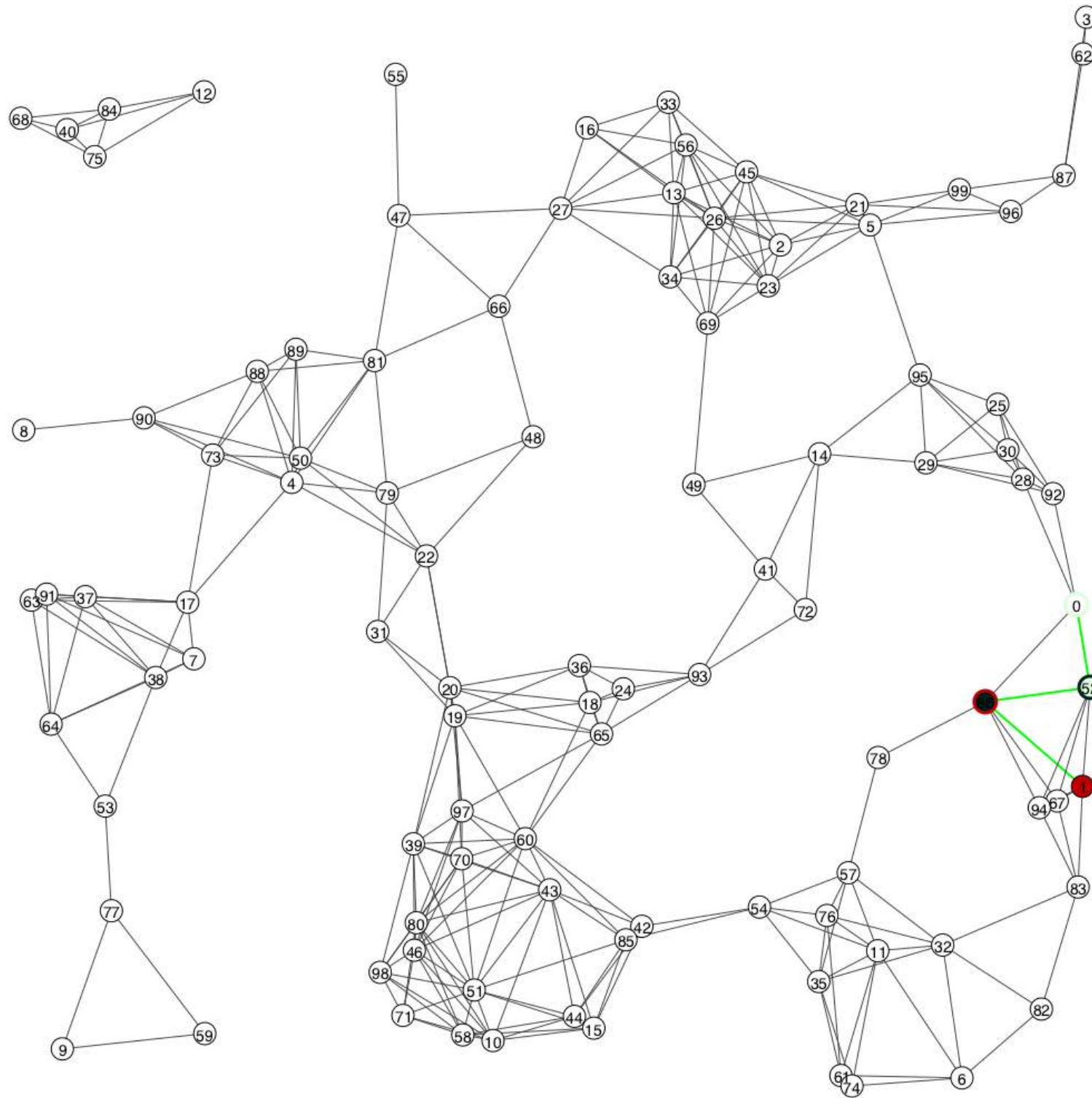
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



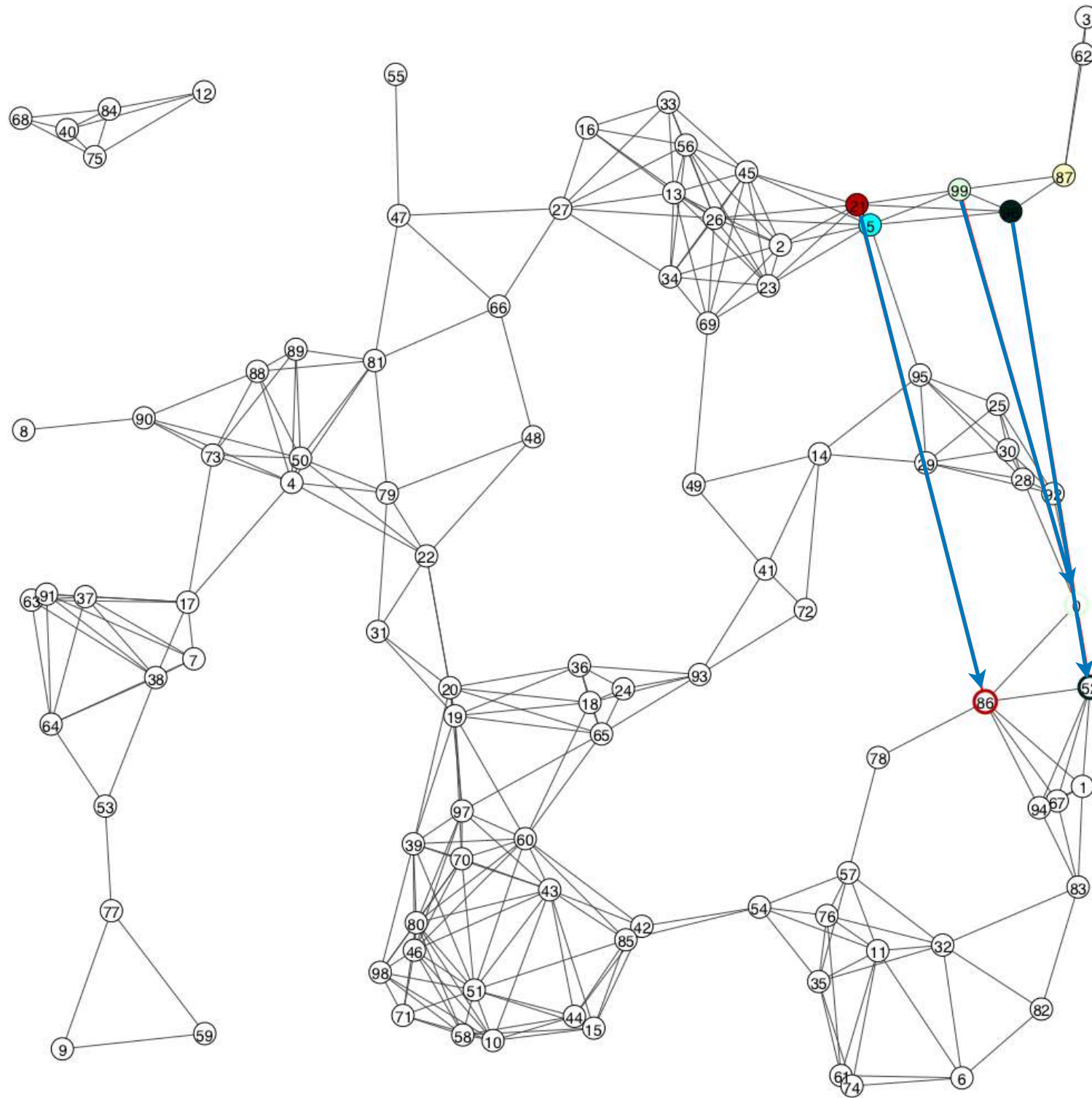
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



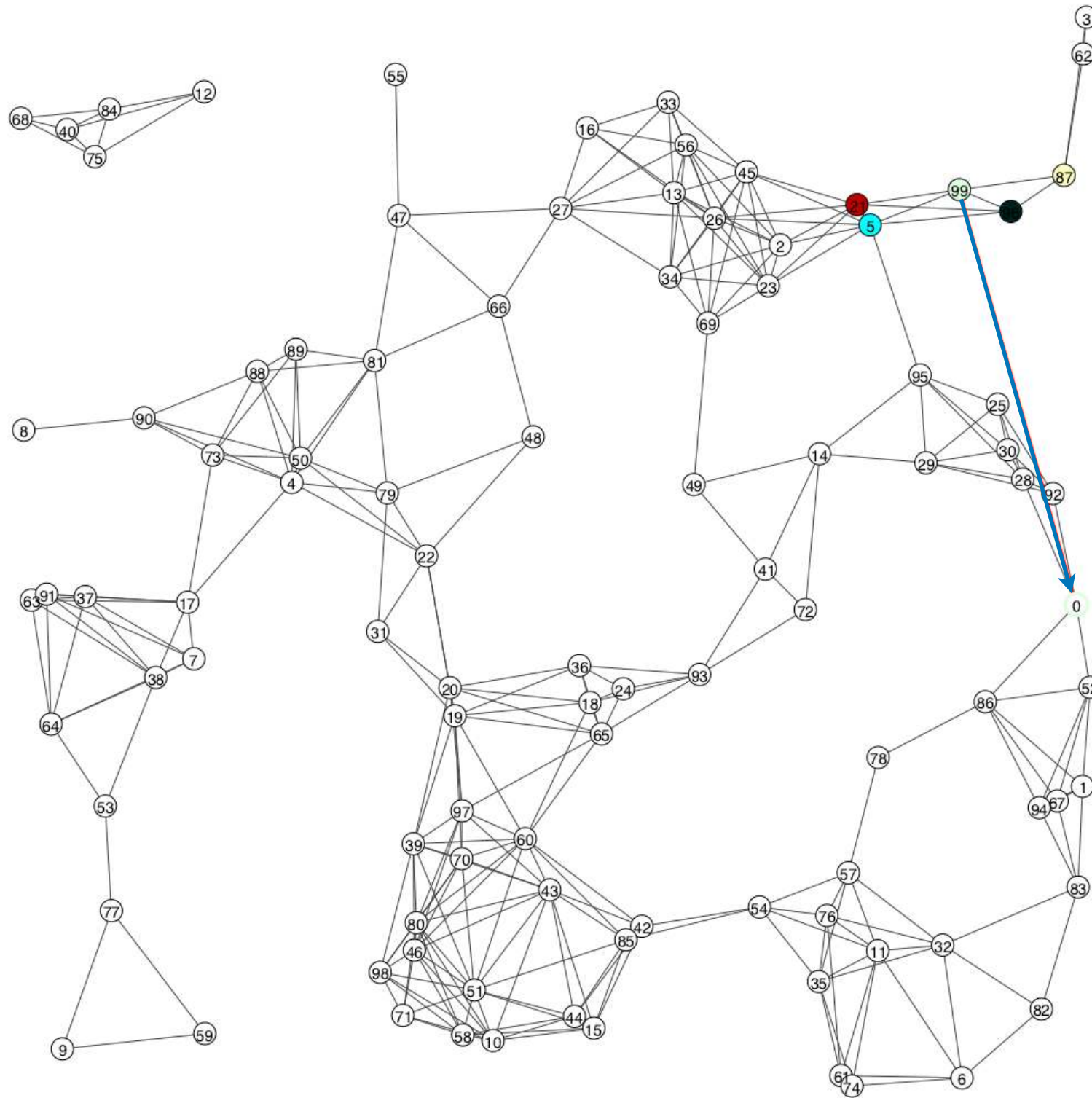
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ $\rightarrow$ $loss = 40\%$ $SNP = 0\%$

Results — Random geometric graph (1)



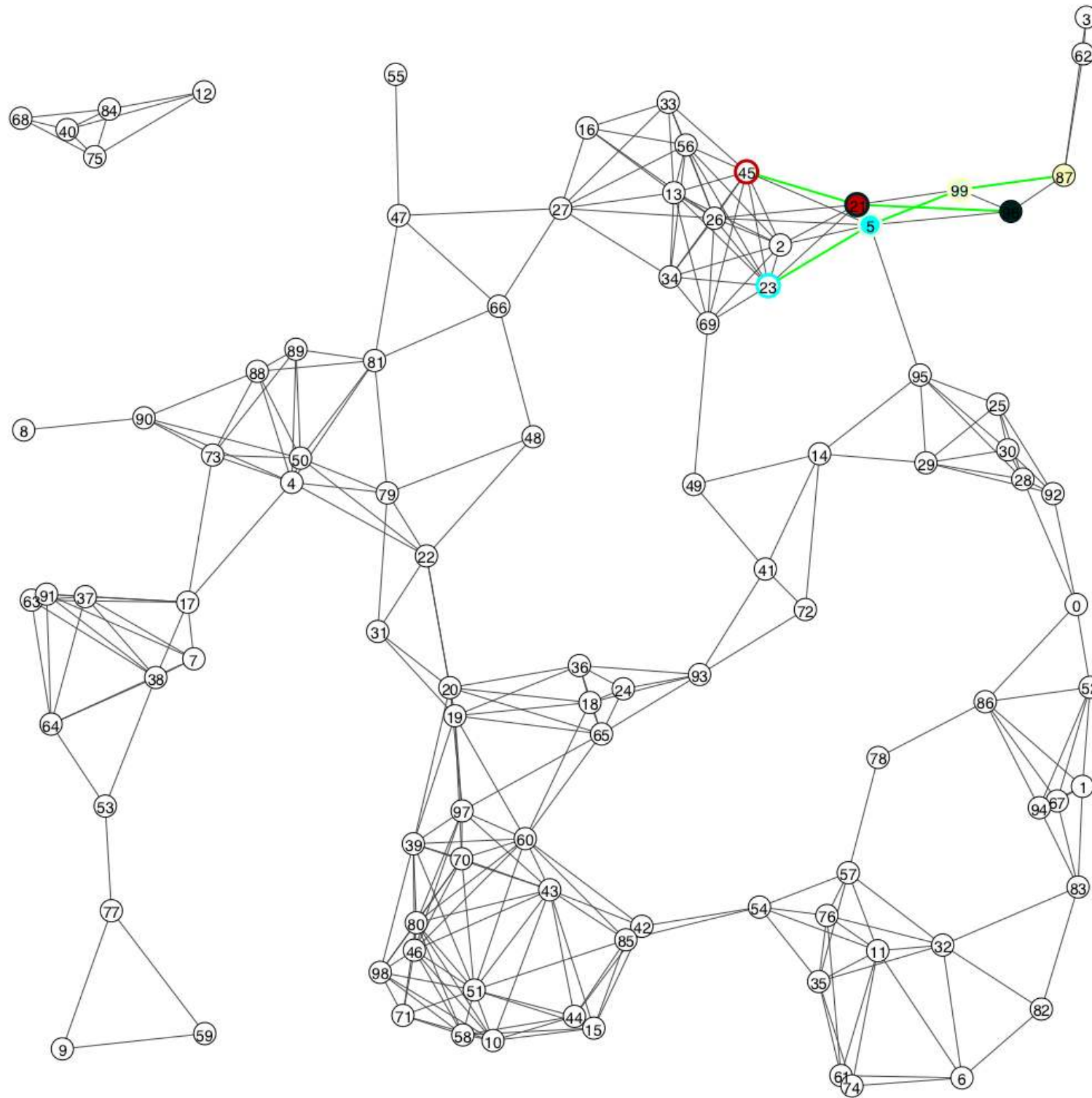
Composition

Results — Random geometric graph (1)



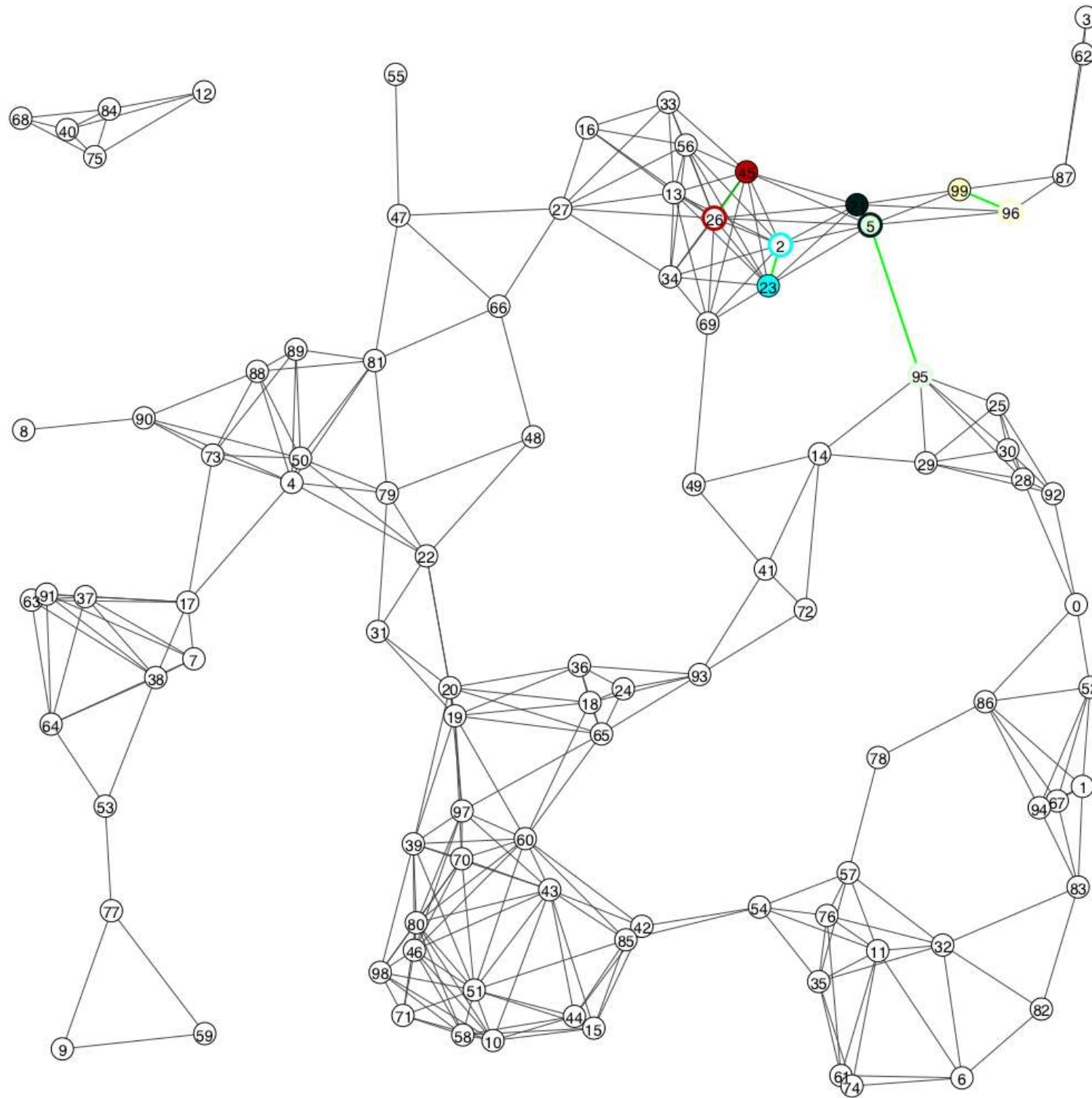
Objective

Results — Random geometric graph (1)



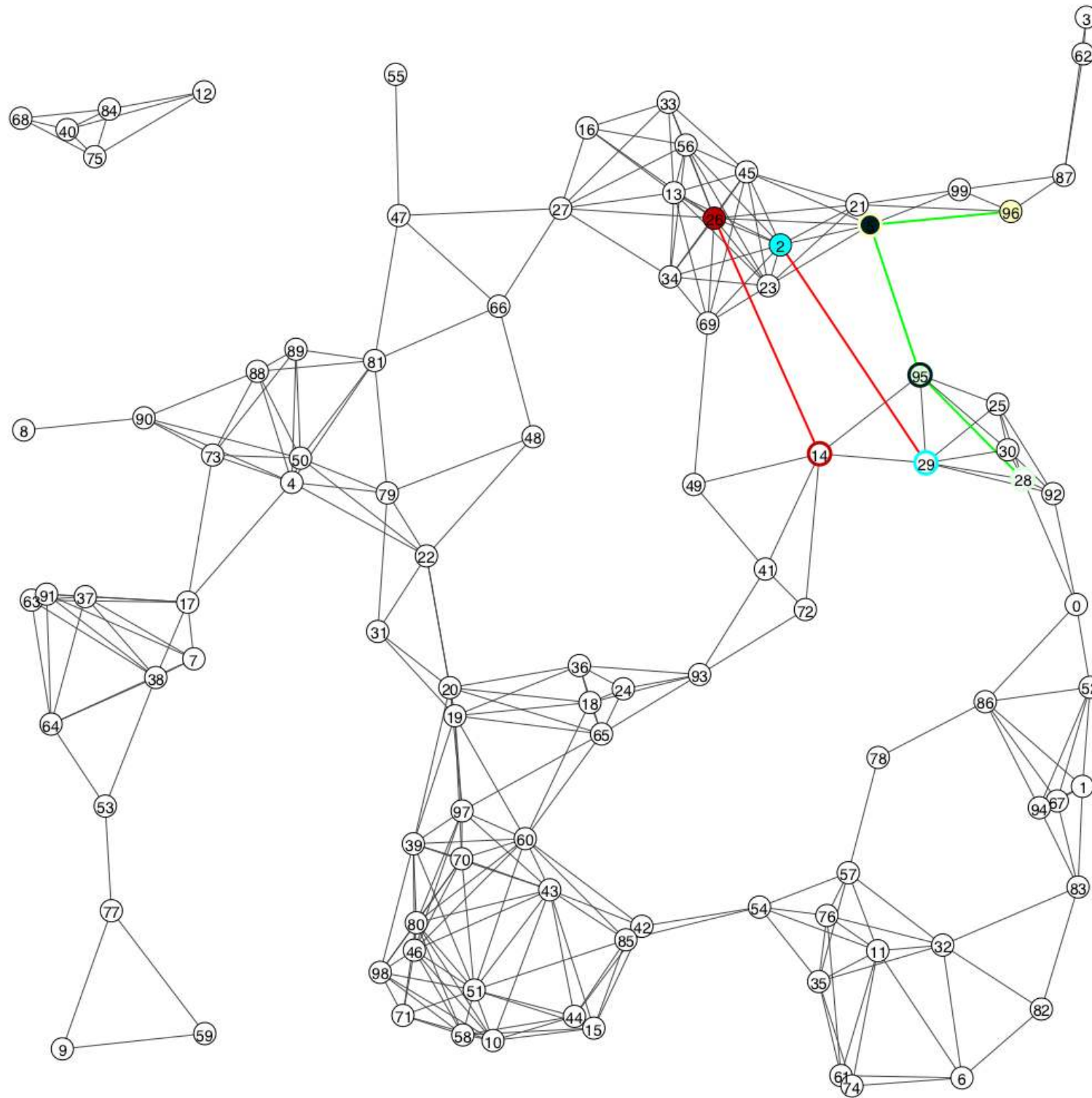
$K \in \{3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 1$ $\rightarrow$ $loss = 0\%$ $SNP = 30\%$

Results — Random geometric graph (1)



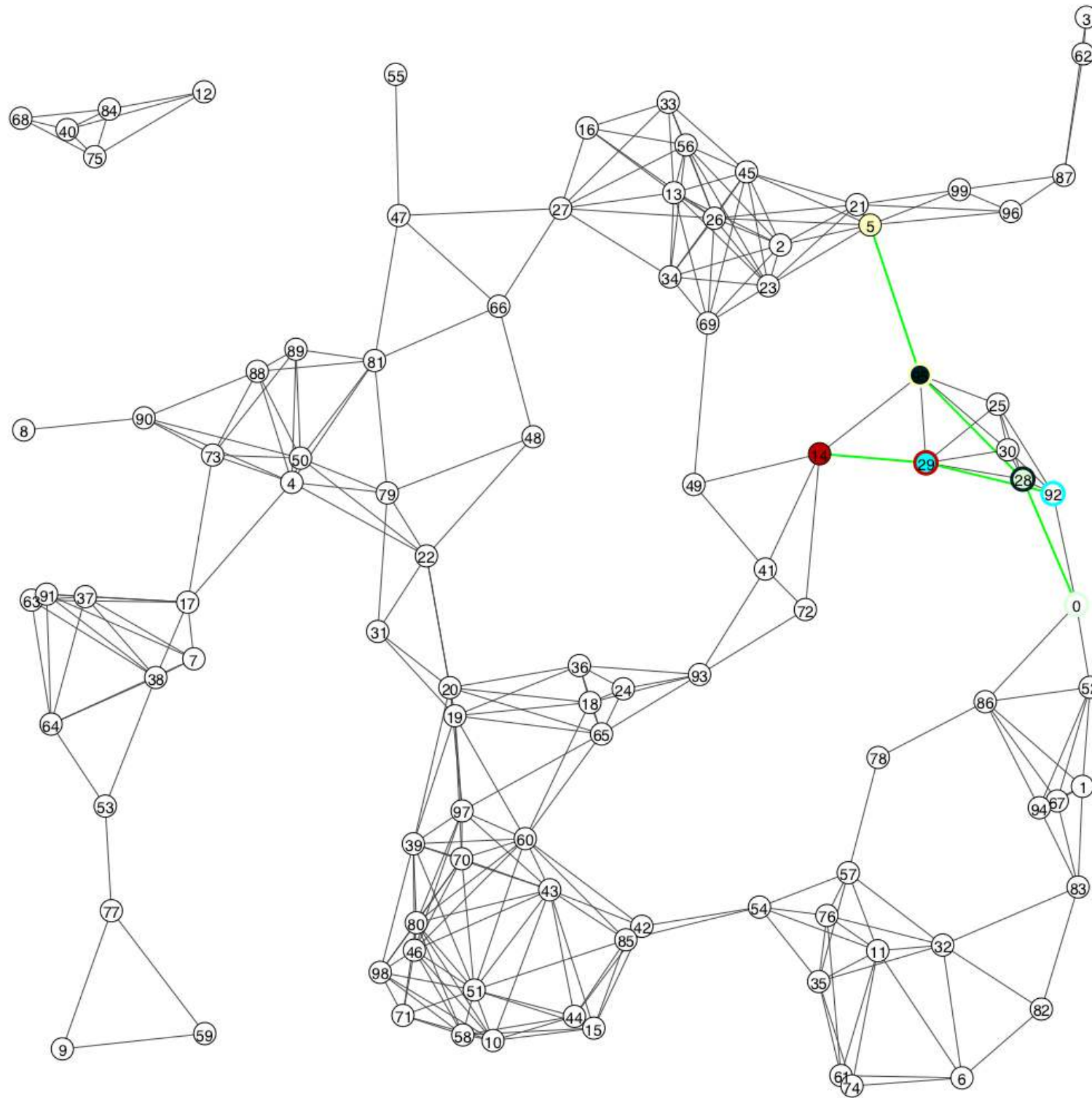
$K \in \{3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 1$ $\rightarrow$ $loss = 0\%$ $SNP = 30\%$

Results — Random geometric graph (1)



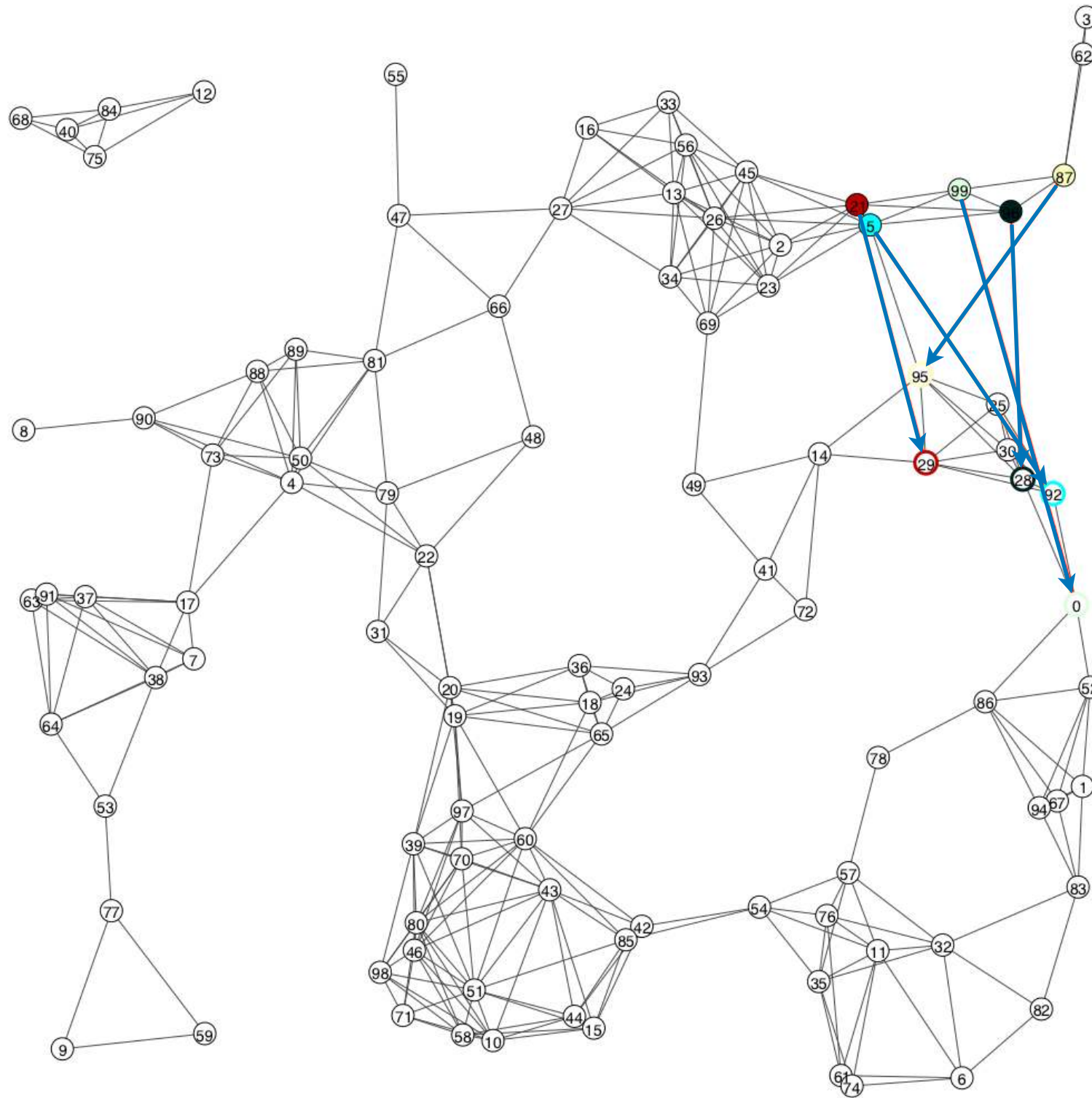
$K \in \{3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 1$ $\rightarrow$ $loss = 0\%$ $SNP = 30\%$

Results — Random geometric graph (1)



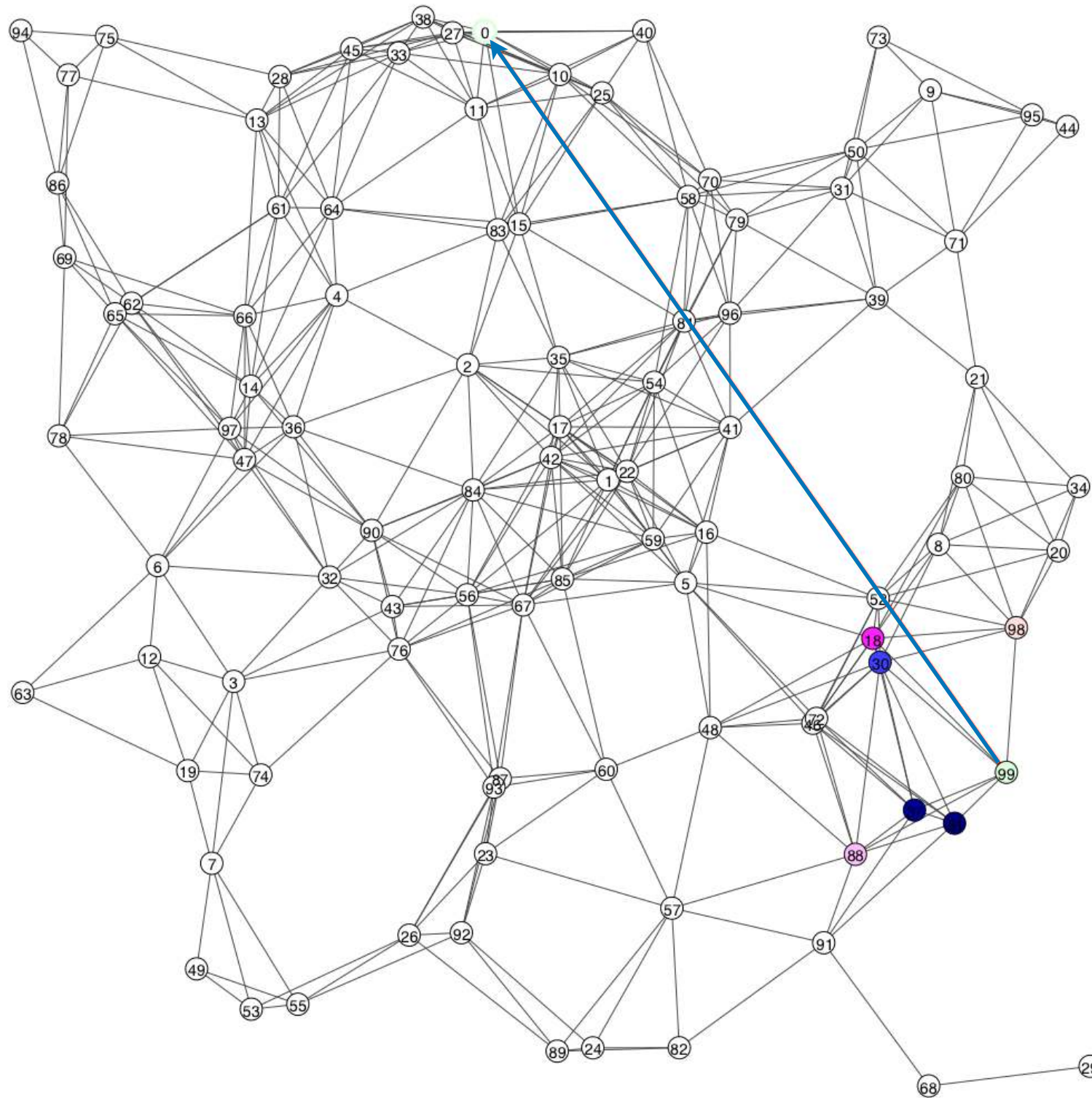
$K \in \{3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 1$ $\rightarrow$ $loss = 0\%$ $SNP = 30\%$

Results — Random geometric graph (1)



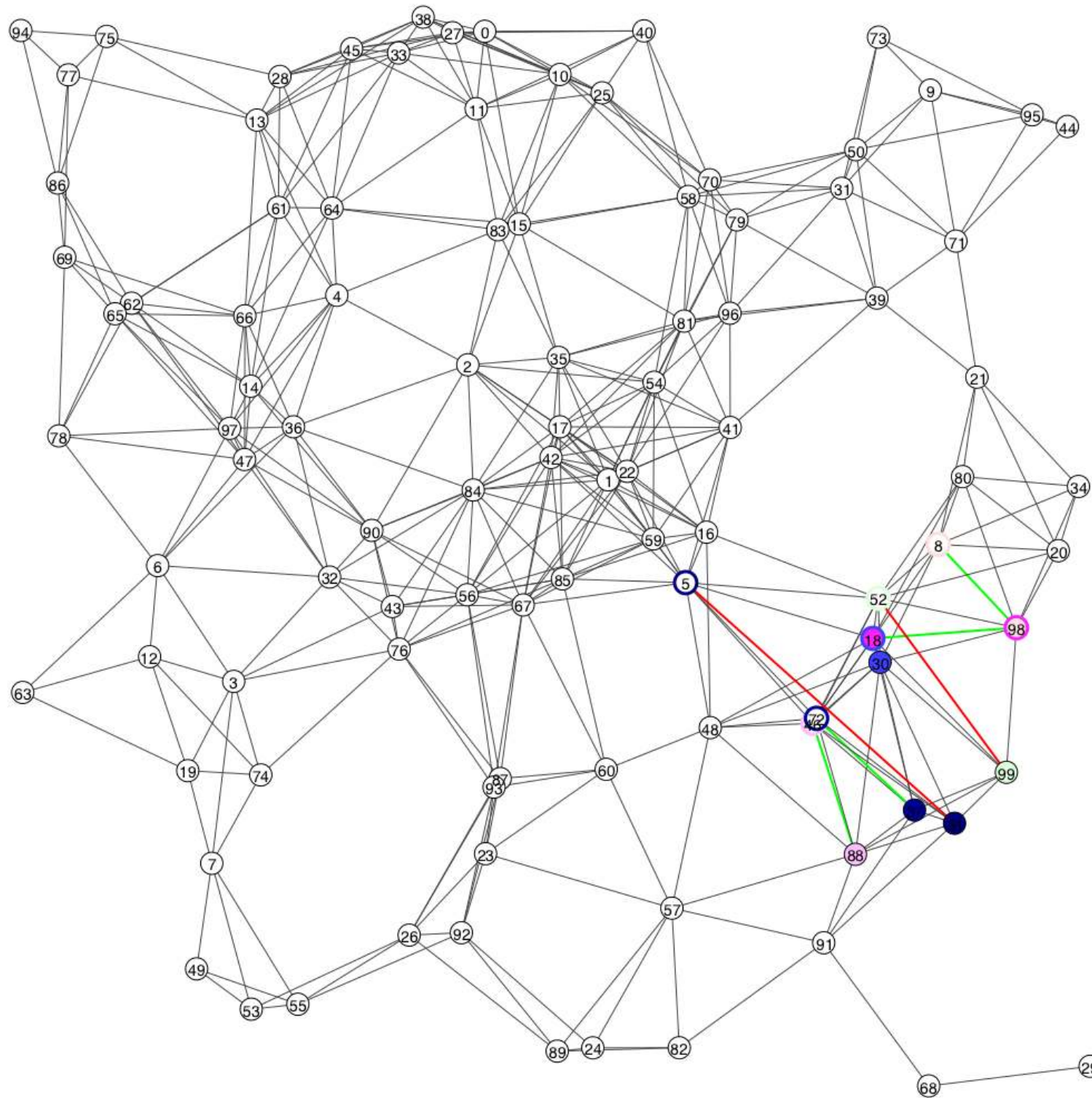
Composition

Results — Random geometric graph (2)



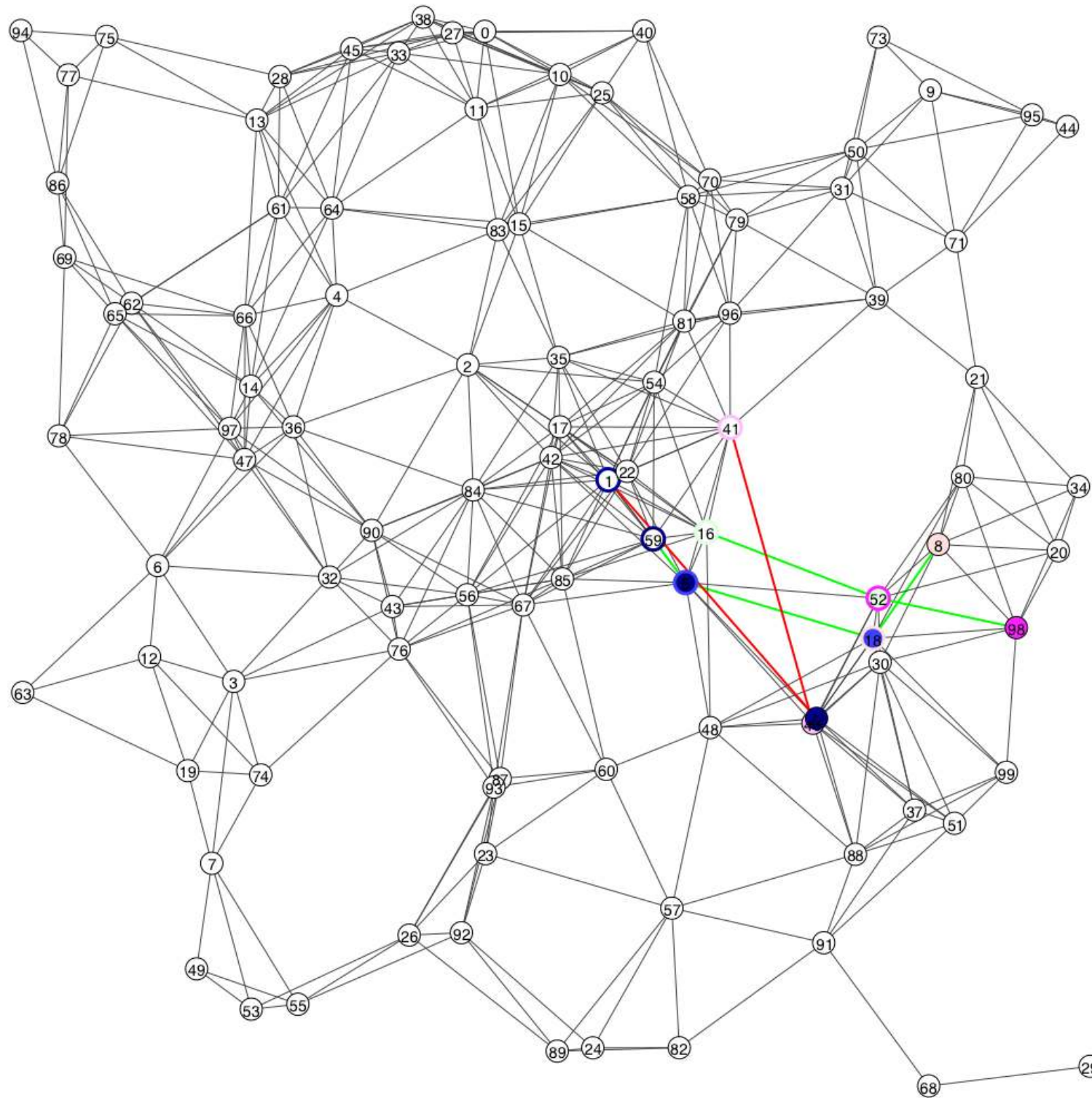
Objective

Results — Random geometric graph (2)



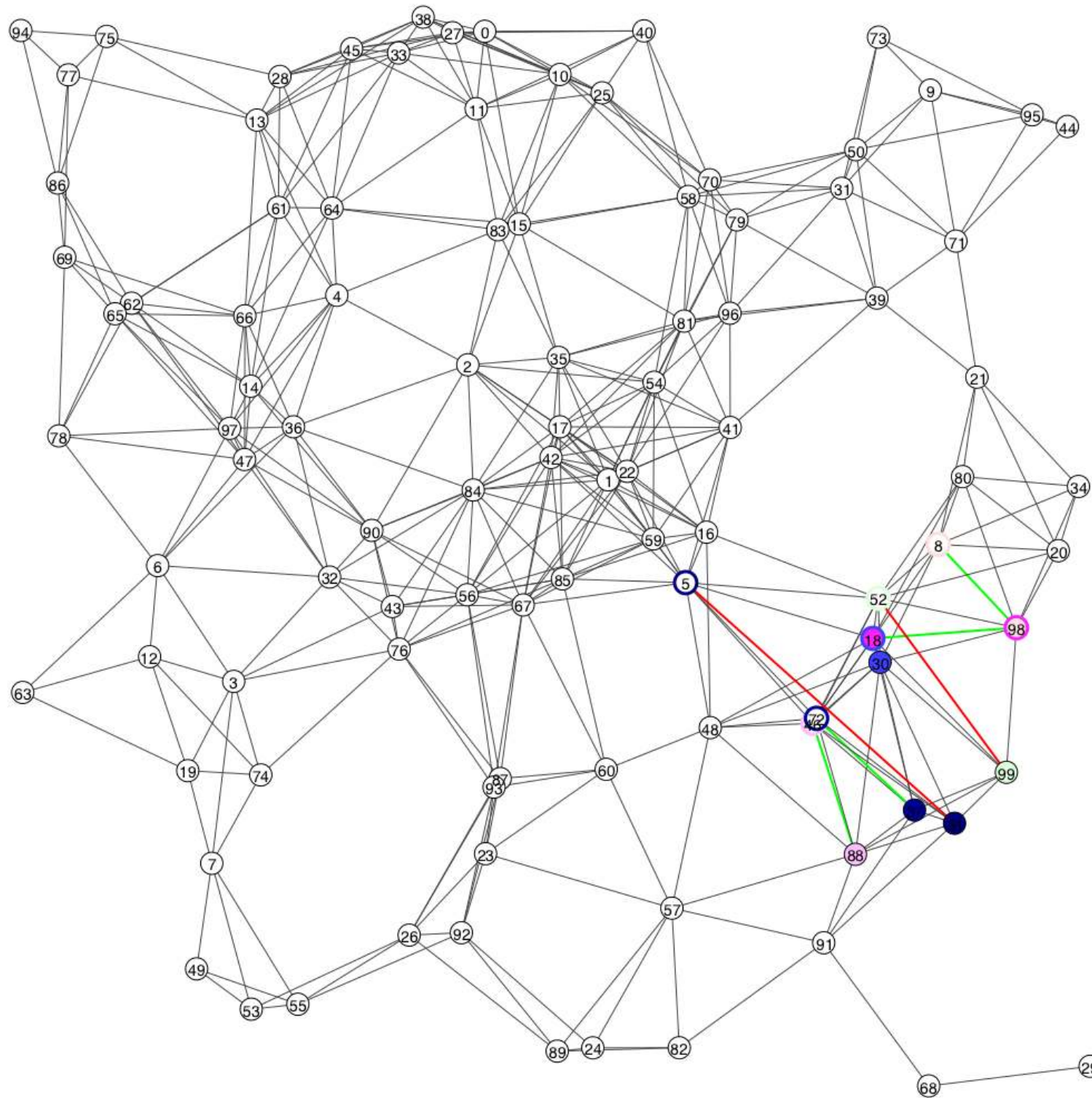
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)



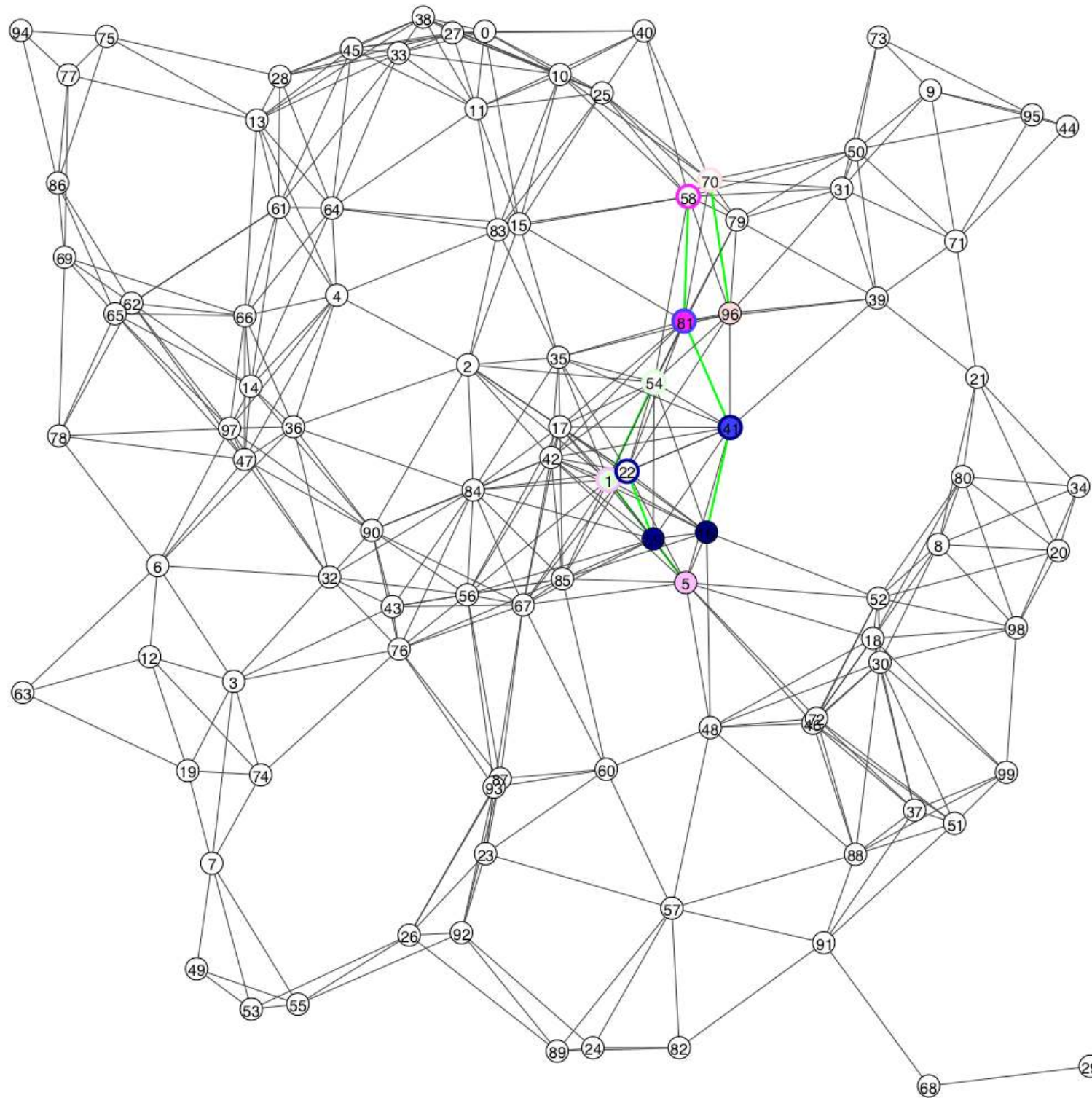
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)



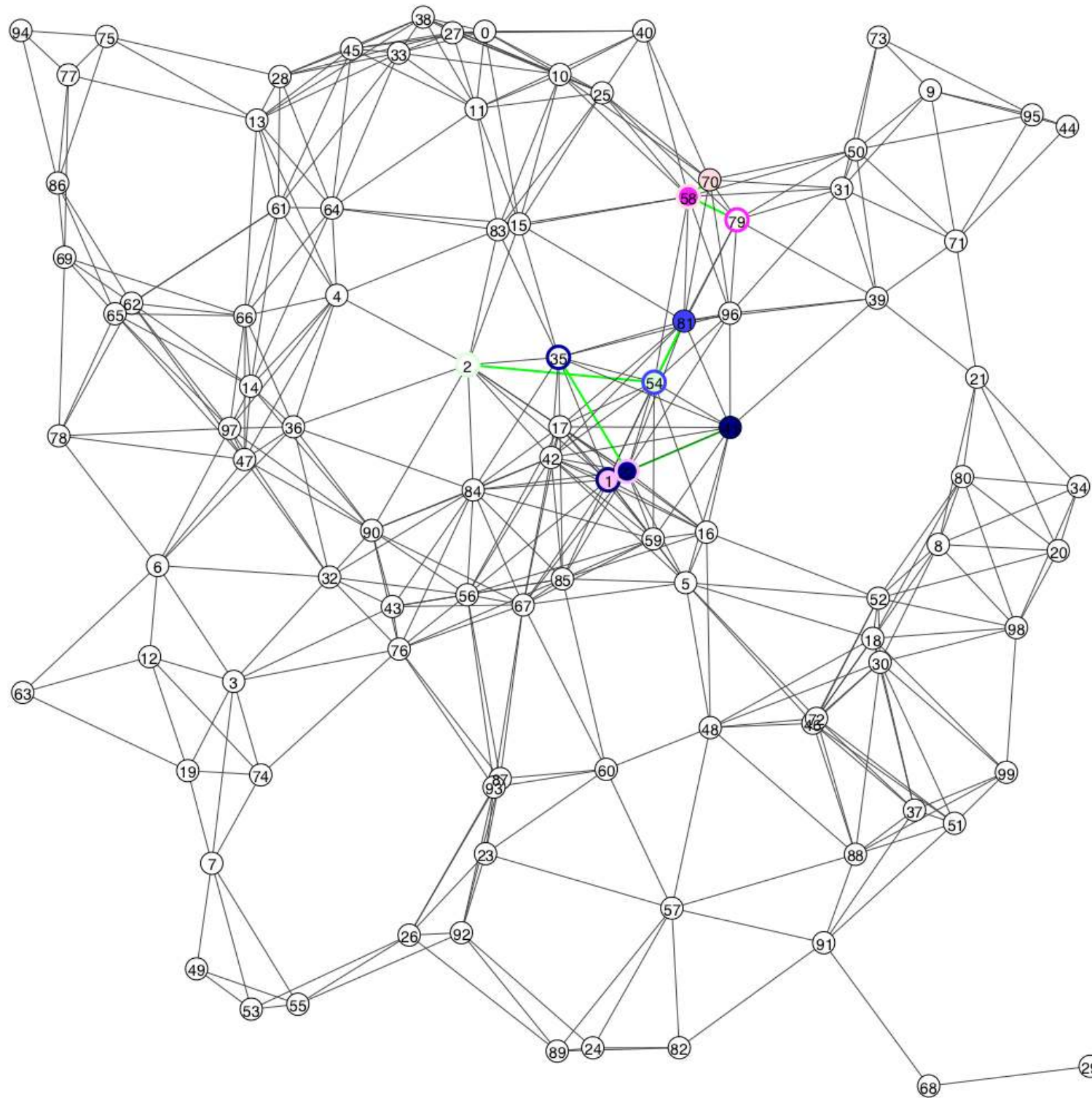
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)



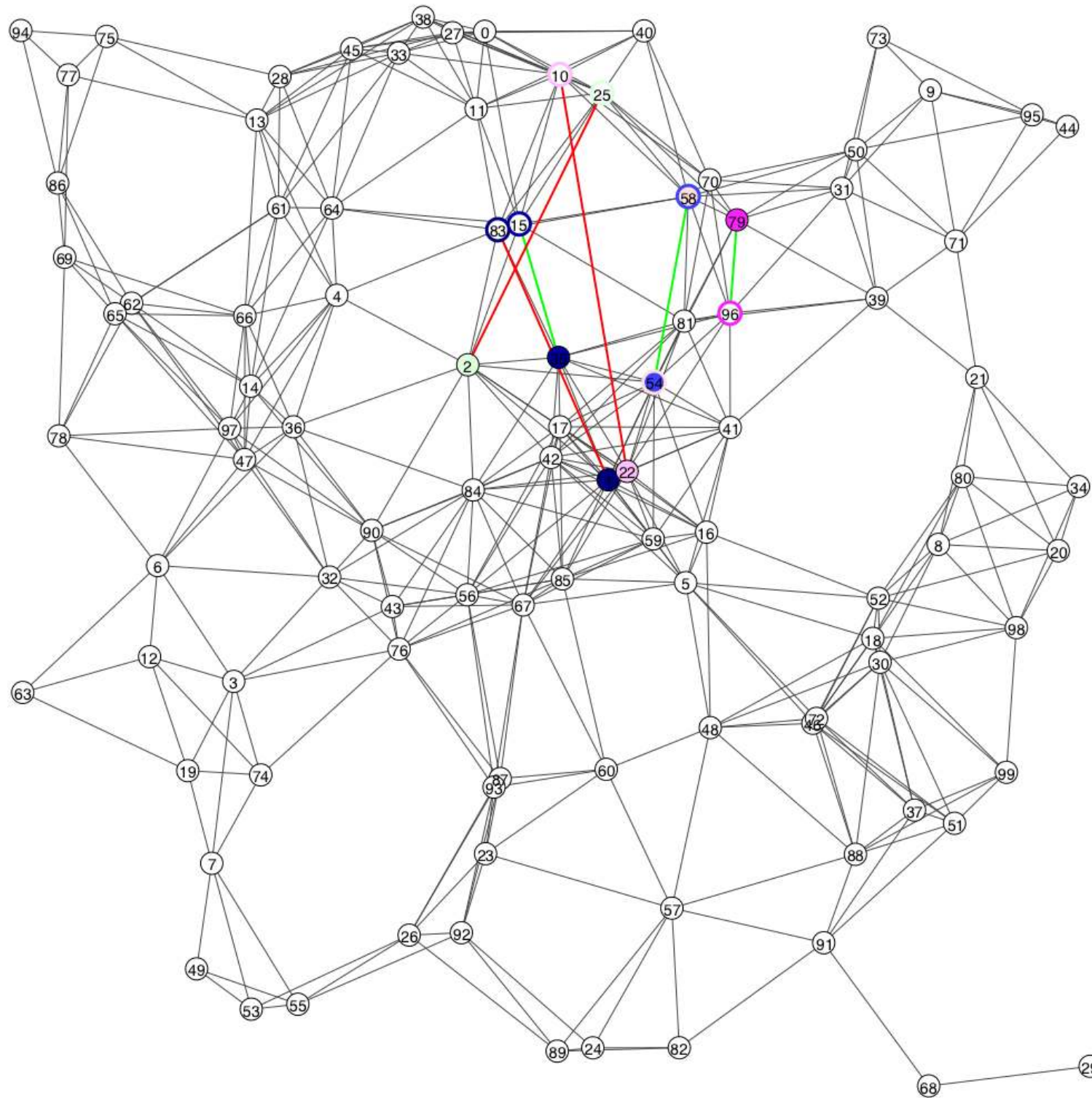
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)



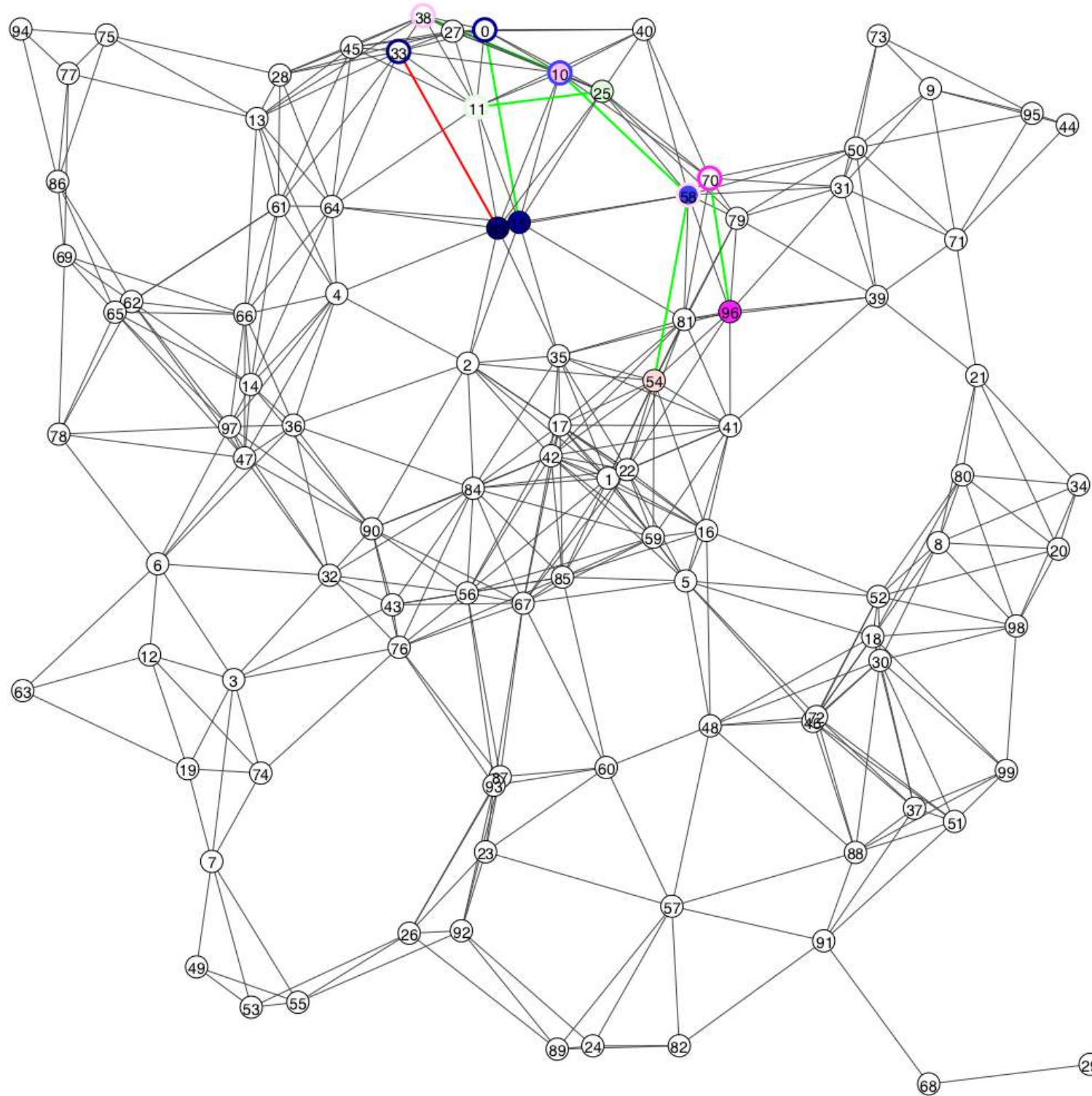
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)



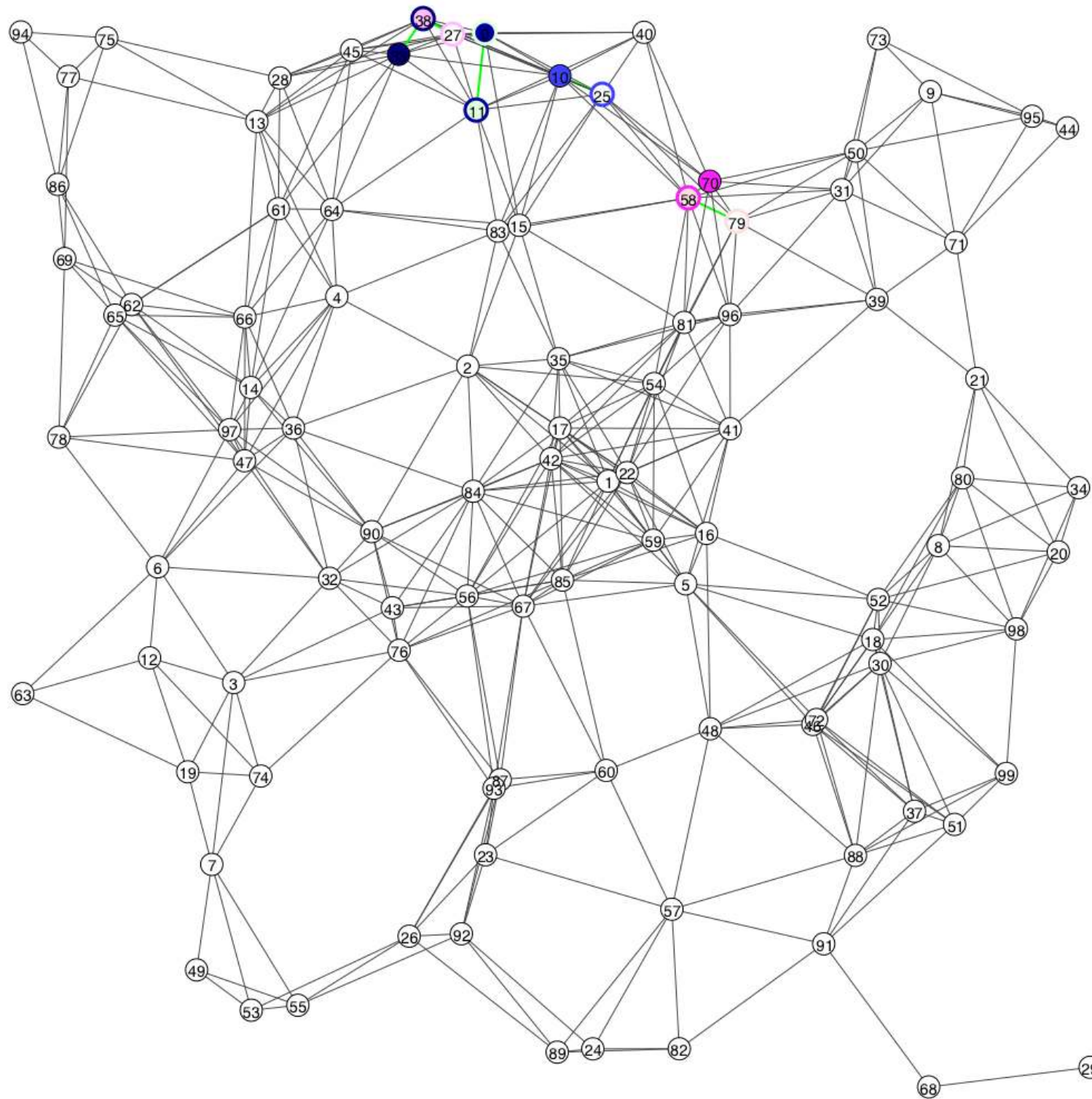
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)



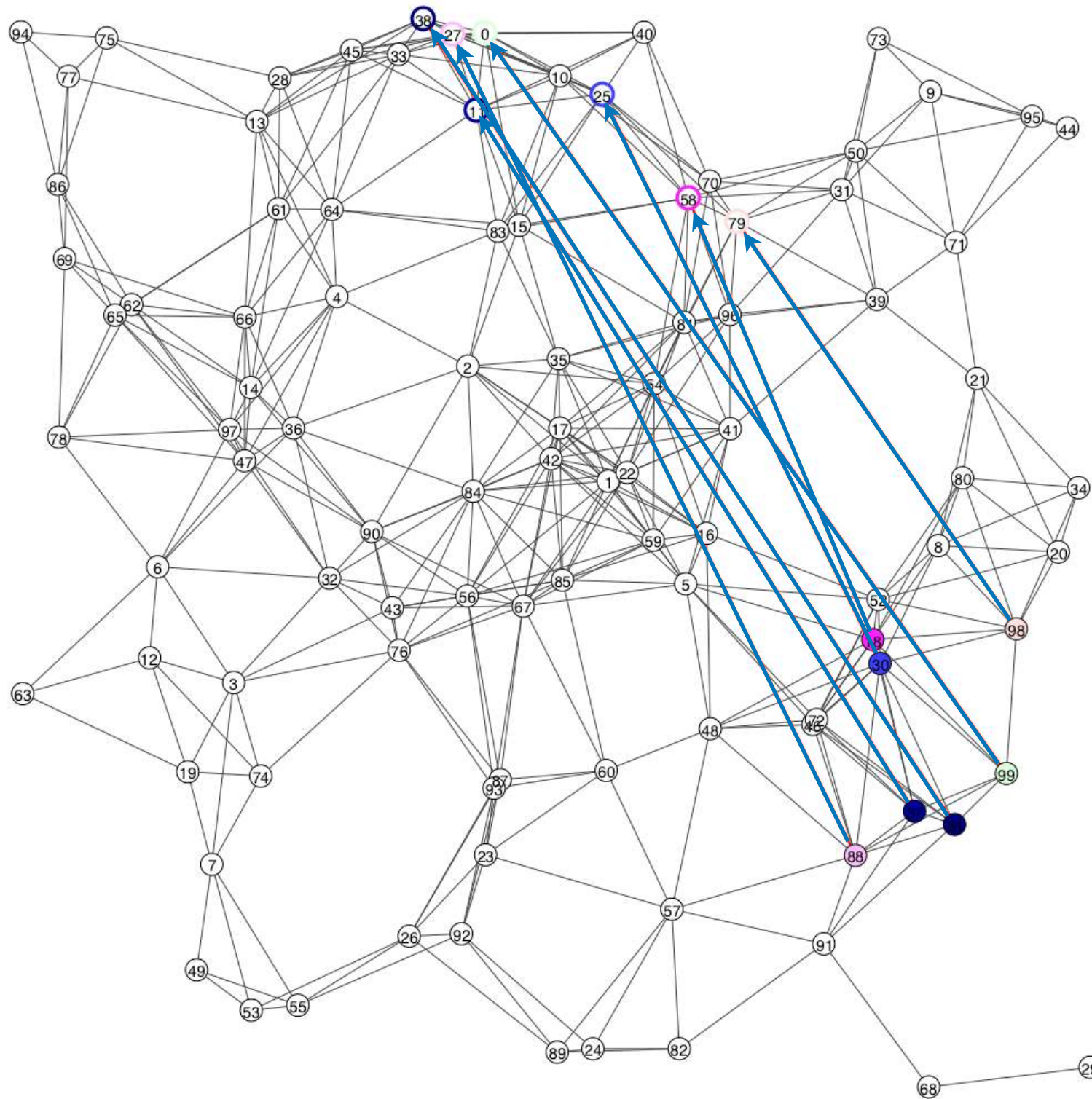
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)



$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ $\rightarrow$ $loss = 0\%$ $SNP = 14.29\%$

Results — Random geometric graph (2)

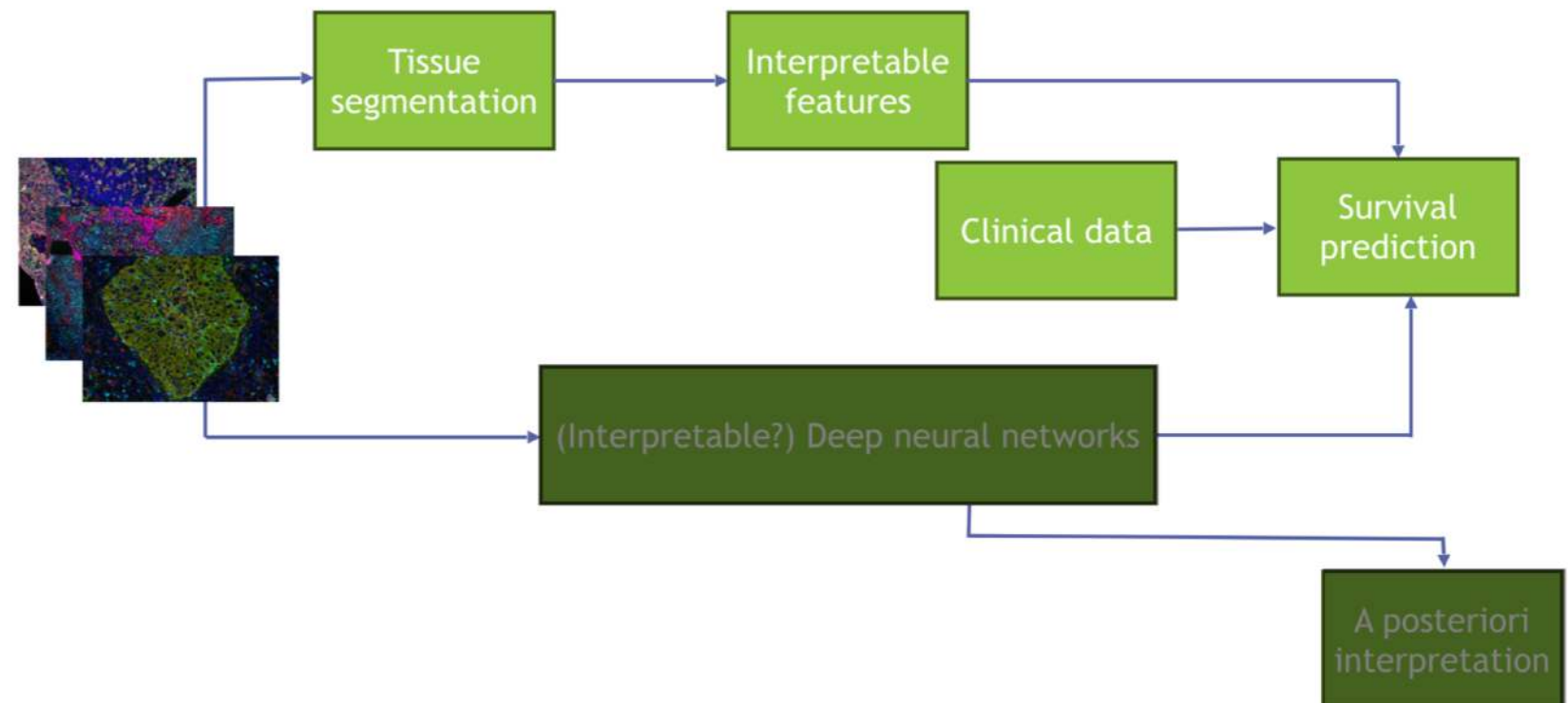
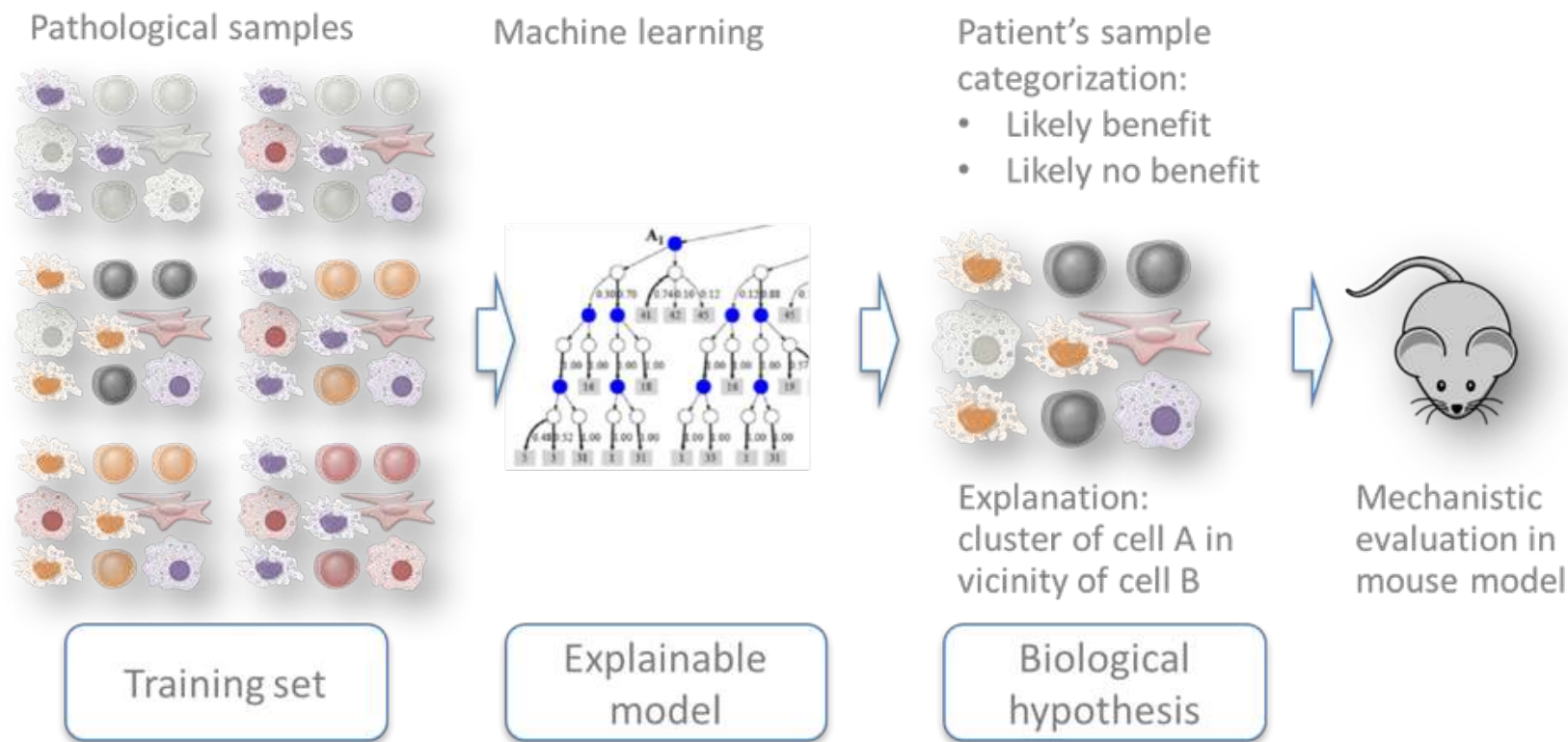


Composition

Outline

- GSP for AI, *a.k.a.* "What I did during my Ph.D. and a bit after"
- **GSP and personalized healthcare**
- GSP and brain signals

Why personalized oncology?



Datasets are becoming larger

SKCM

Experimental Strategy	Cases (n=470)	Files (n=12 724)
■ Diagnostic Slide	<u>433</u> 	<u>475</u> 
■ Tissue Slide	<u>469</u> 	<u>475</u> 

BRCA

Experimental Strategy	Cases (n=1 098)	Files (n=31 511)
■ Diagnostic Slide	<u>1 062</u> 	<u>1 133</u> 
■ Tissue Slide	<u>1 093</u> 	<u>1 978</u> 

COAD

Experimental Strategy	Cases (n=461)	Files (n=14 270)
■ Diagnostic Slide	<u>451</u> 	<u>459</u> 
■ Tissue Slide	<u>460</u> 	<u>983</u> 

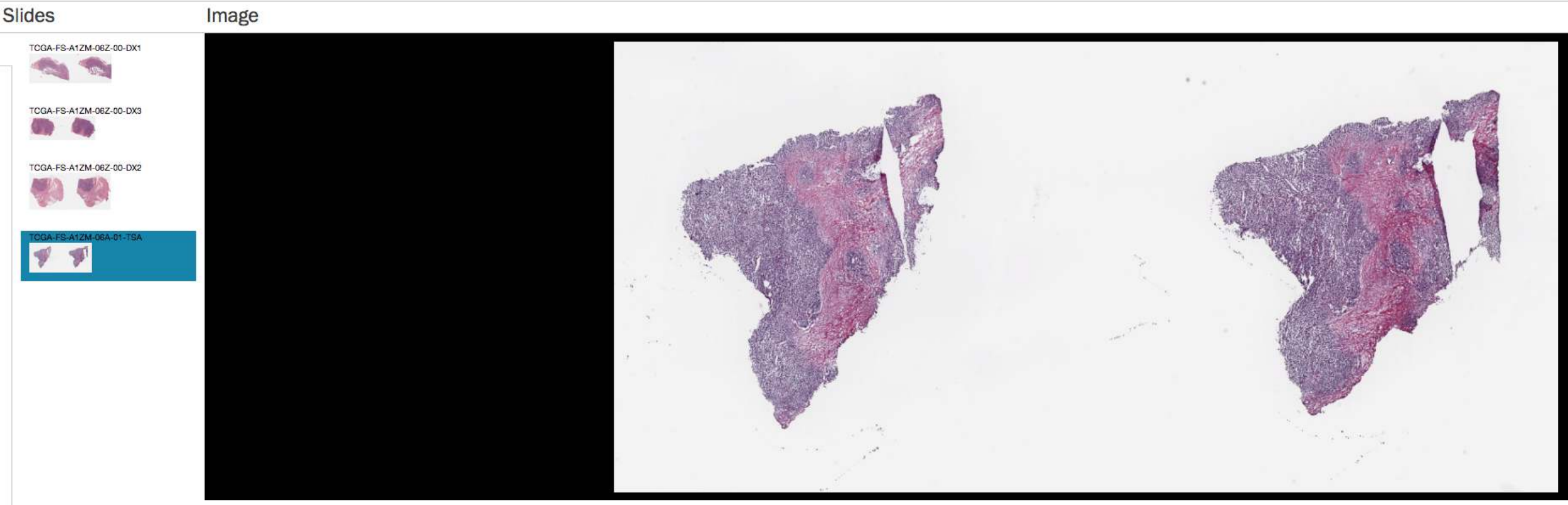
LUAD

Experimental Strategy	Cases (n=585)	Files (n=17 051)
■ Diagnostic Slide	<u>478</u> 	<u>541</u> 
■ Tissue Slide	<u>514</u> 	<u>1 067</u> 

PRAD

Experimental Strategy	Cases (n=500)	Files (n=14 287)
■ Diagnostic Slide	<u>403</u> 	<u>449</u> 
■ Tissue Slide	<u>490</u> 	<u>723</u> 

Datasets are becoming larger

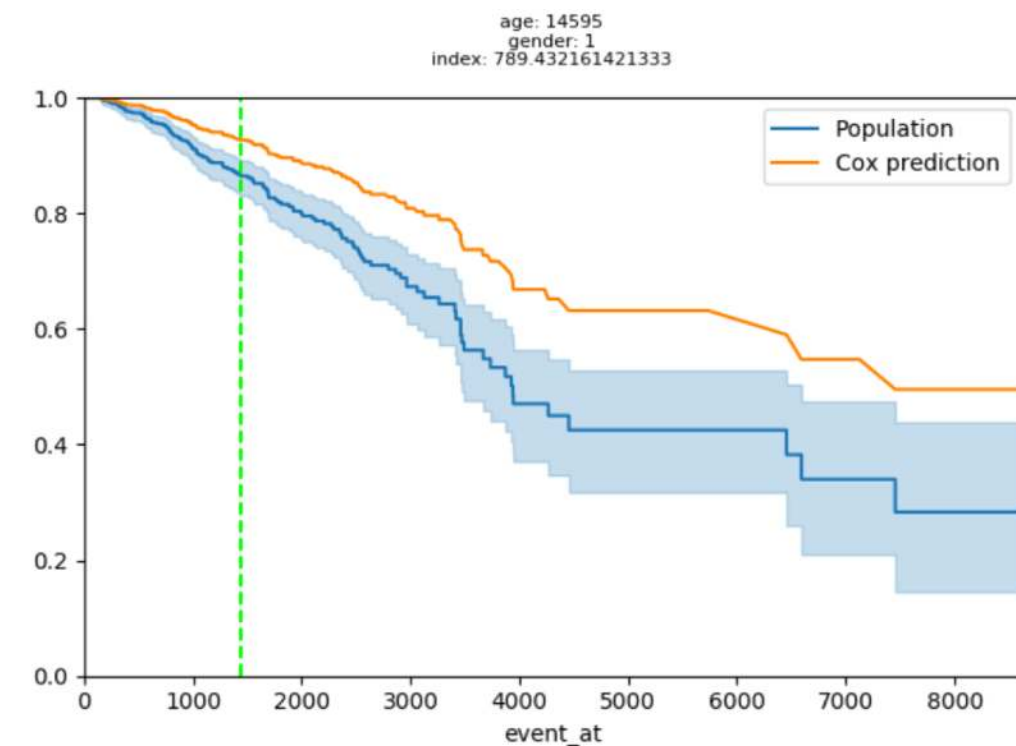
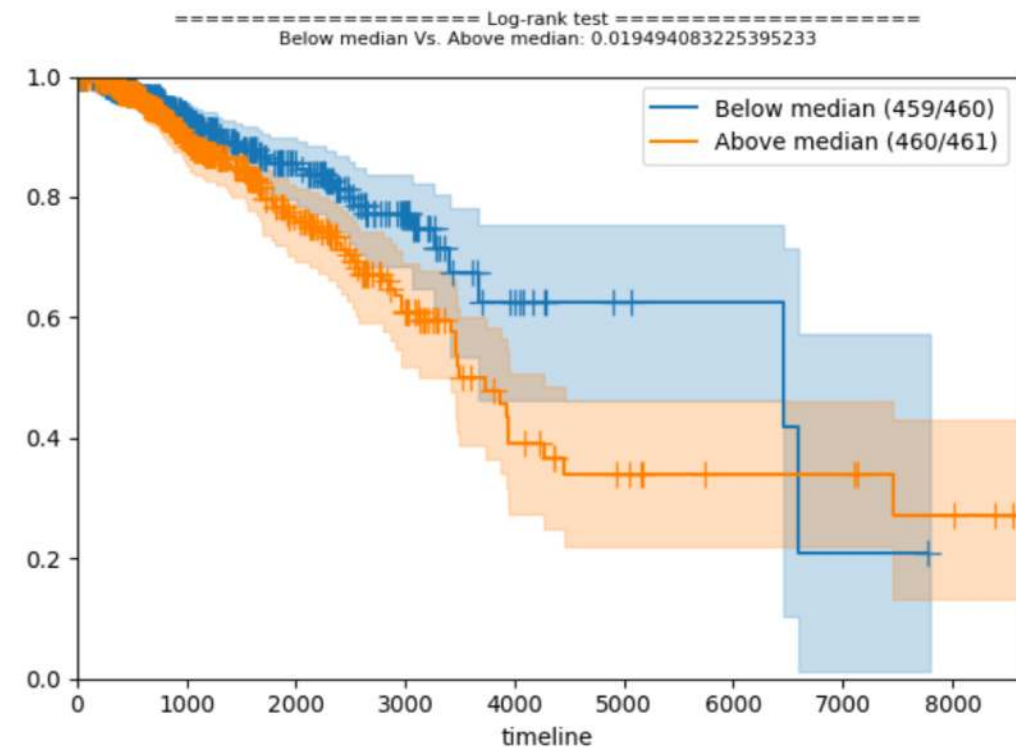


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Submitter_id	TCGA-FS-A1ZM-06A-01-TSA
Percent_lymphocyte_infiltration	--
Percent_tumor_nuclei	75
Percent_monocyte_infiltration	--
Percent_normal_cells	--
Percent_stromal_cells	30
Percent_eosinophil_infiltration	--
Slide_id	b045f429-078b-4431-8691-3a9604b818ce
Percent_neutrophil_infiltration	--
Section_location	TOP
Percent_granulocyte_infiltration	--
Percent_necrosis	--
Percent_inflam_infiltration	--
Number_proliferating_cells	--
Percent_tumor_cells	70

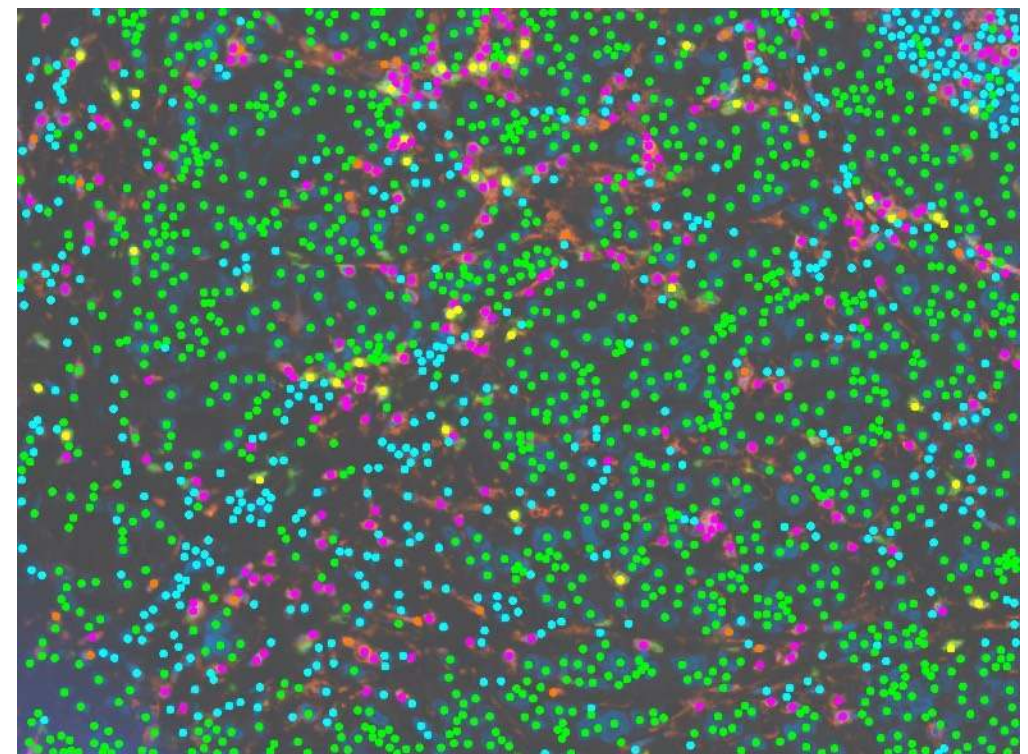
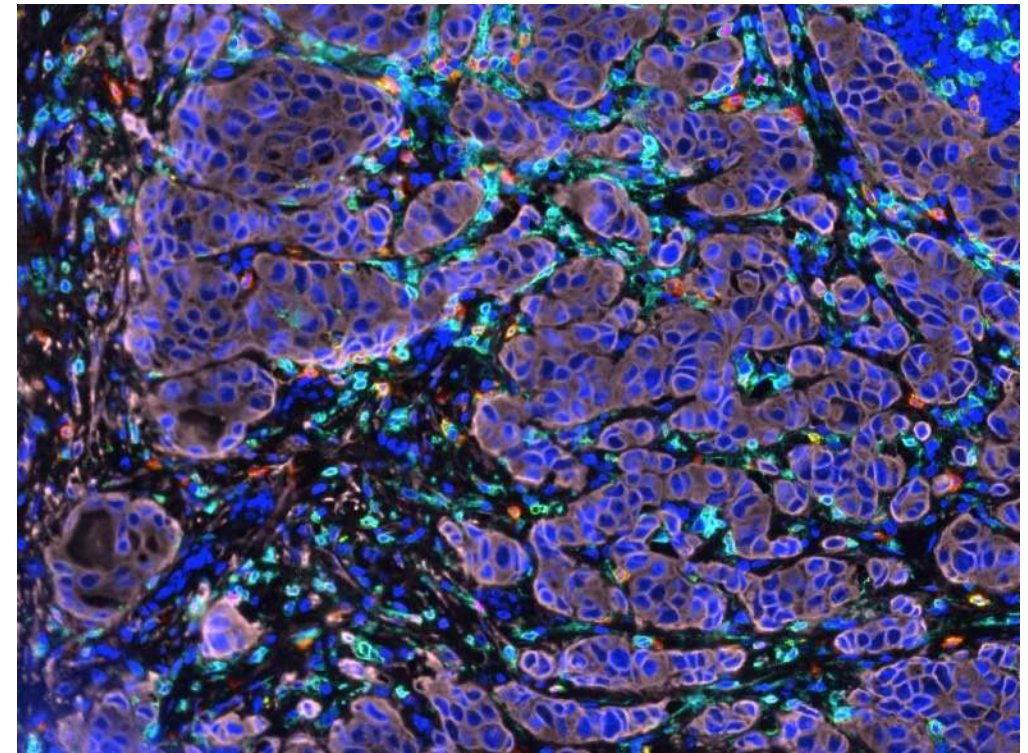
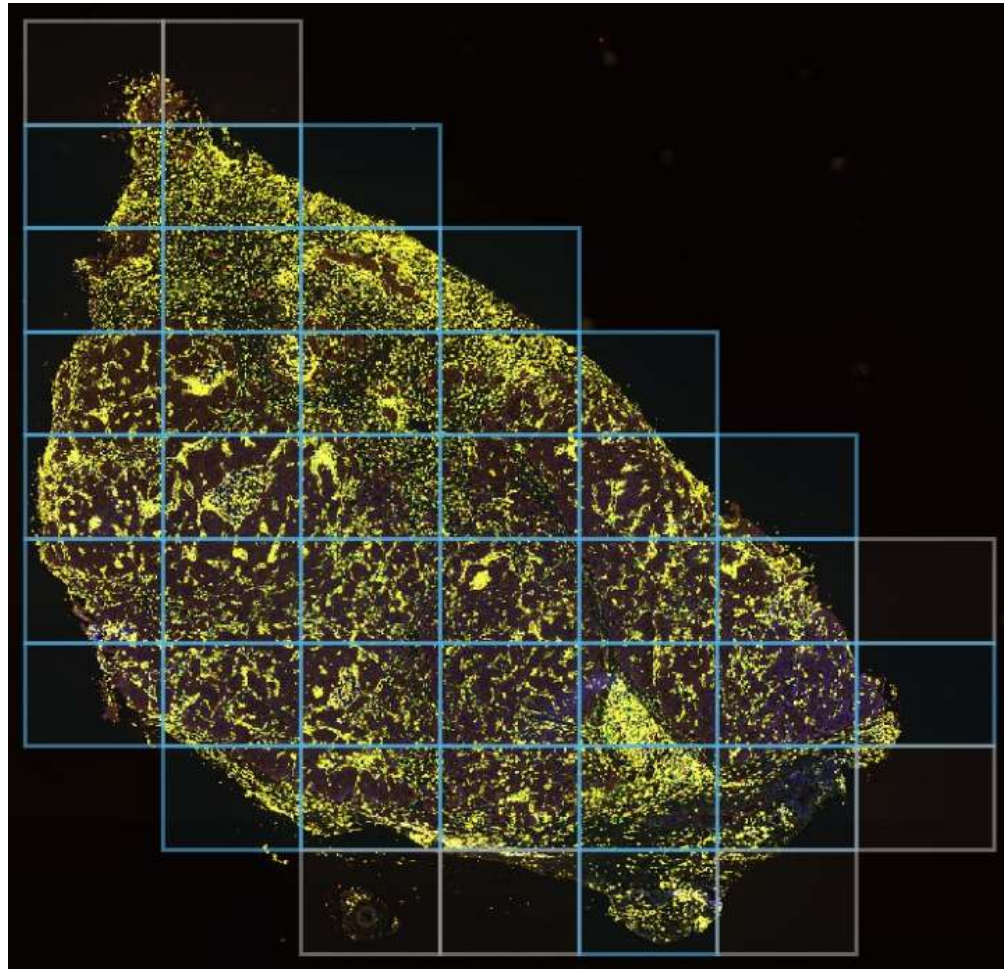
UUID	8b74dba2-0f4c-5e4e-8de1-62397a19dcdb
Classification Of Tumor	not reported
Alcohol Intensity	--
Age At Diagnosis	74 years 182 days
Days To Birth	-27210
Days To Death	--
Days To Last Follow Up	3080
Days To Last Known Disease Status	--
Days To Recurrence	--
Last Known Disease Status	not reported
Morphology	8720/3
Primary Diagnosis	C77.9
Prior Malignancy	not reported
Progression Or Recurrence	not reported
Site Of Resection Of Biopsy	C77.9
Tissue Or Organ Of Origin	C77.9
Tumor Grade	not reported
Tumor Stage	stage iii
Vital Status	alive

Datasets are becoming larger

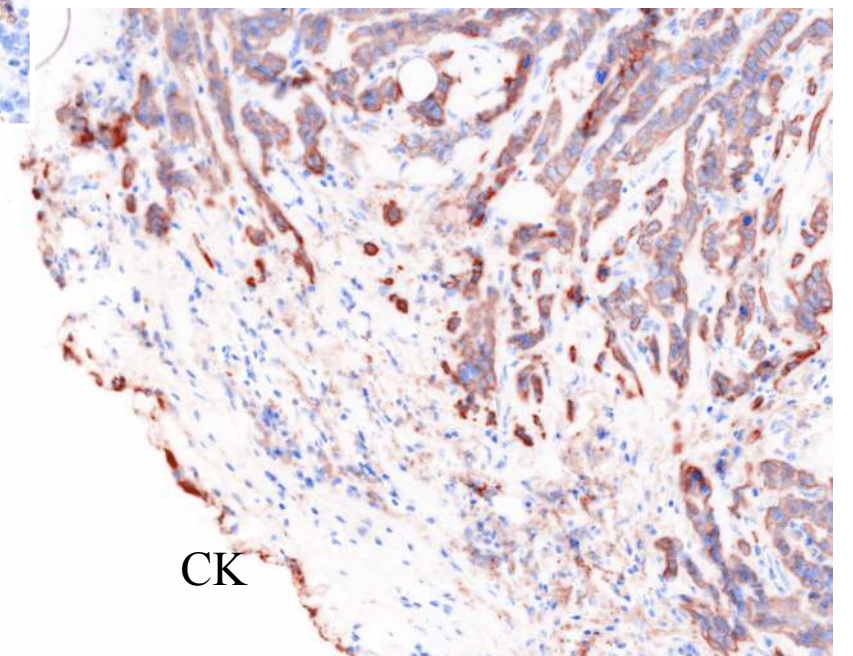
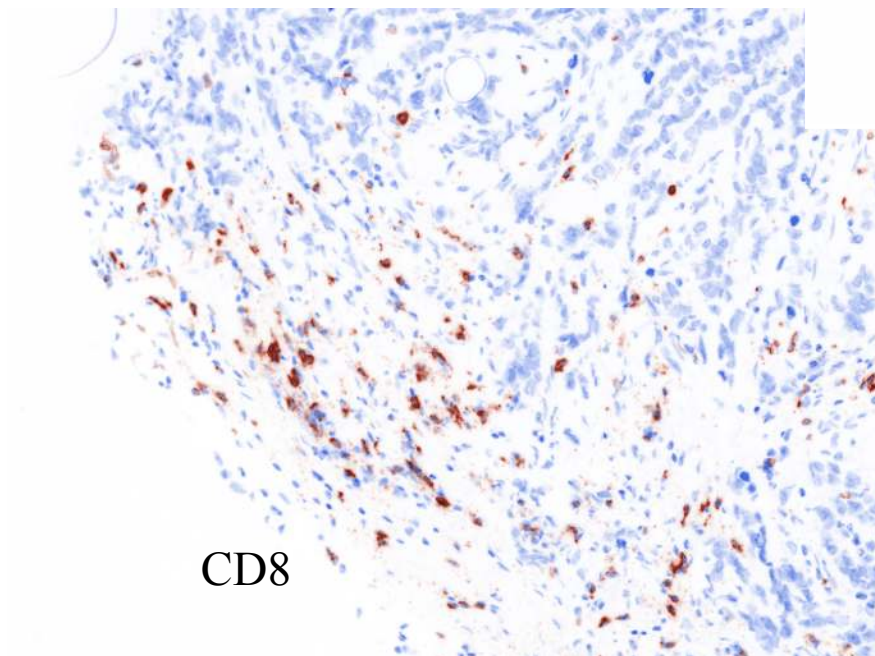
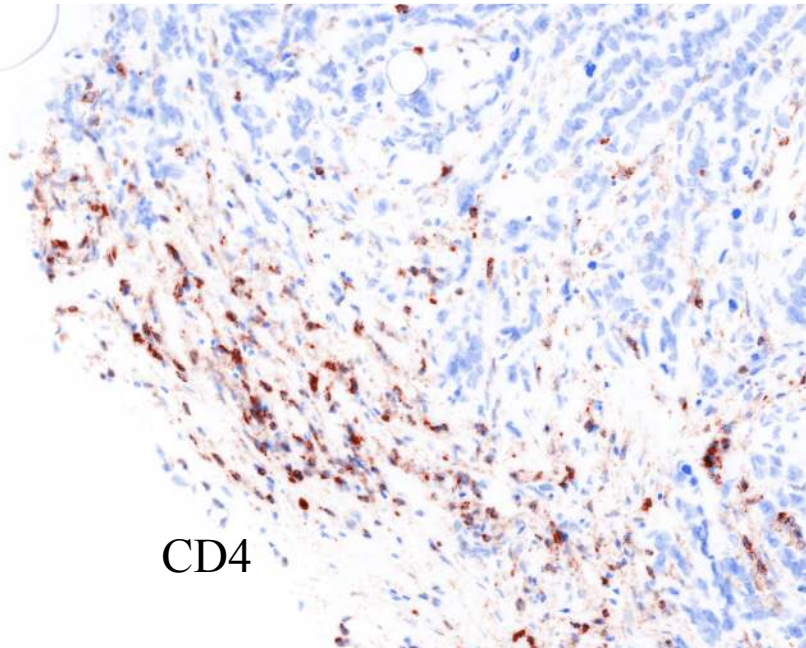
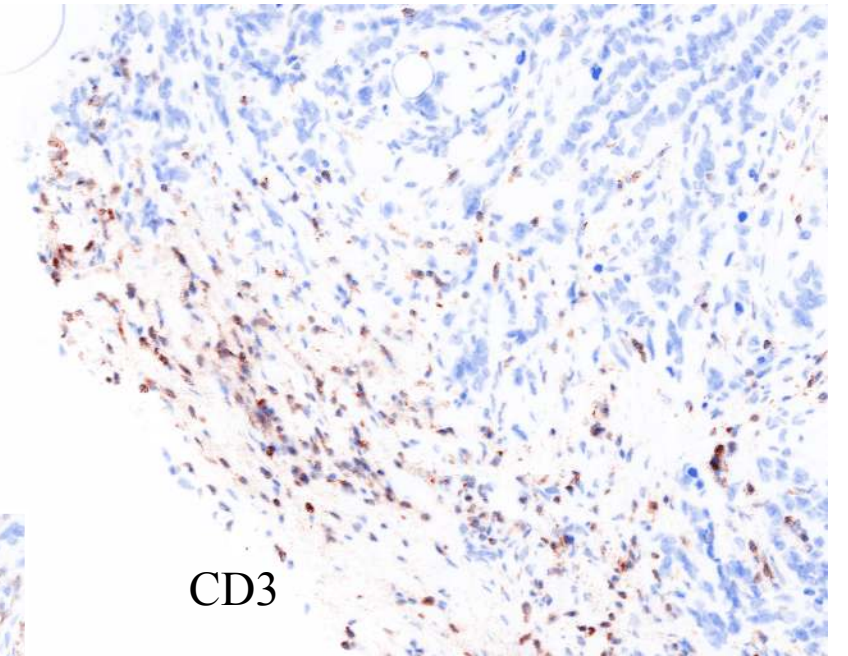
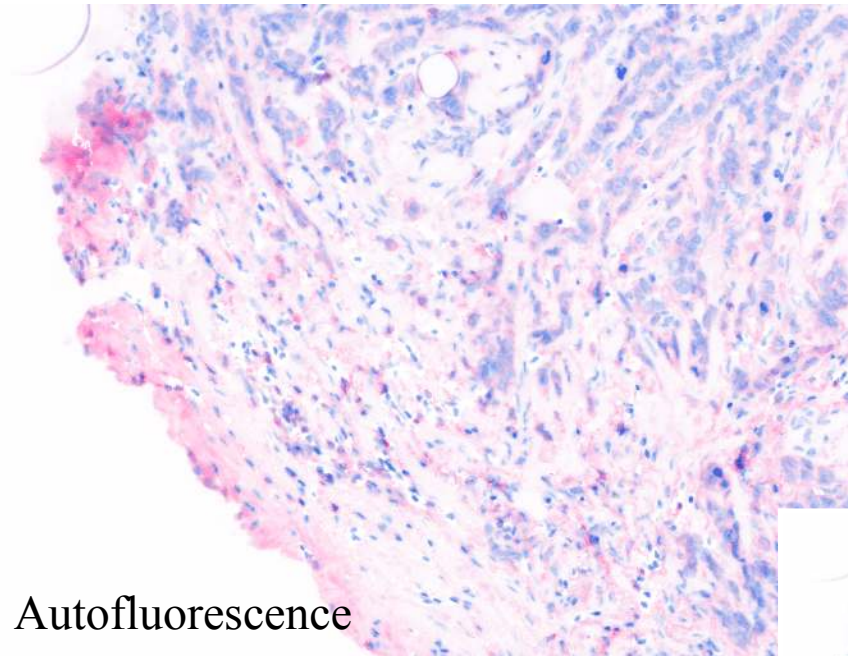
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Classification Of Tumor	not reported
Alcohol Intensity	--
Age At Diagnosis	74 years 182 days
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Vital Status	alive



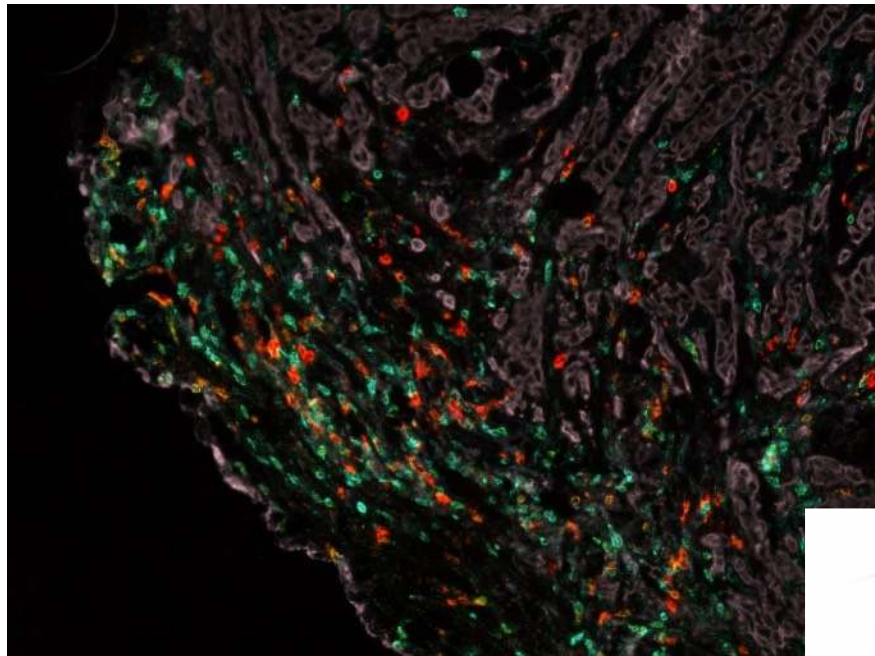
Techniques are improving



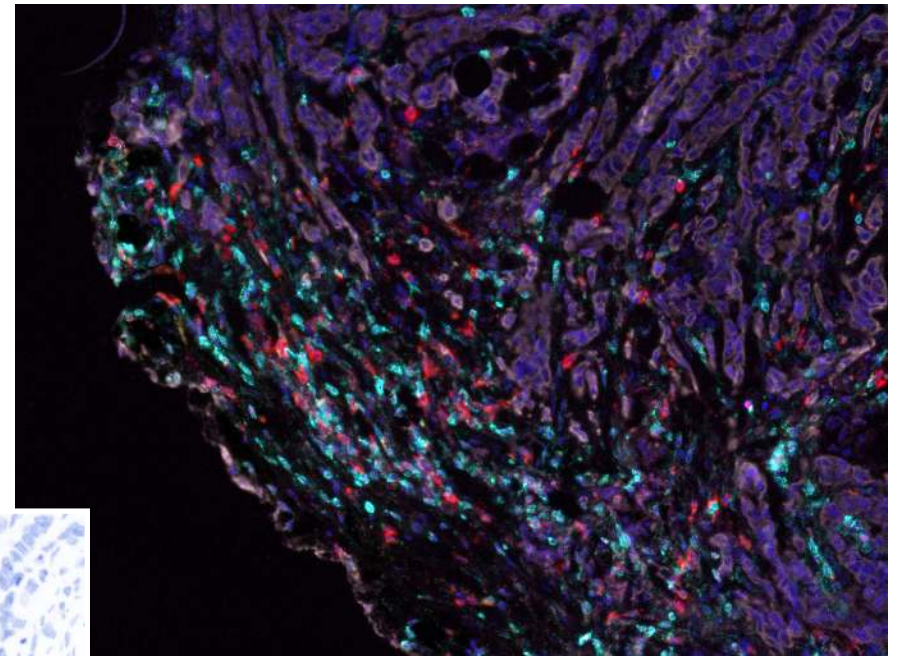
Techniques are improving



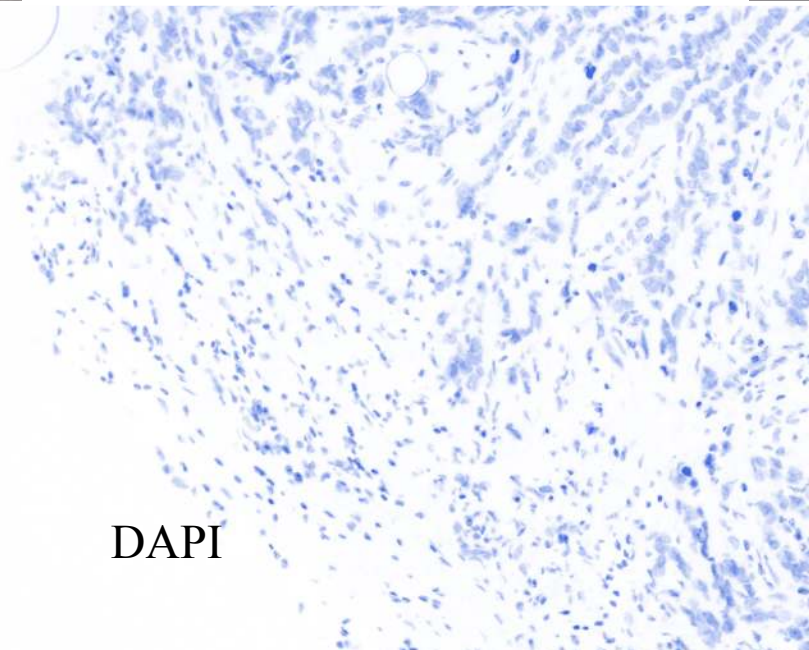
Techniques are improving



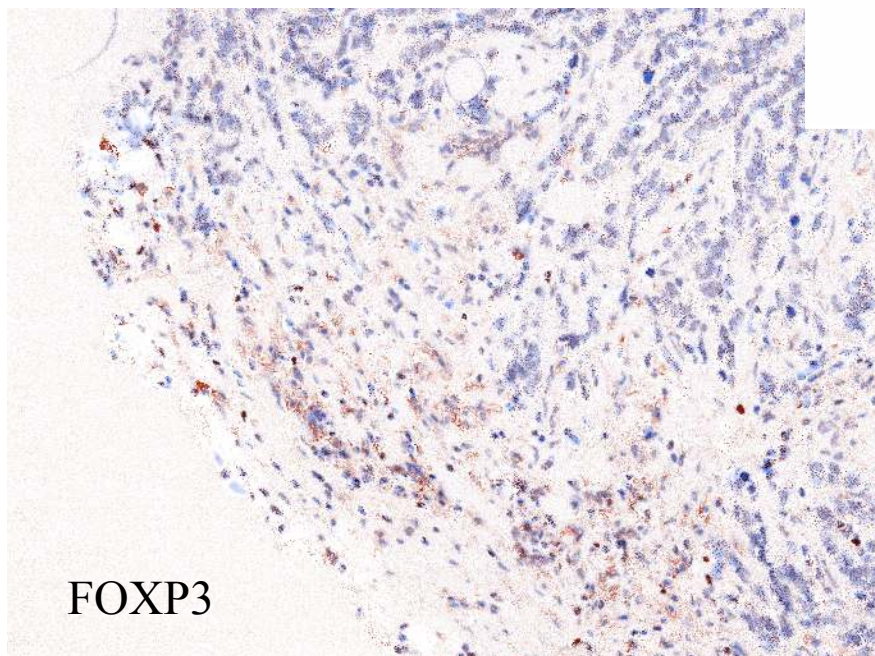
Composite



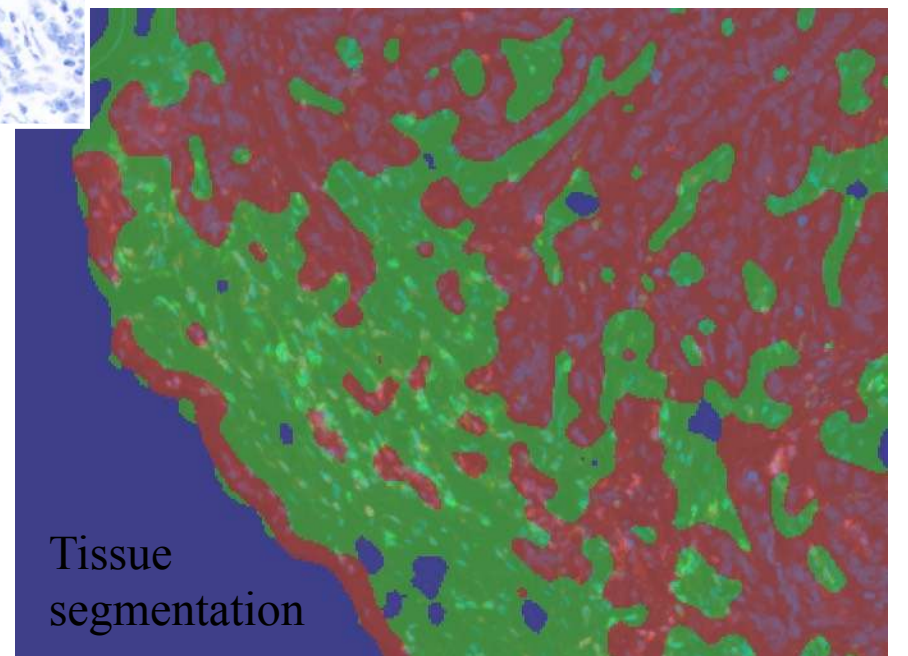
Composite



DAPI

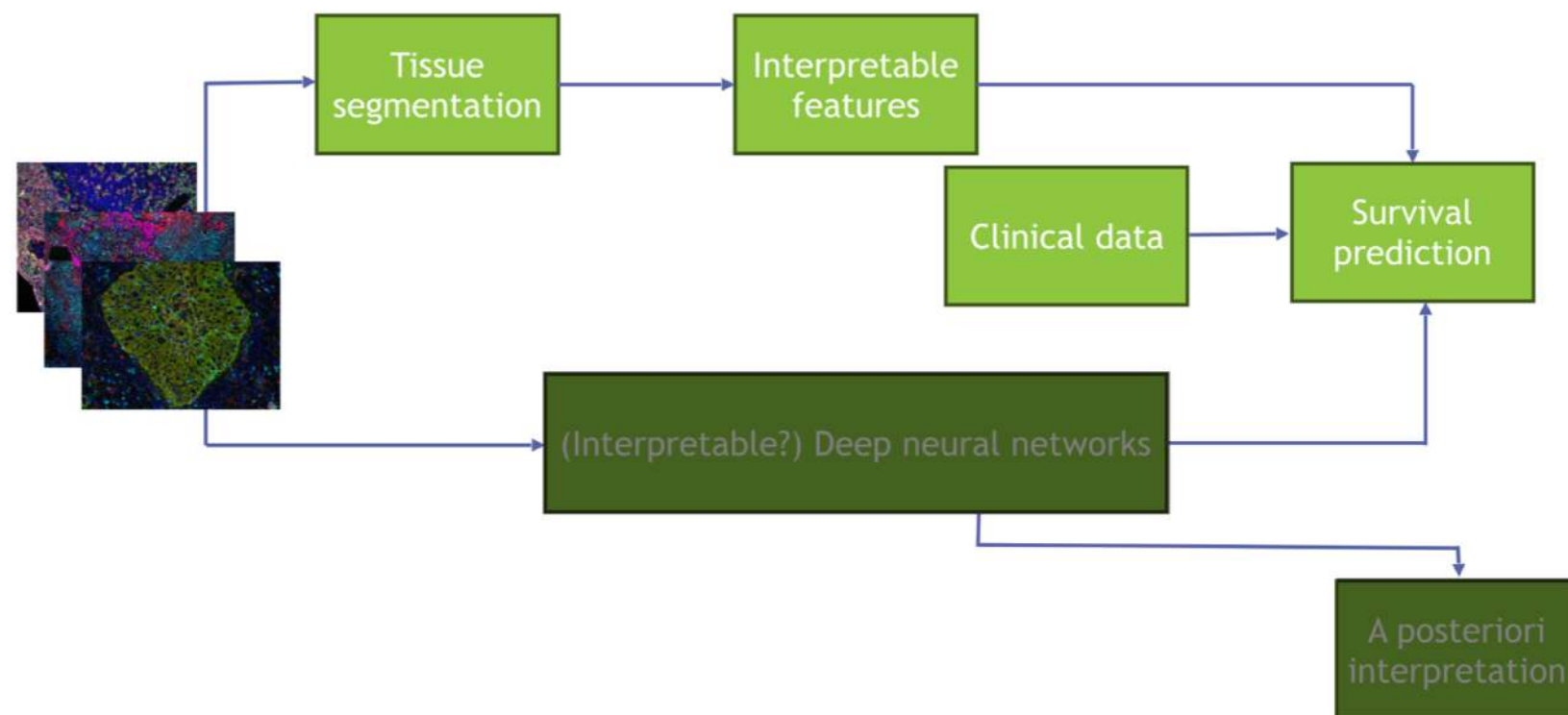


FOXP3

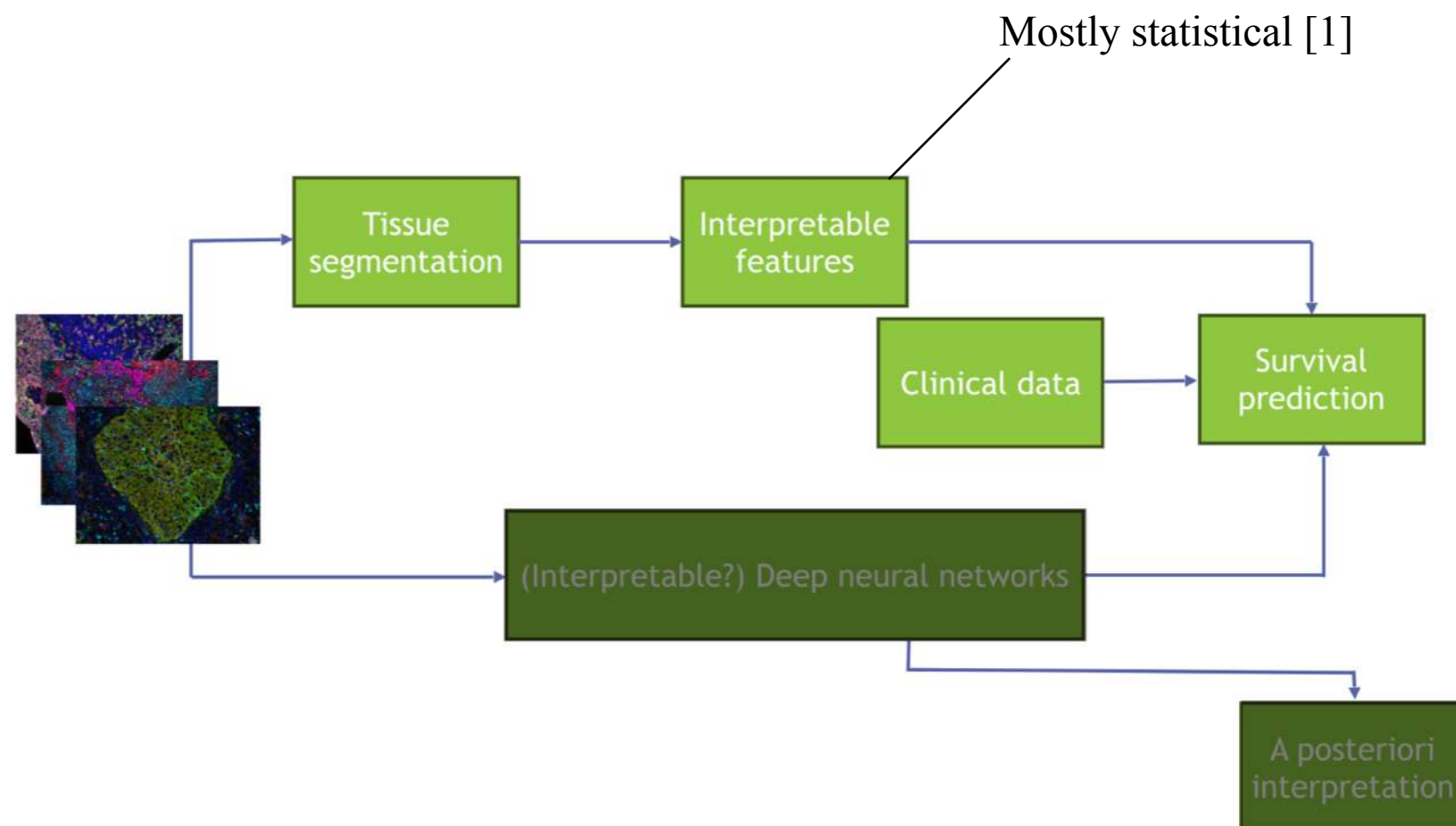
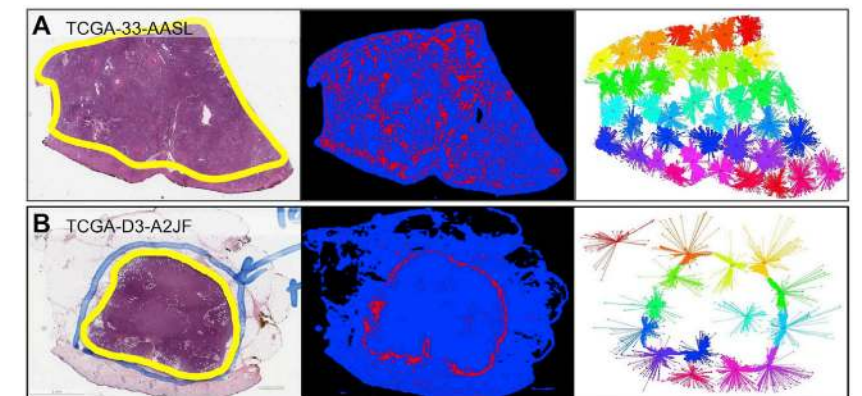


Tissue
segmentation

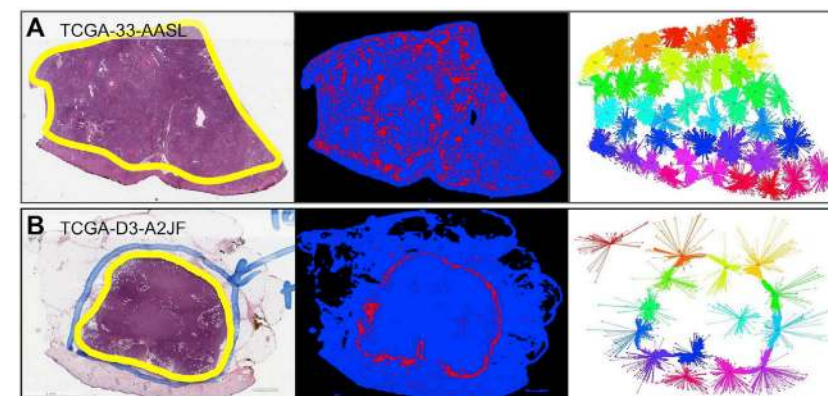
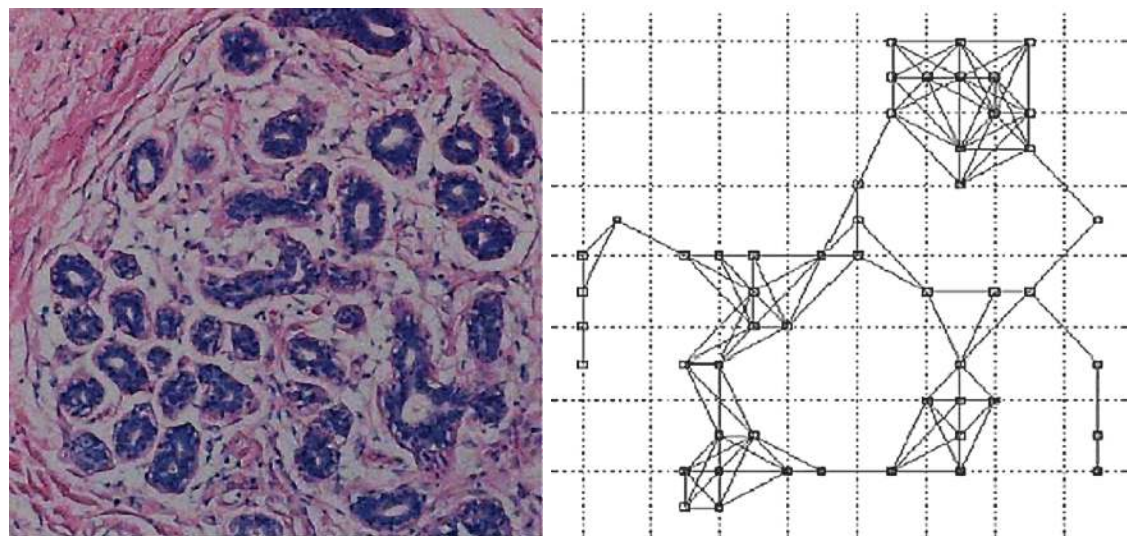
What is the link with GSP?



What is the link with GSP?

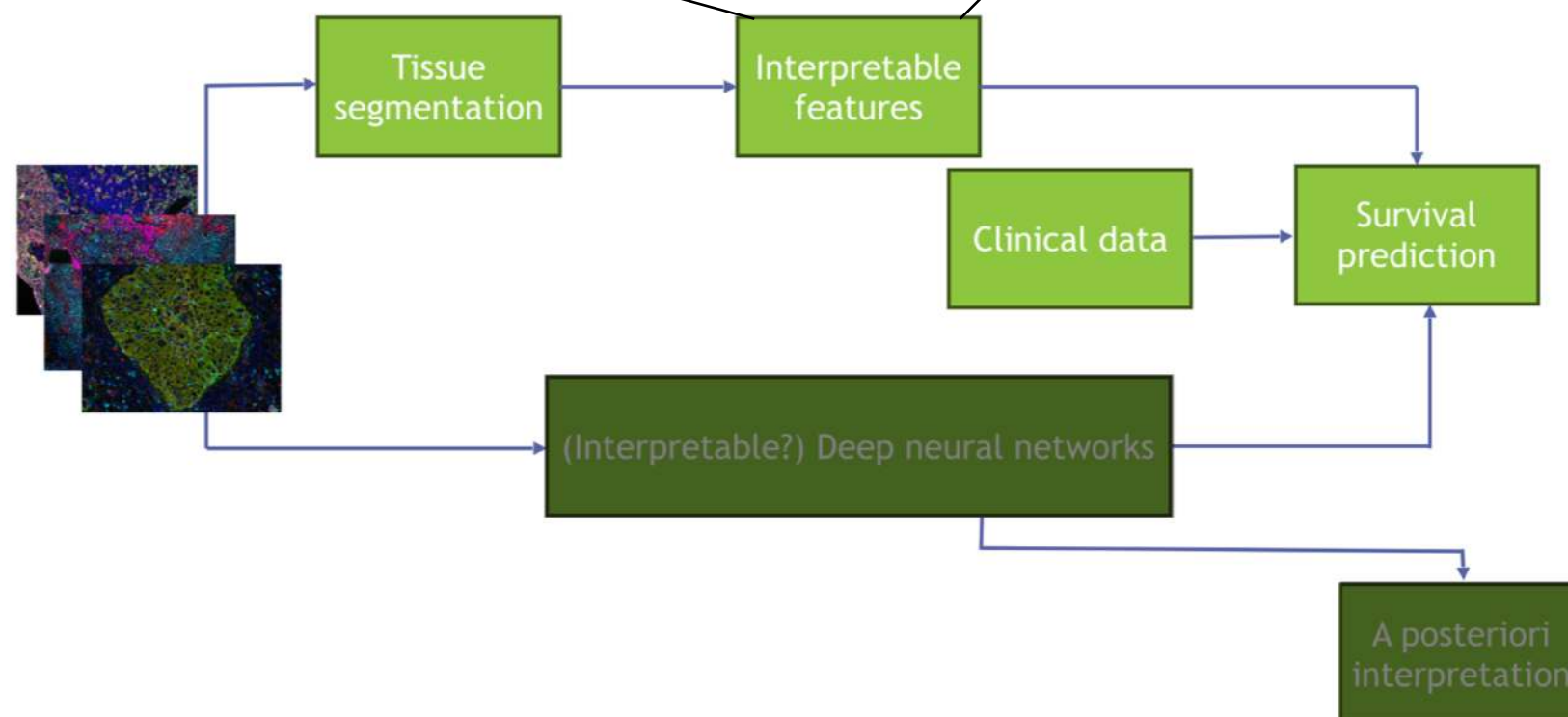


What is the link with GSP?



Graph features [2]

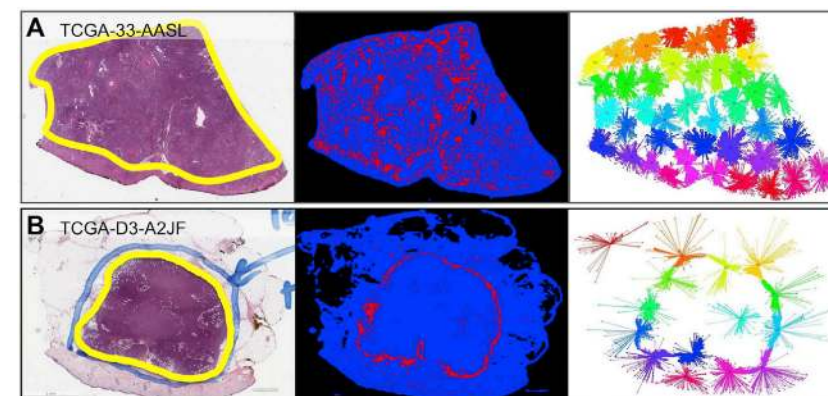
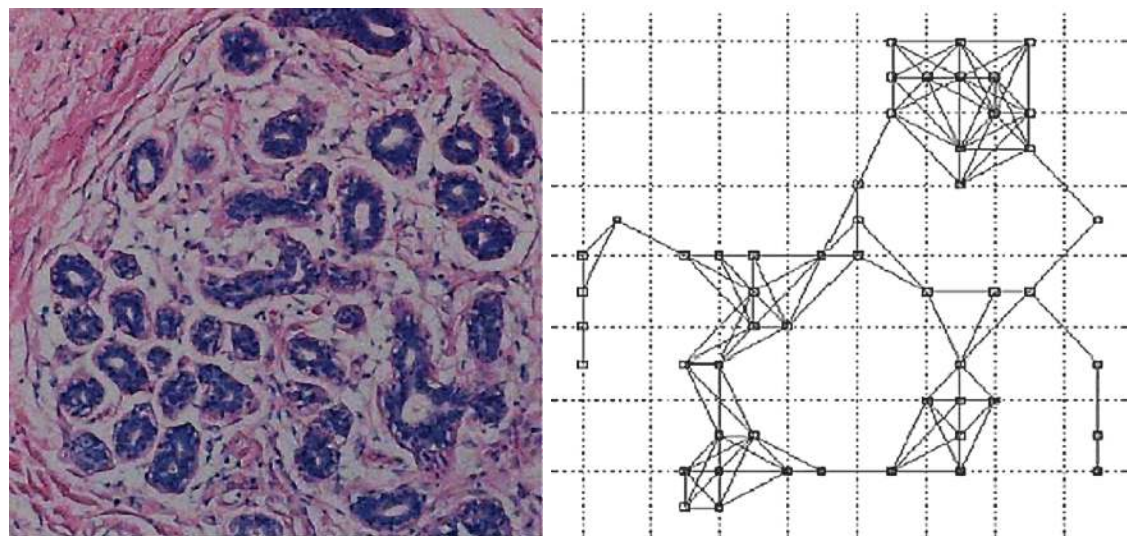
Mostly statistical [1]



[1] Saltz et al. - Spatial Organization and Molecular Correlation of Tumor-Infiltrating Lymphocytes Using Deep Learning on Pathology Images (2018)

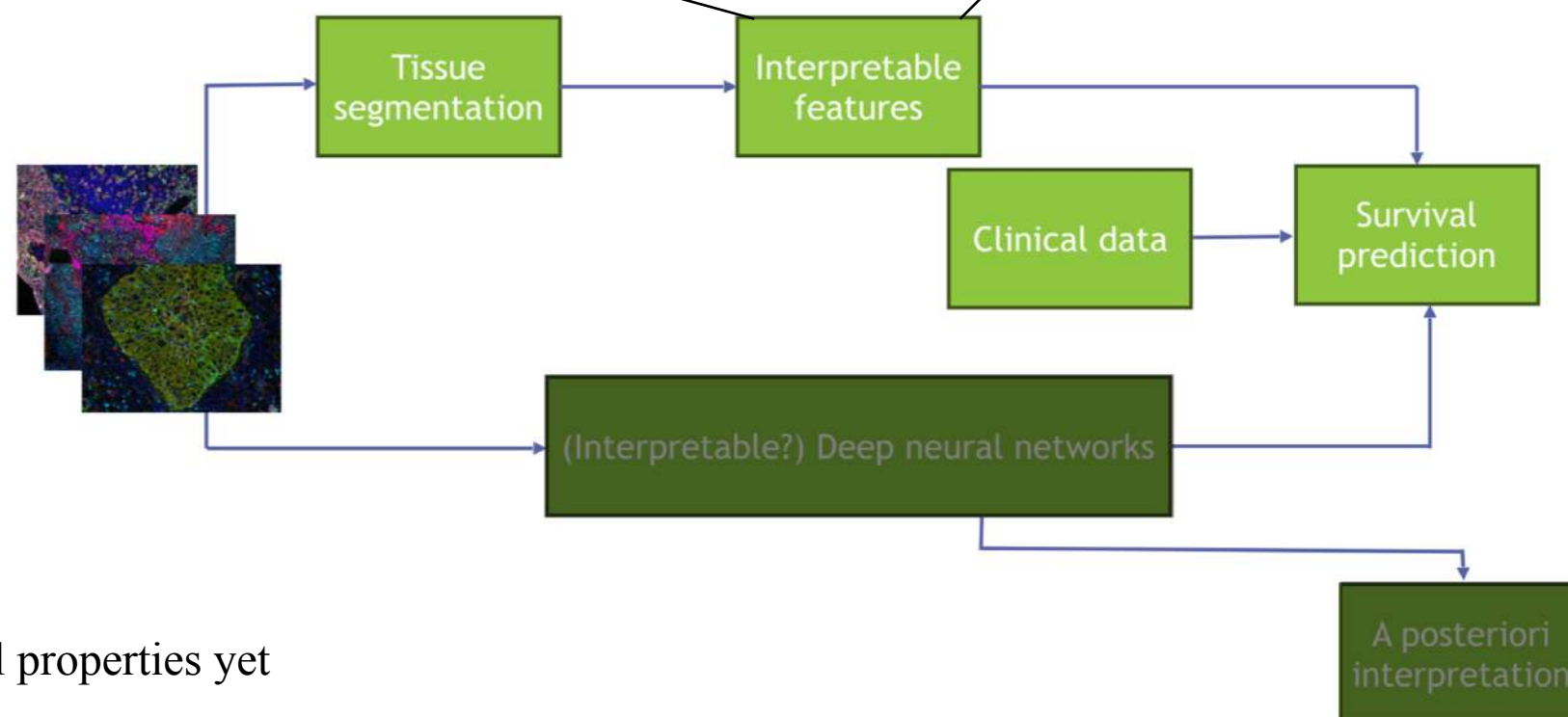
[2] Yener - Cell-Graphs: Image-Driven Modeling of StructureFunction Relationship (2017)

What is the link with GSP?



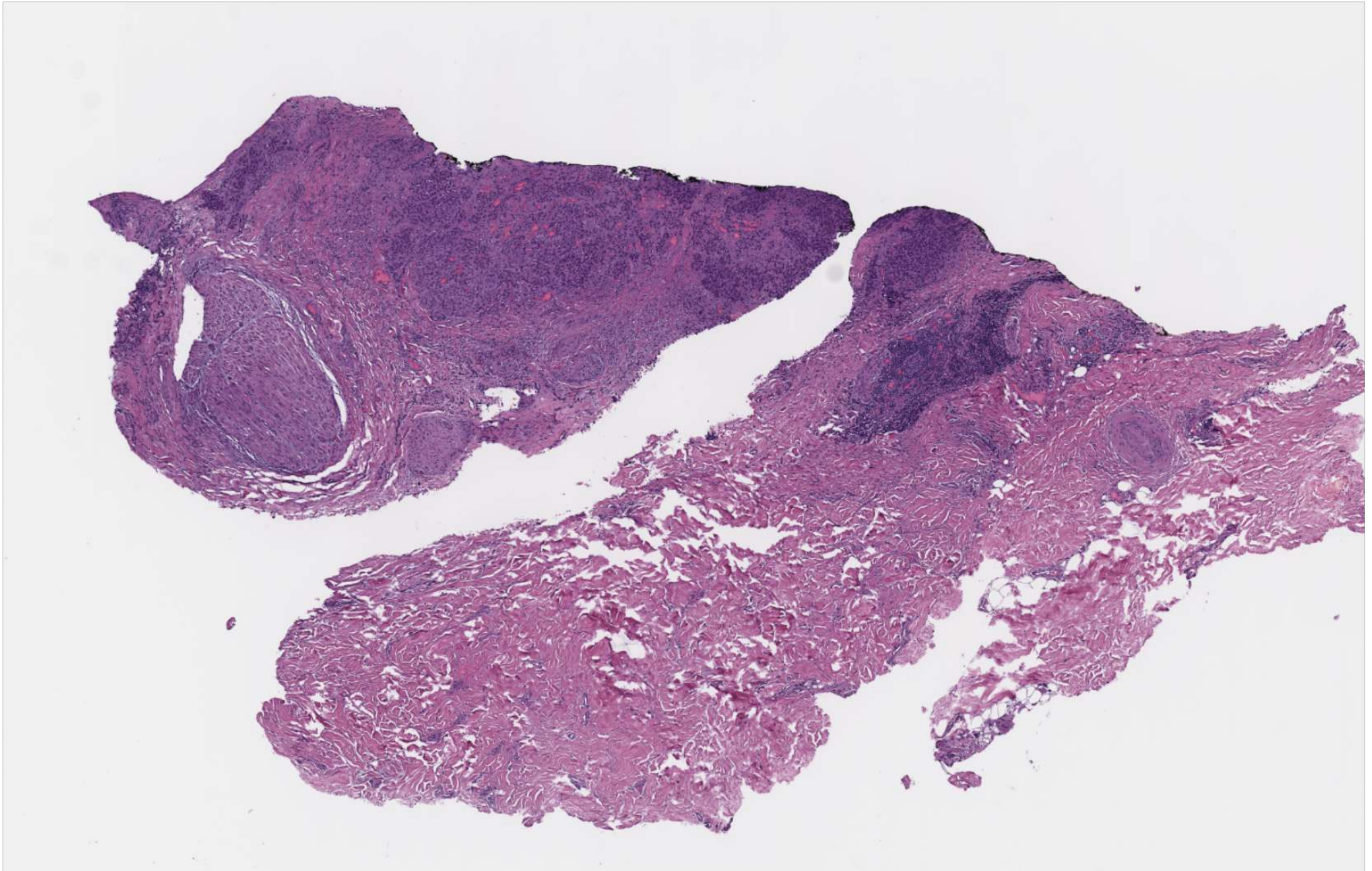
Graph features [2]

Mostly statistical [1]



No work uses signal properties yet

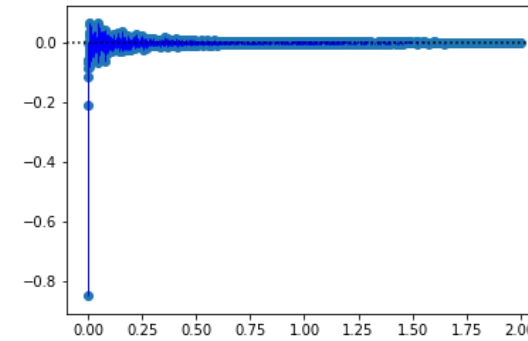
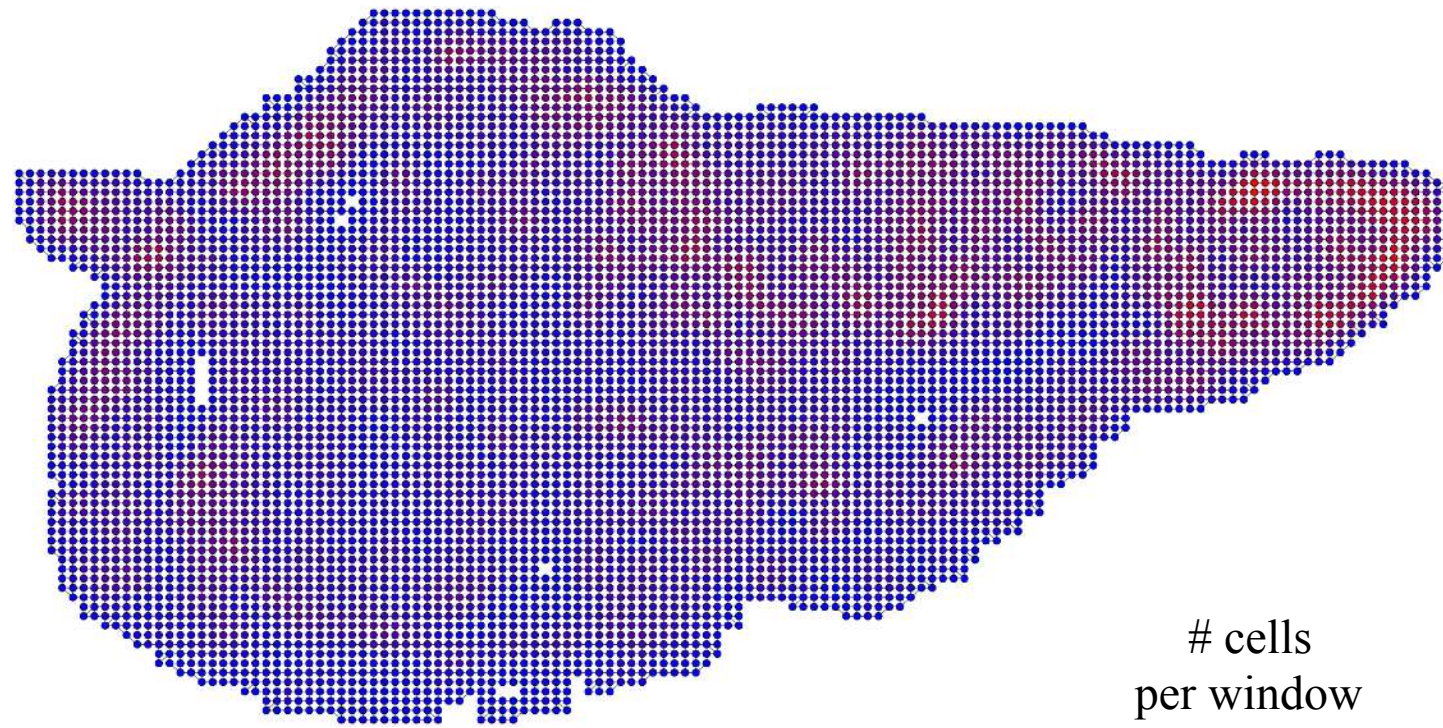
Our approach



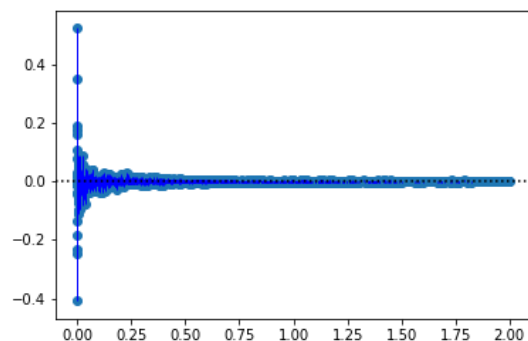
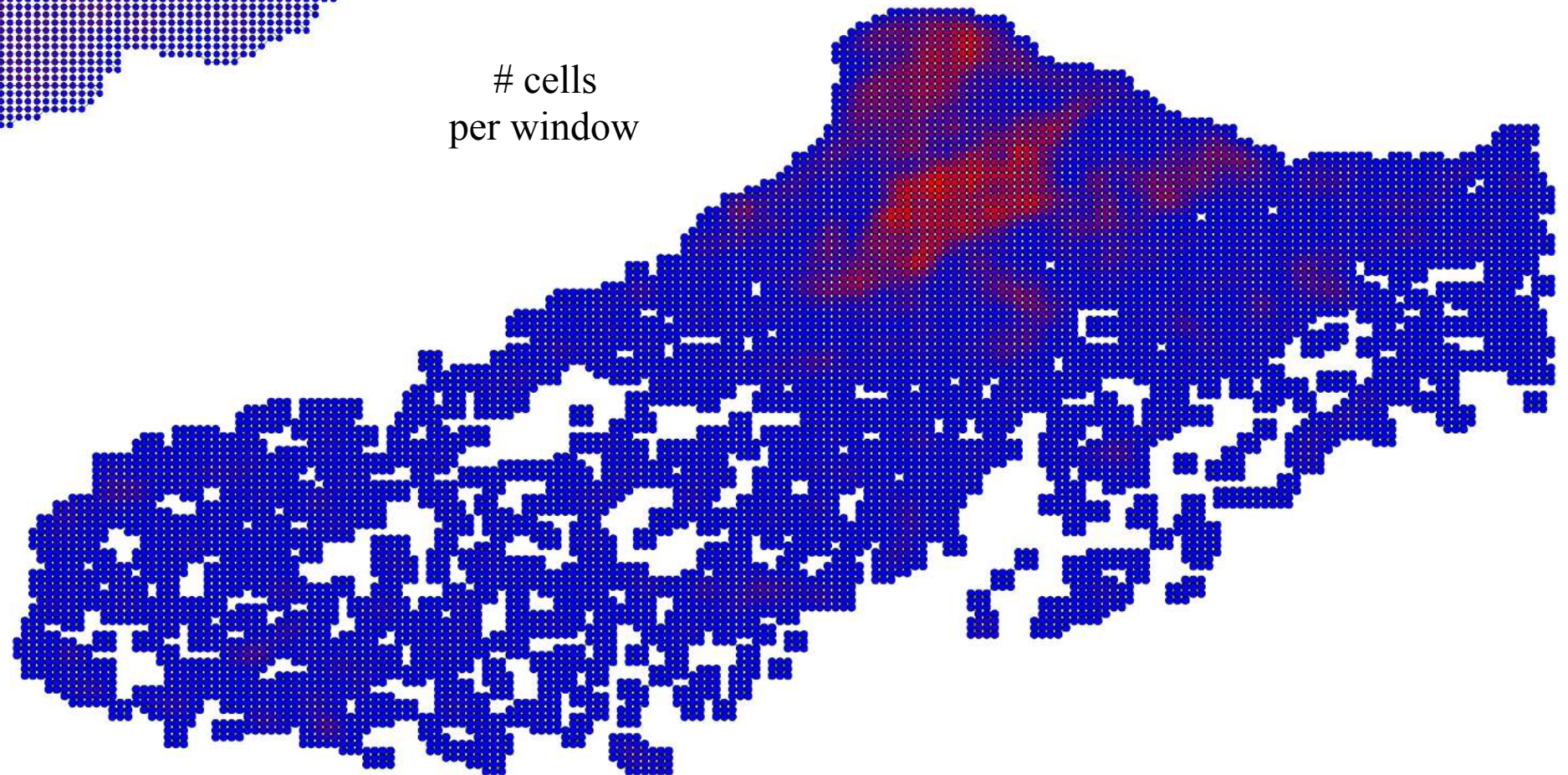
Our approach



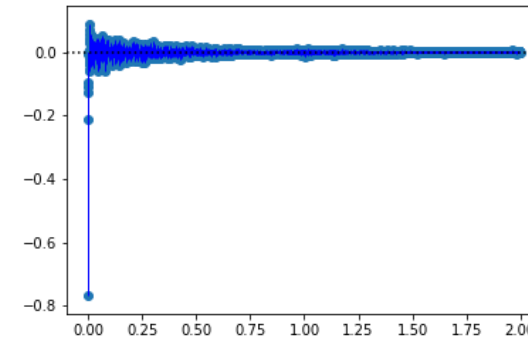
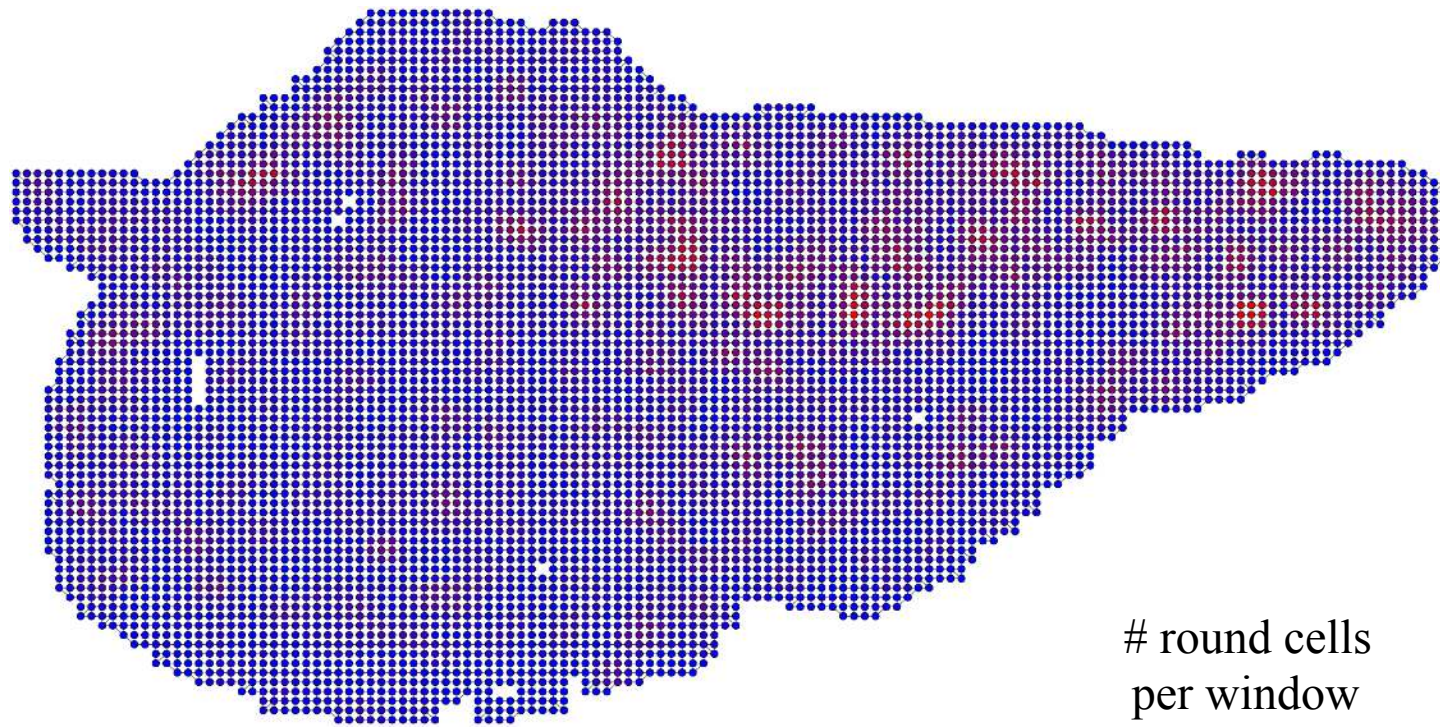
Our approach



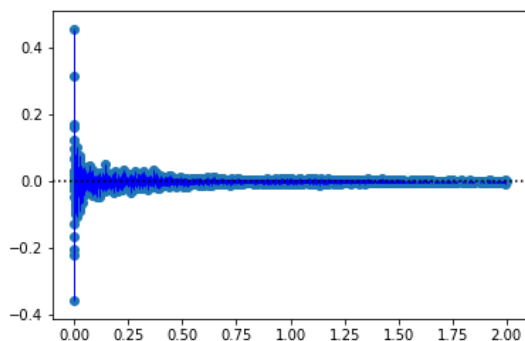
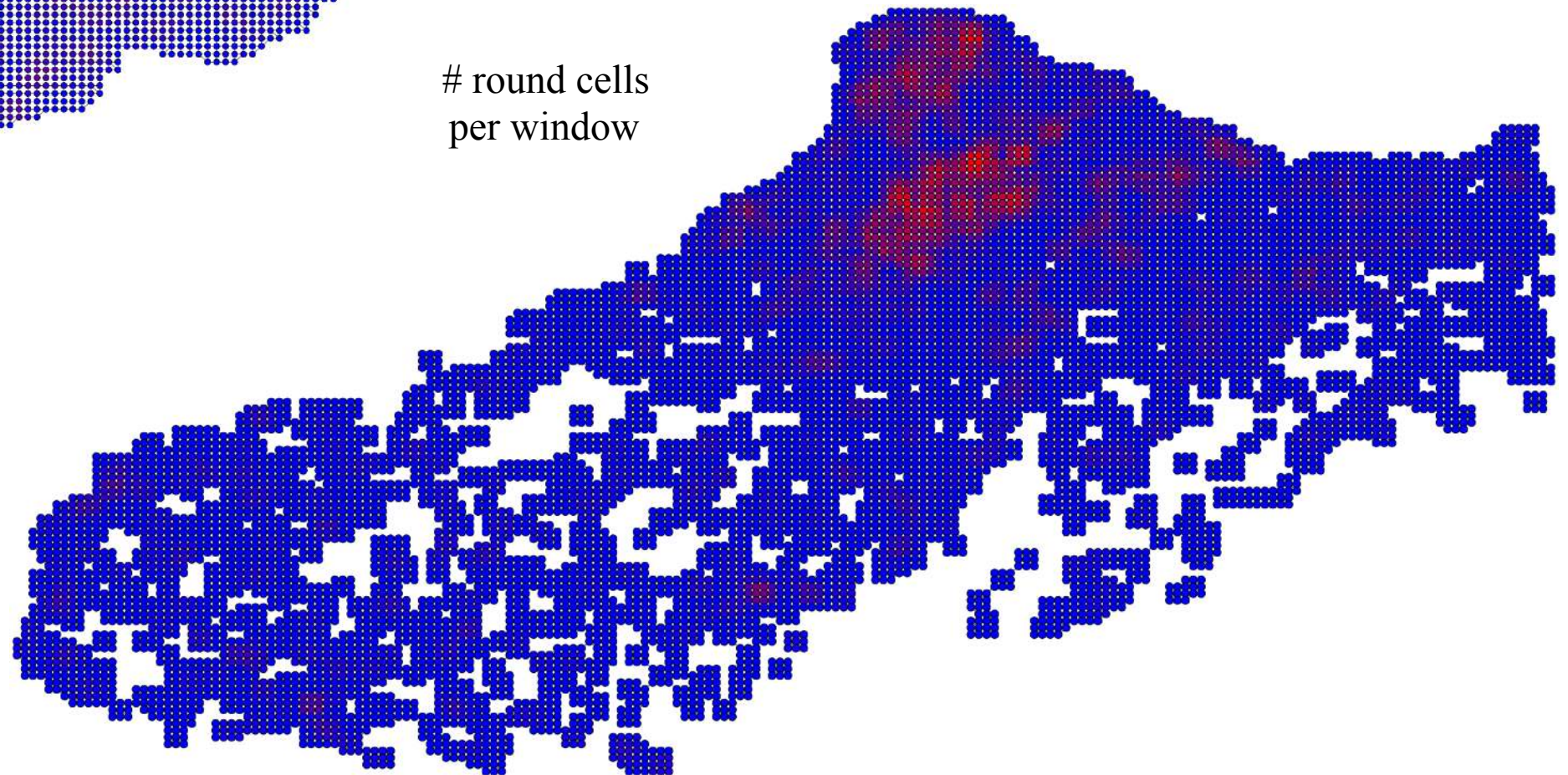
cells
per window



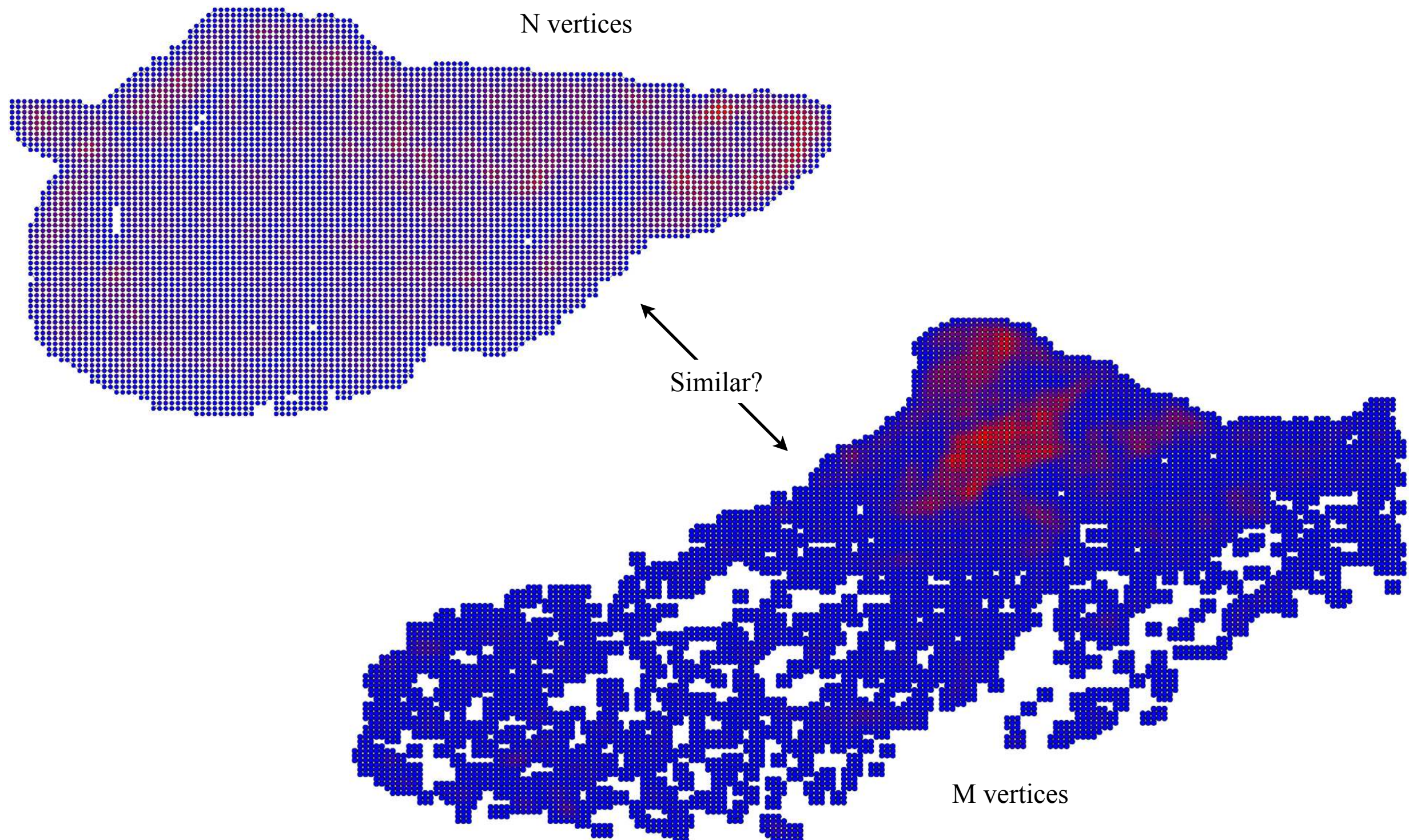
Our approach



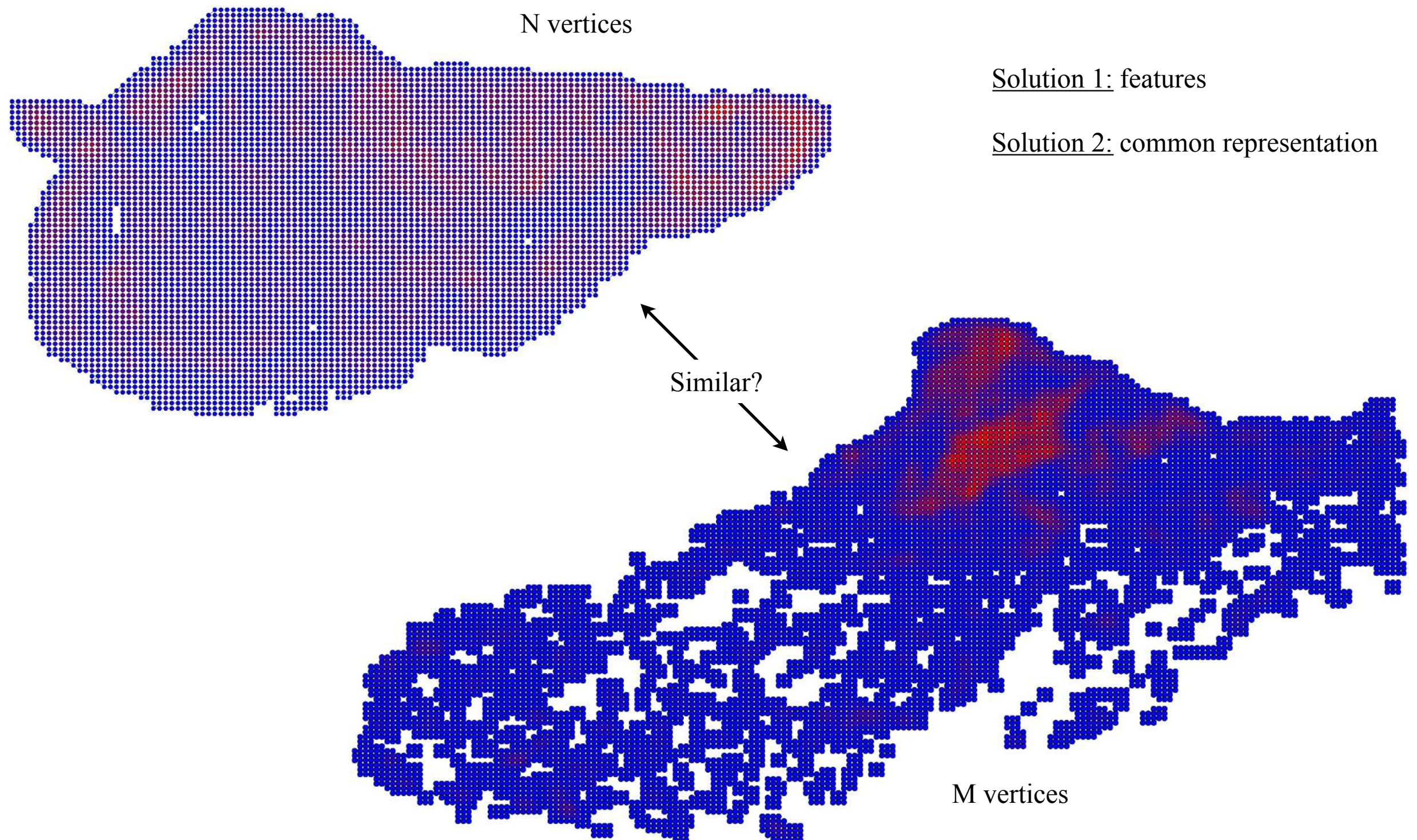
round cells
per window



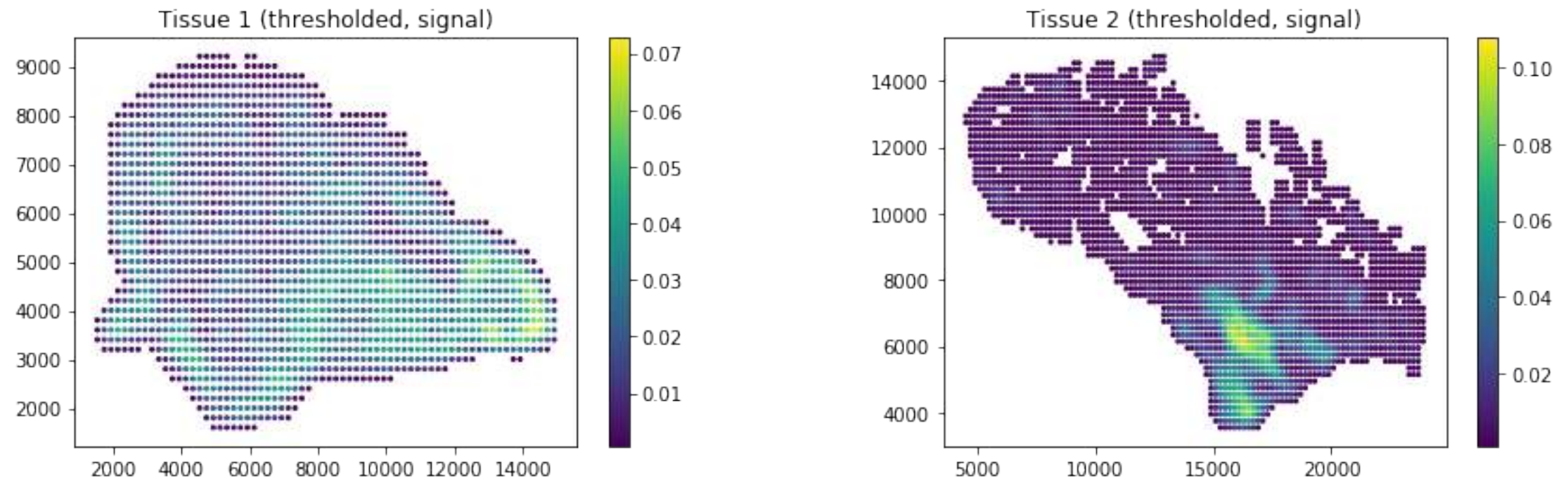
An interesting GSP problem



An interesting GSP problem

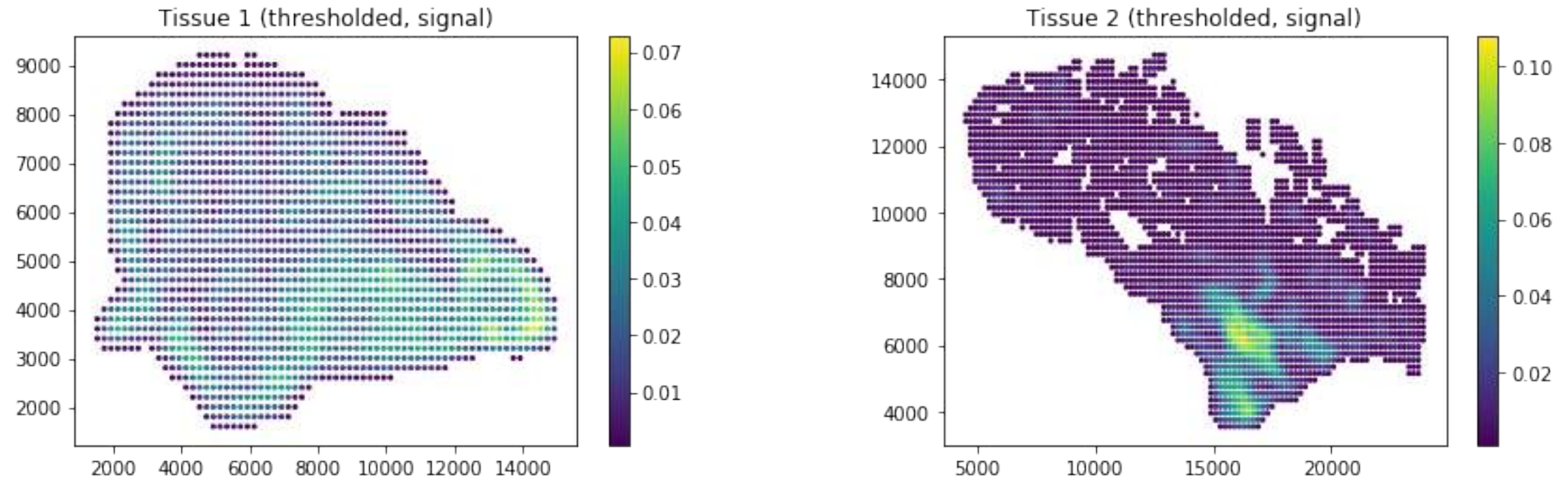


An interesting GSP problem

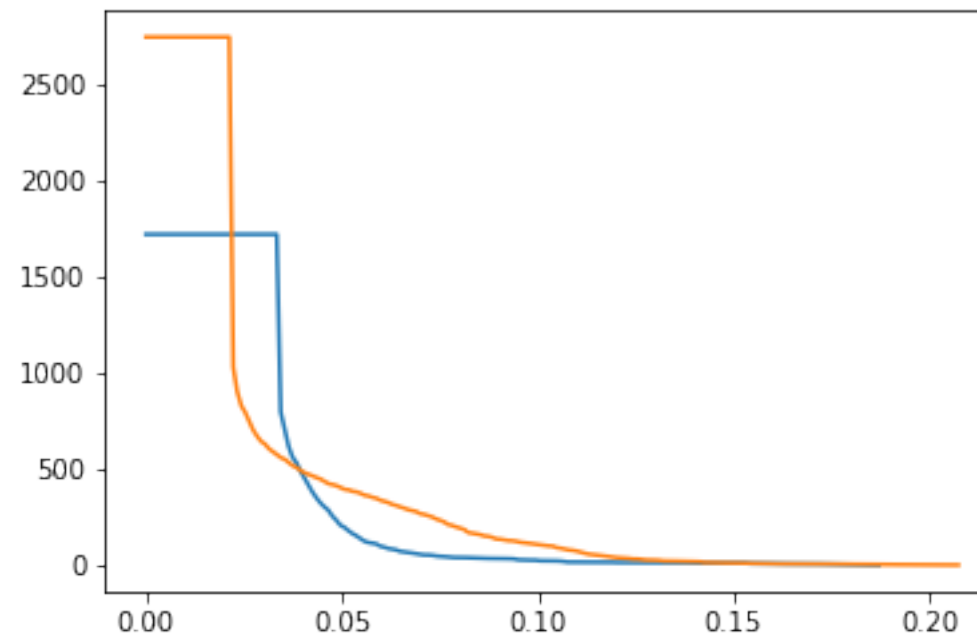


Learned with smoothness prior

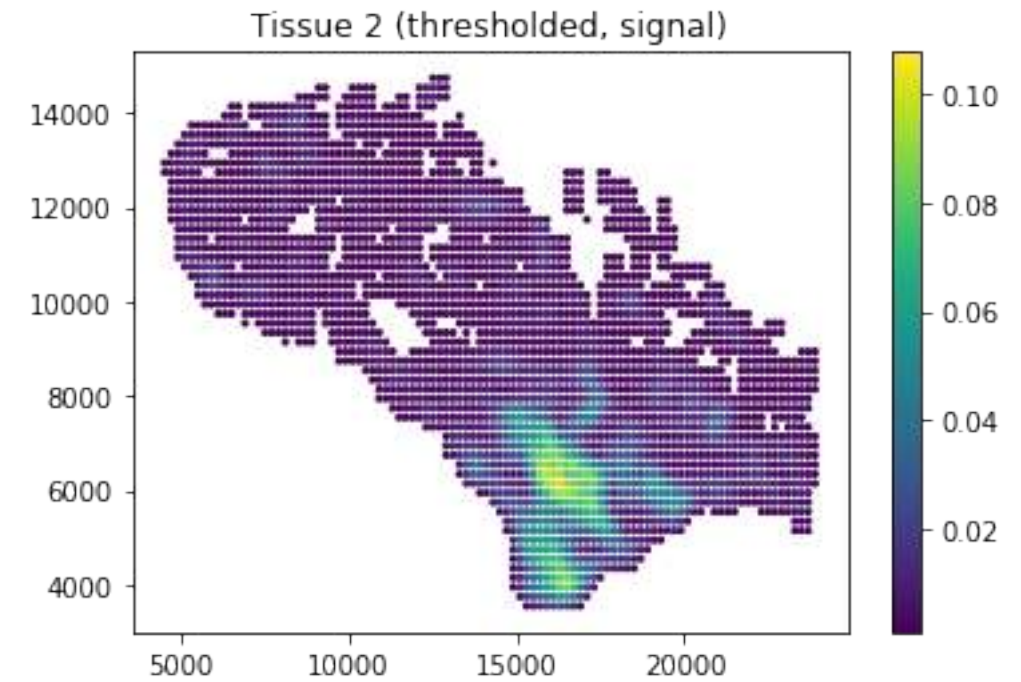
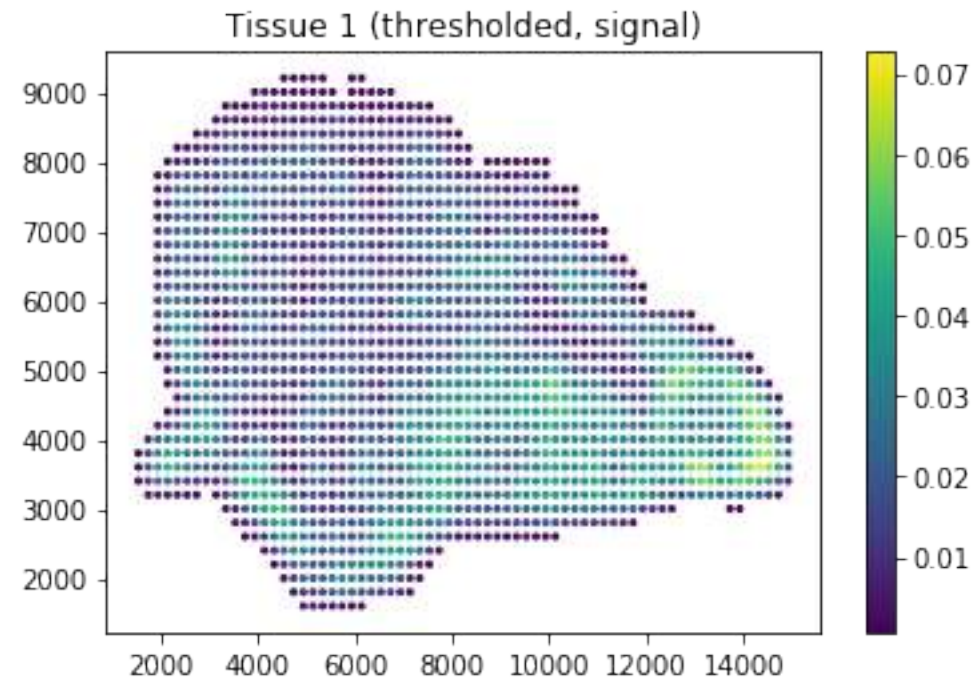
An interesting GSP problem



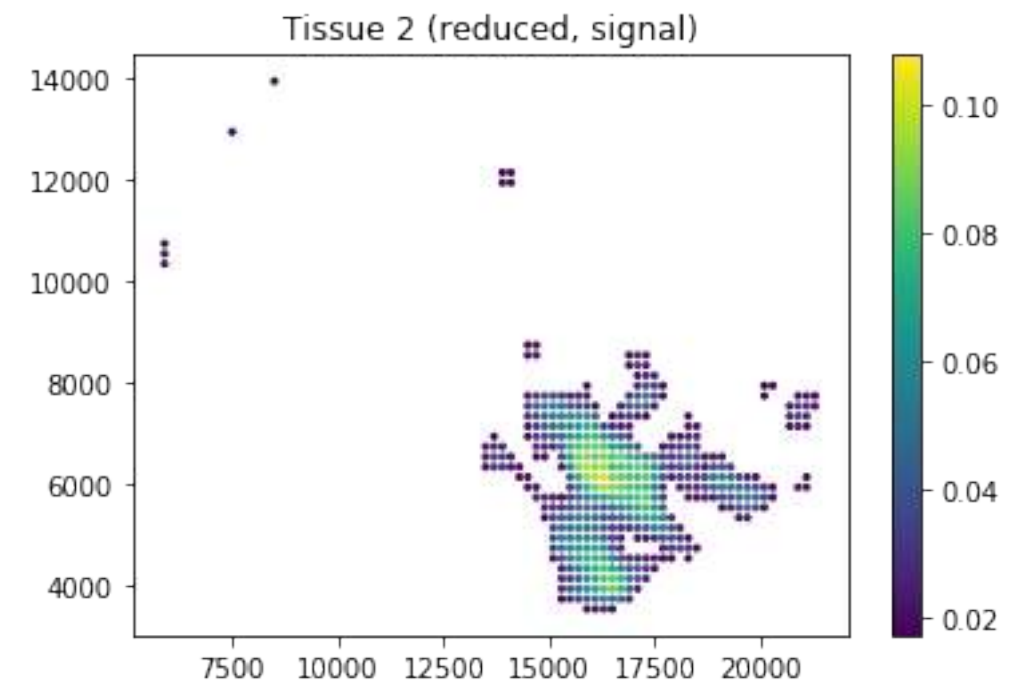
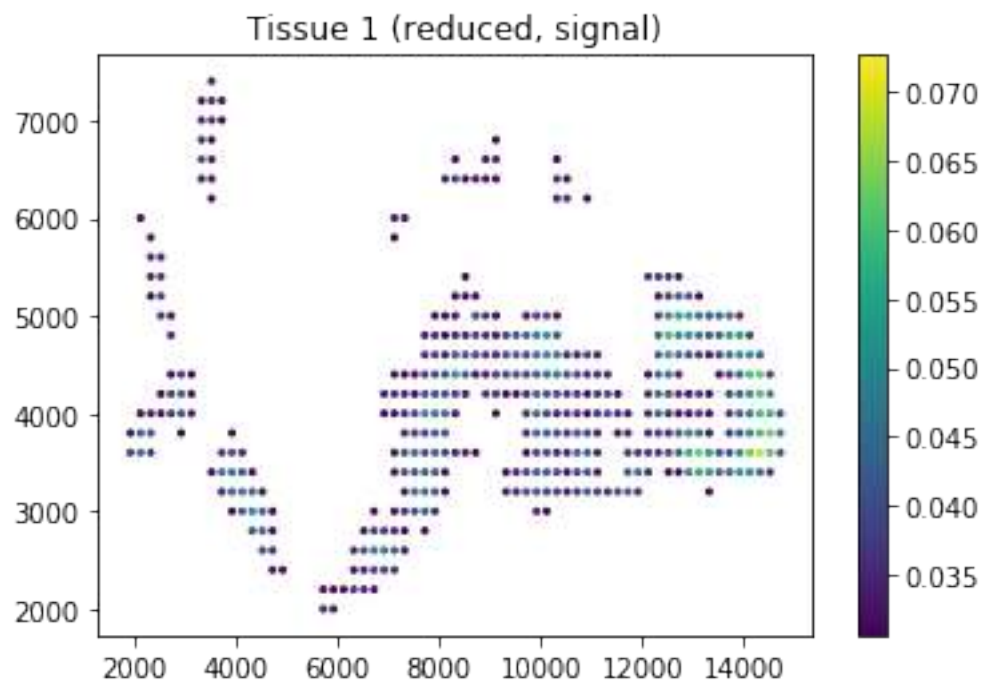
Learned with smoothness prior



An interesting GSP problem

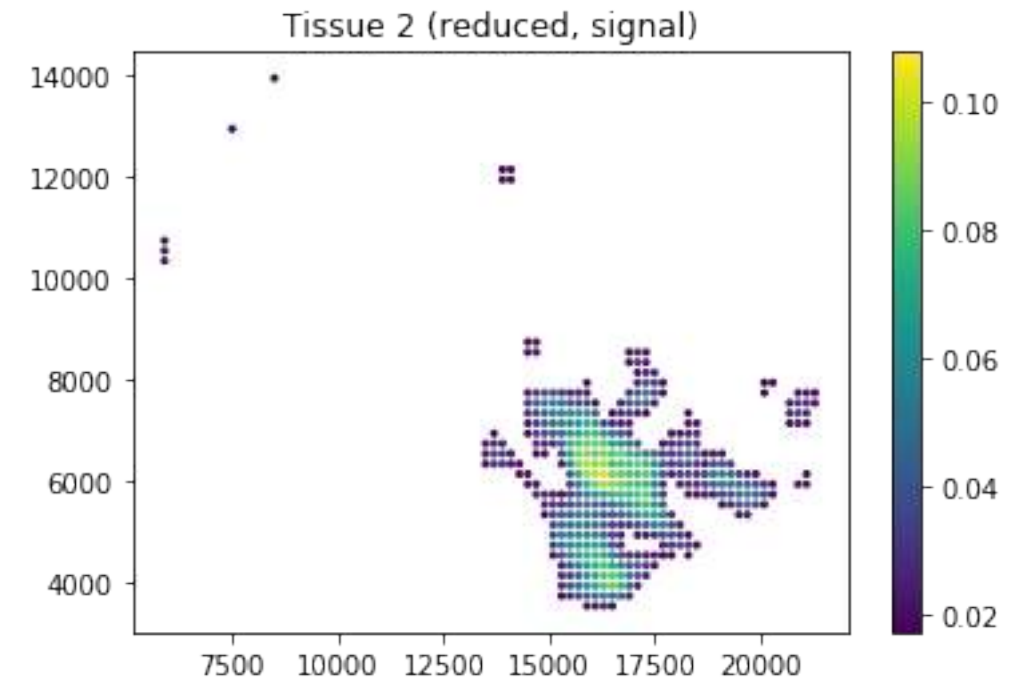
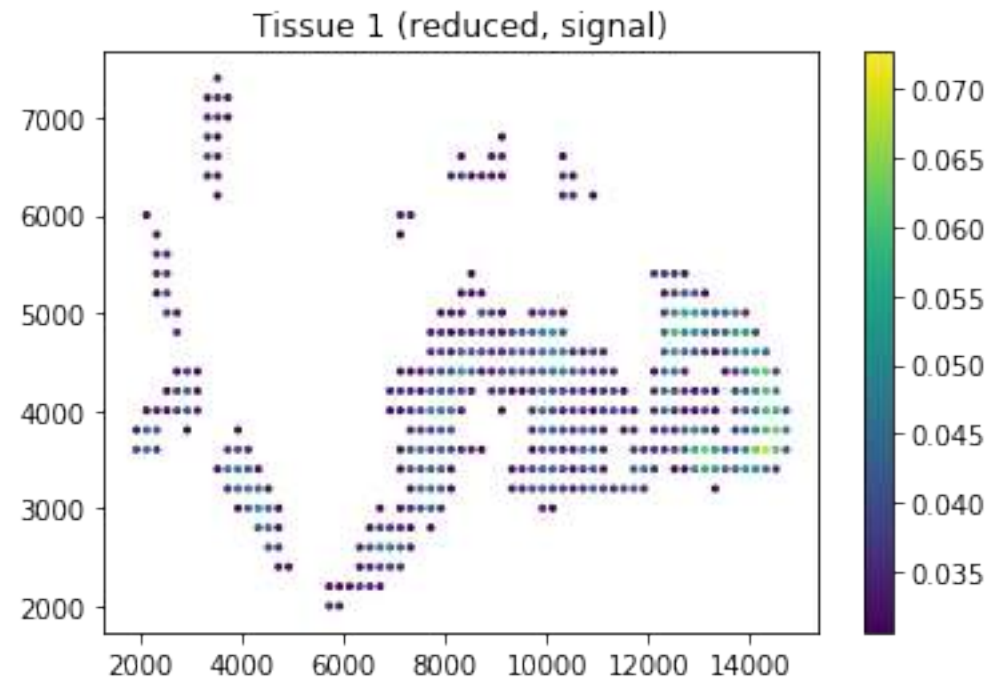


Learned with smoothness prior



Slightly different vertex count

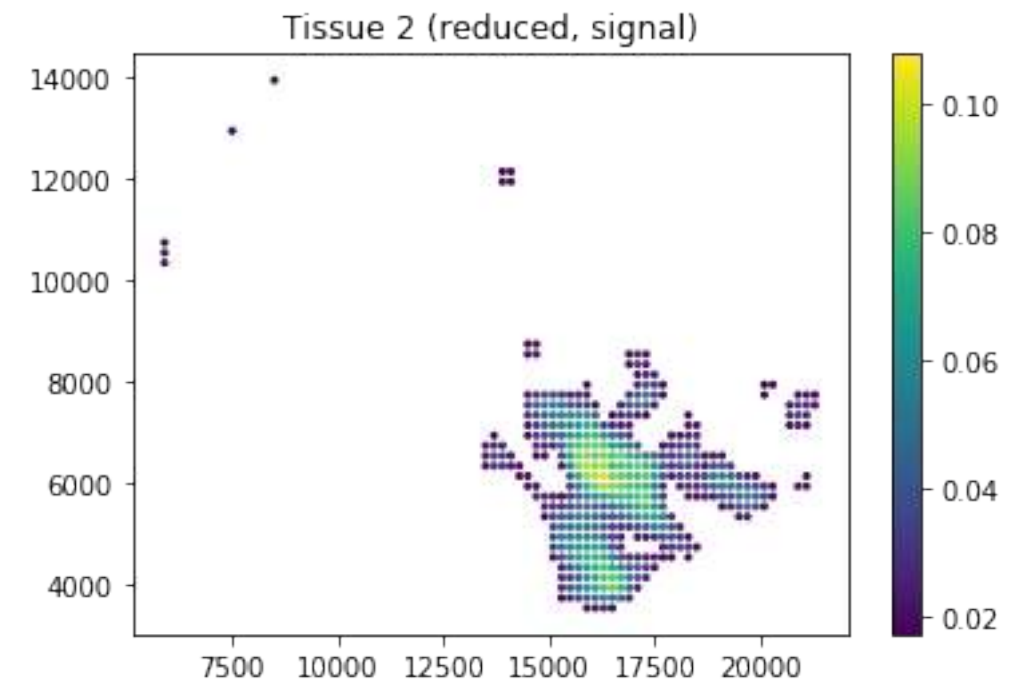
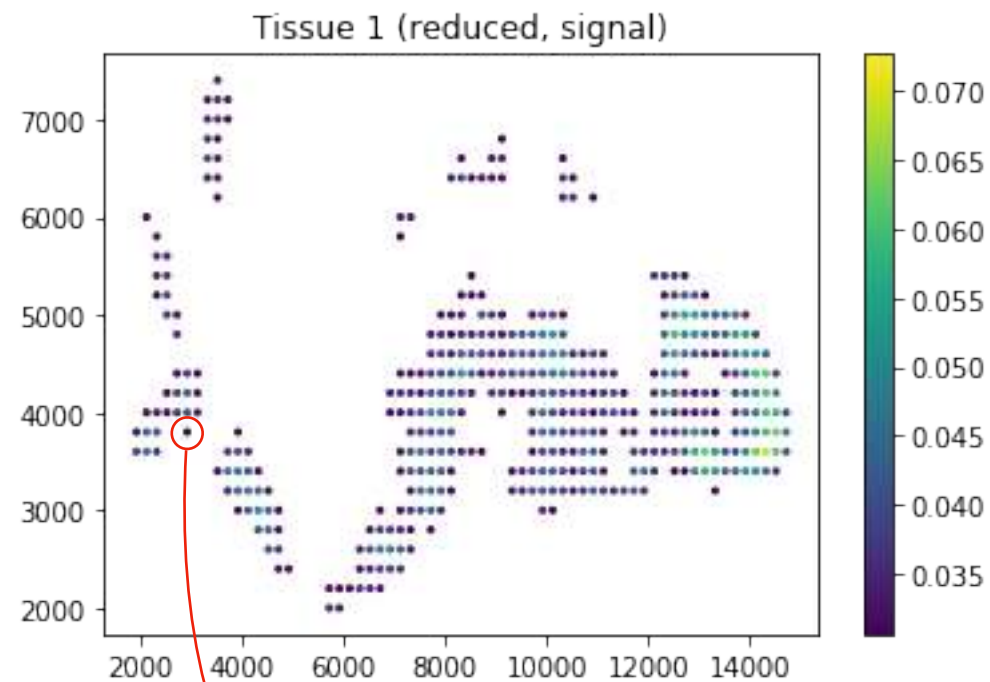
An interesting GSP problem



$$d_{\text{geo}}(\mathbf{A}_1, \mathbf{A}_2) = \sqrt{\sum_{1 \leq i \leq N} |\log \lambda_i|^2} \quad [1]$$

$\downarrow$ $\downarrow$ $\downarrow$
 Laplacians Generalized eigenvalues

An interesting GSP problem

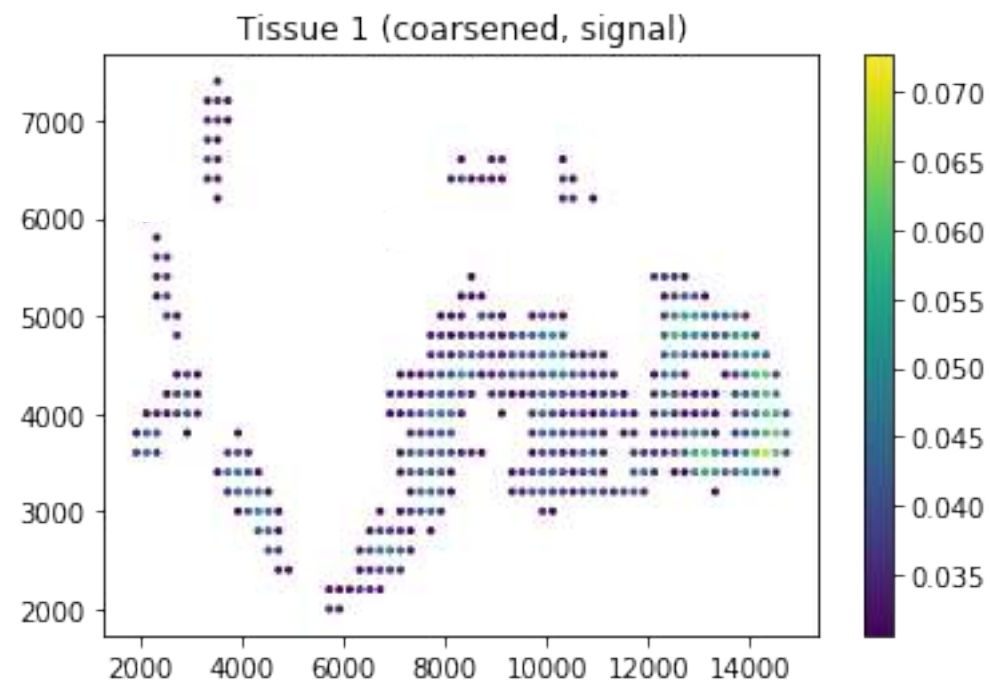
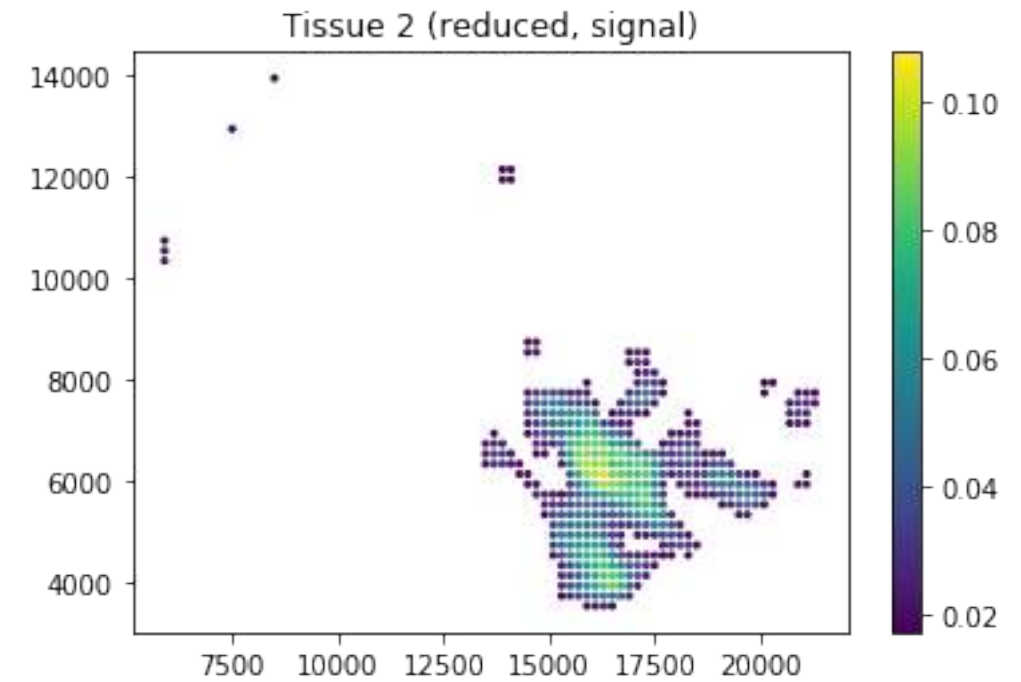
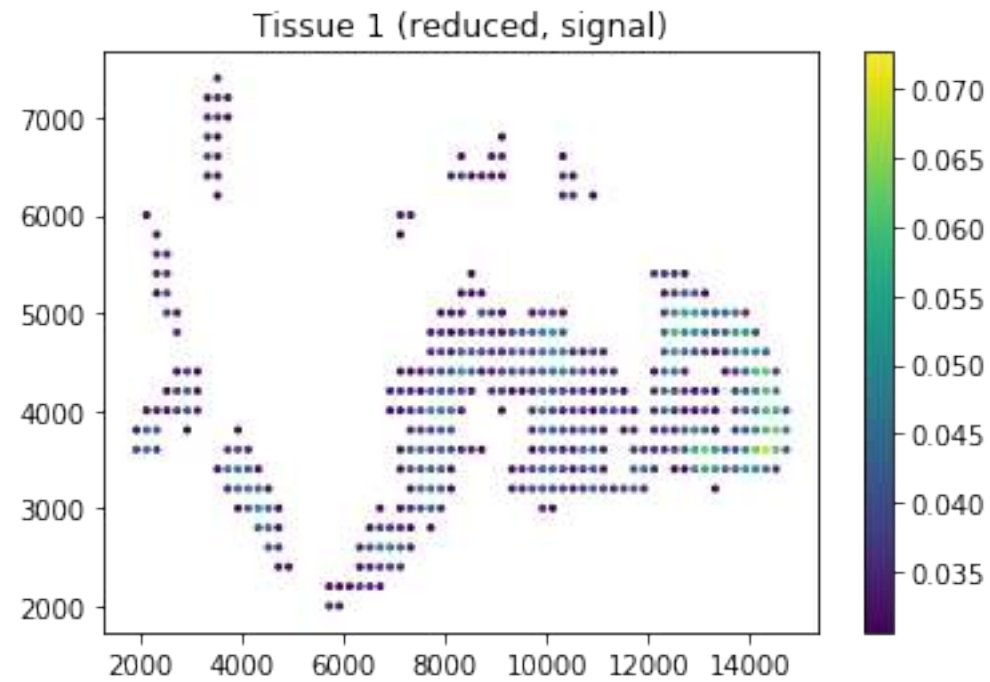


Disconnect the
minimizer

$$d_{\text{geo}}(\mathbf{A}_1, \mathbf{A}_2) = \sqrt{\sum_{1 \leq i \leq N} |\log \lambda_i|^2} \quad [1]$$

$\downarrow$ $\downarrow$ $\downarrow$
 Laplacians Generalized eigenvalues

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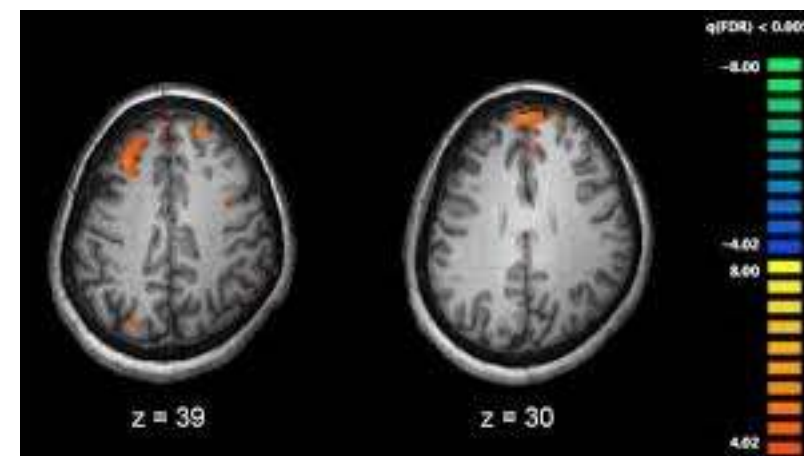
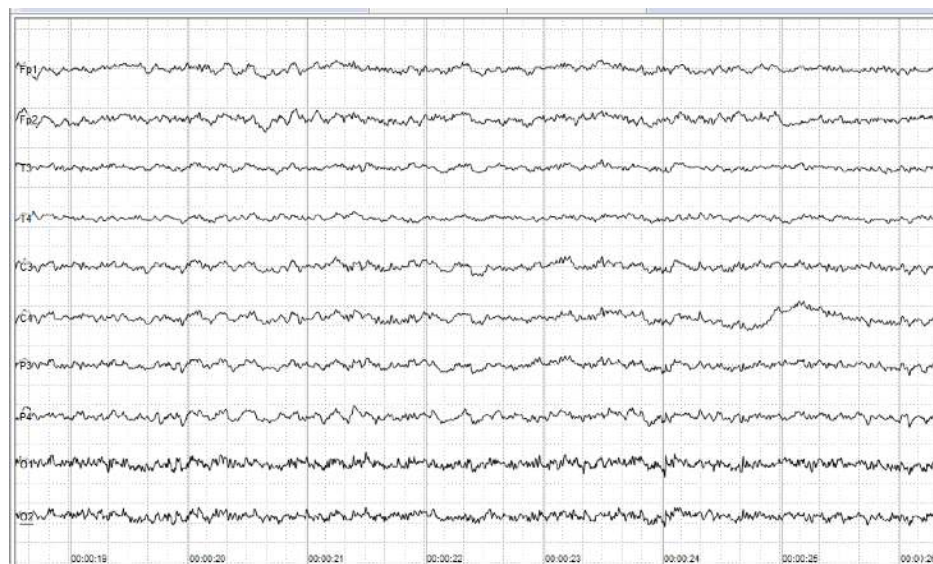


Project on the other graph
with optimal transport
between Gaussians?

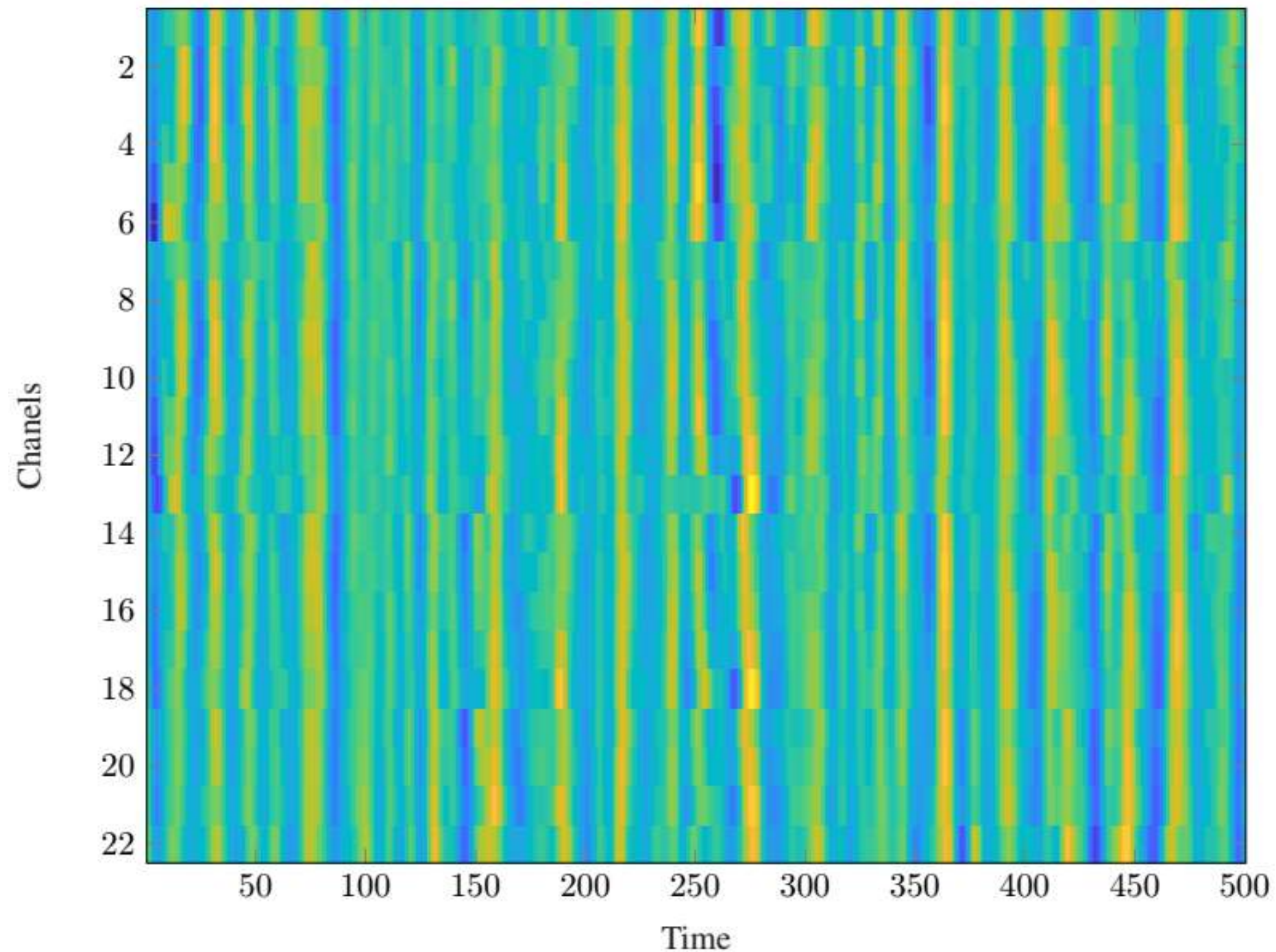
Outline

- GSP for AI, *a.k.a.* "What I did during my Ph.D. and a bit after"
- GSP and personalized healthcare
- **GSP and brain signals**

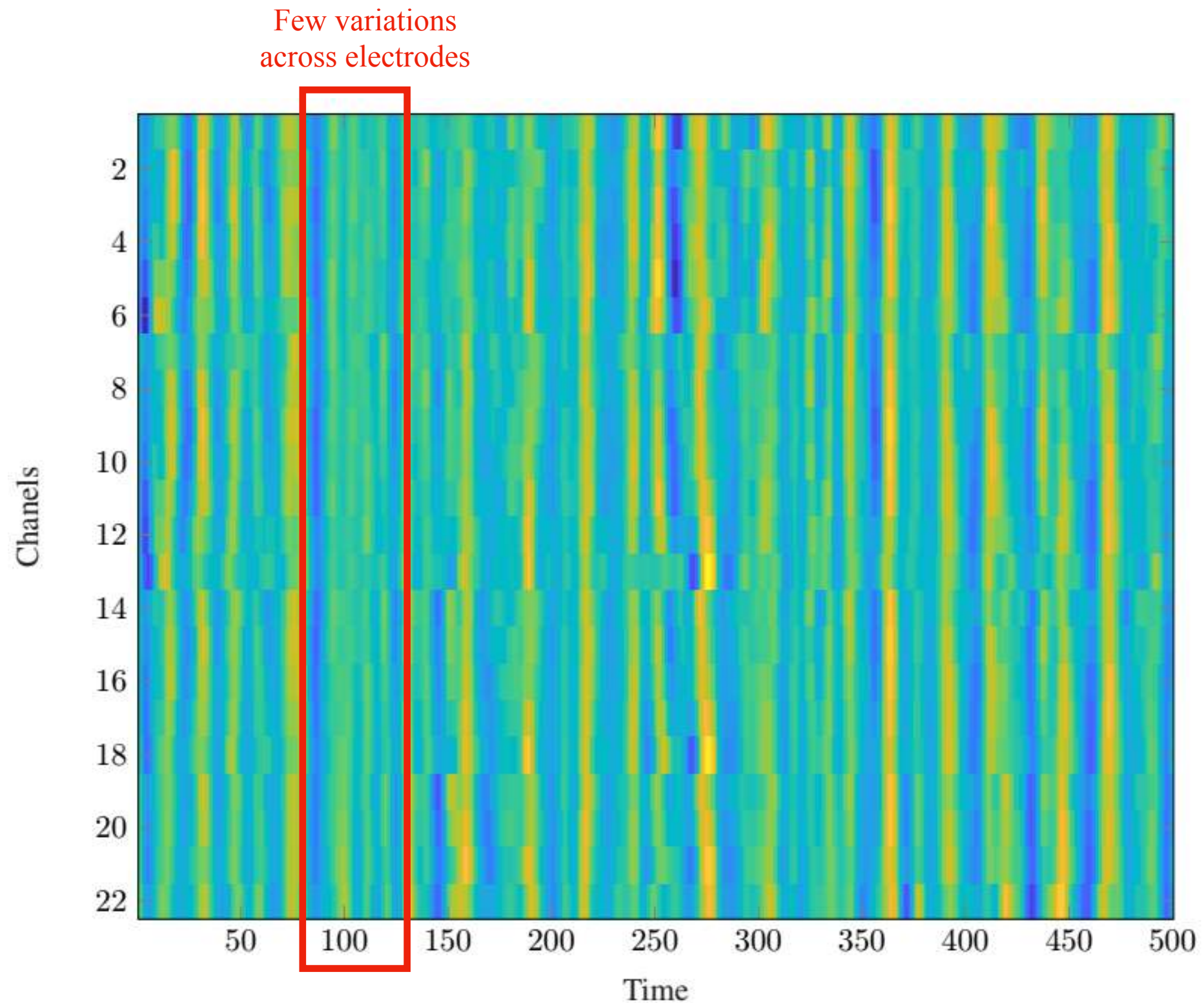
EEG and fMRI measure brain activity



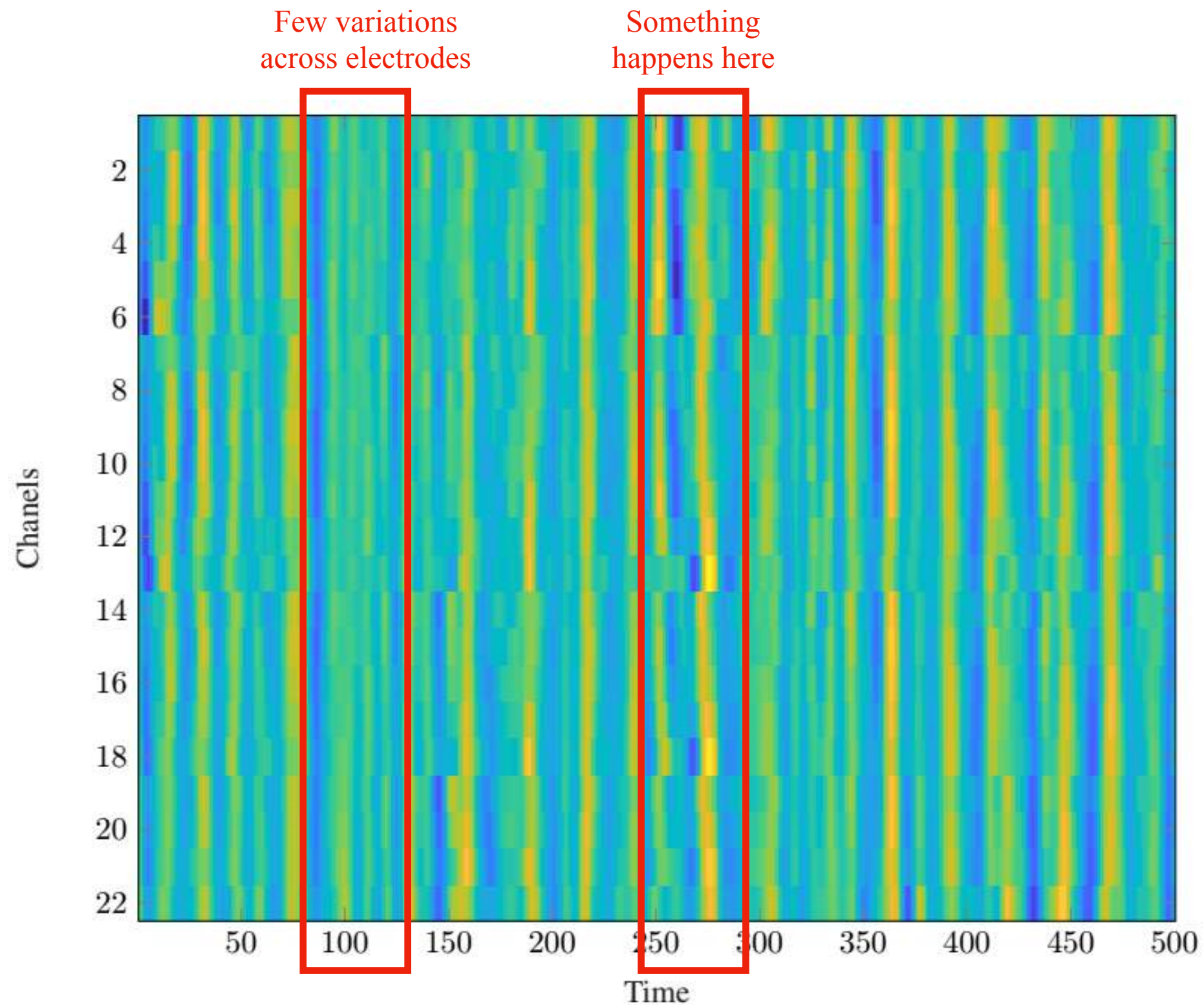
Task seen as non-stationary behavior



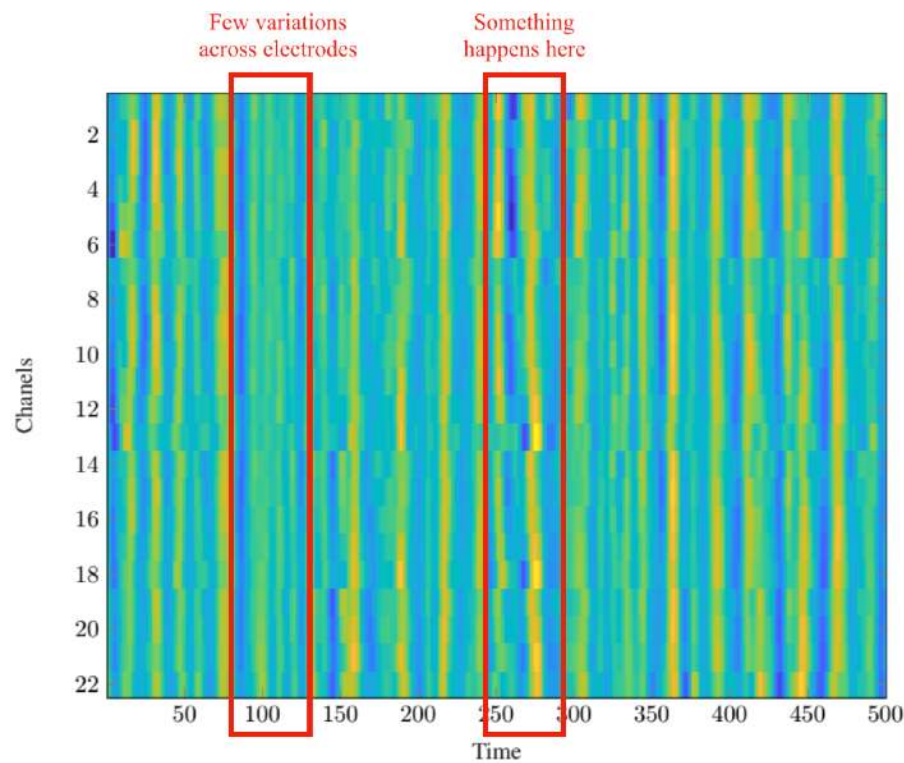
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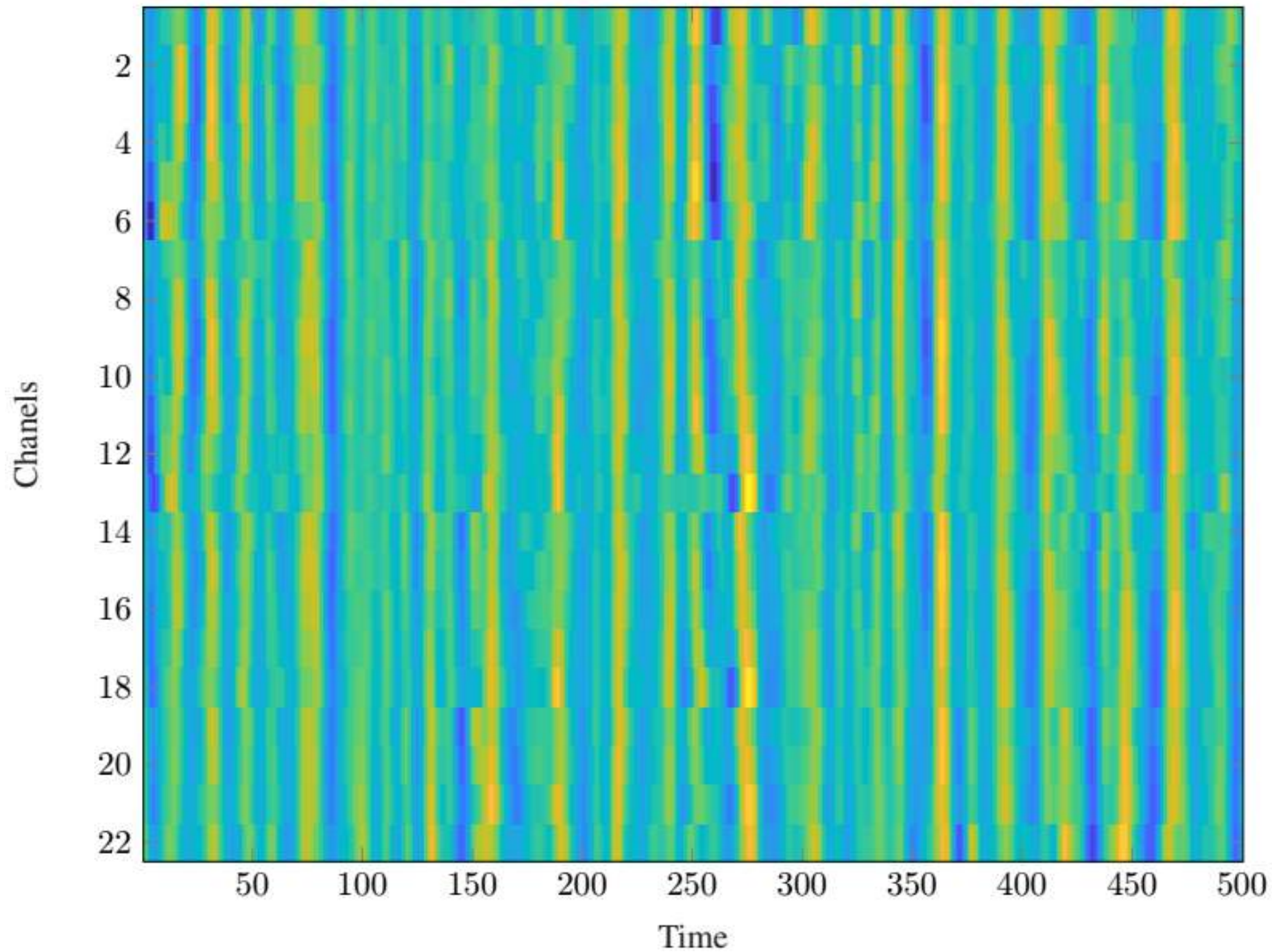
How do we measure stationarity?



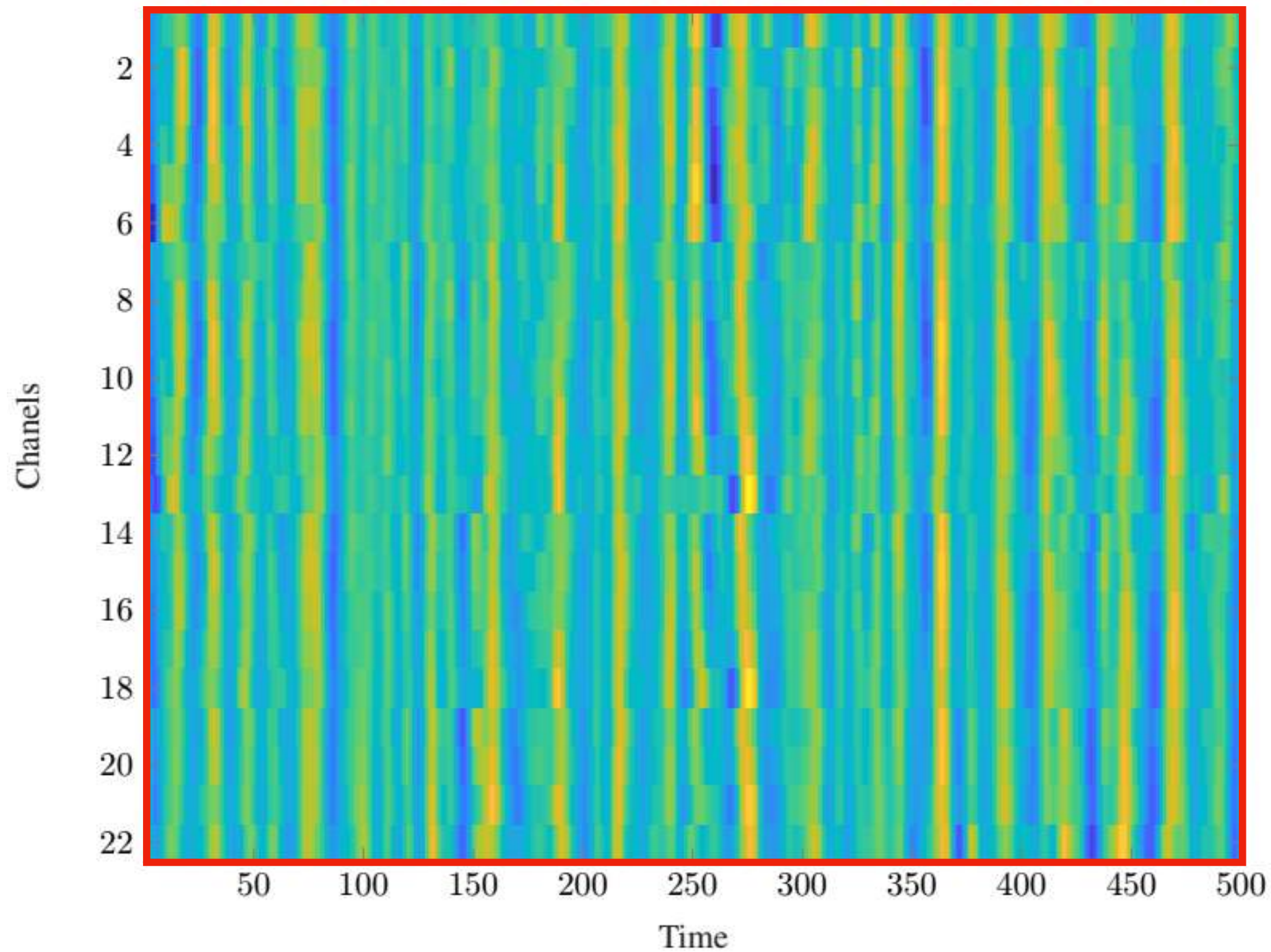
$$\begin{aligned}
 \tilde{\Sigma} &= \sum_{i=1}^M \mathbf{X}(i) \mathbf{X}(i)^\top \\
 &= \sum_{k \in \mathbf{k}} \sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i) = k}} \mathbf{T}^k \mathbf{Y}(i) \mathbf{Y}(i)^\top \mathbf{T}^{k^\top} \\
 &= \sum_{k \in \mathbf{k}} \mathbf{T}^k \left(\sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i) = k}} \mathbf{Y}(i) \mathbf{Y}(i)^\top \right) \mathbf{T}^{k^\top} \\
 \Sigma &= \sum_{k \in \mathbf{k}} \mathbf{T}^k \mathbb{E}_{\mathbf{Y}} \left[\sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i) = k}} \mathbf{Y}(i) \mathbf{Y}(i)^\top \right] \mathbf{T}^{k^\top} \\
 &= \sum_{k \in \mathbf{k}} |\{i, \mathbf{k}(i) = k\}| \mathbf{T}^{2k},
 \end{aligned}$$

The covariance matrix of signals that are stationary on a graph shares the same eigenvectors as the graph itself

How do we measure stationarity?

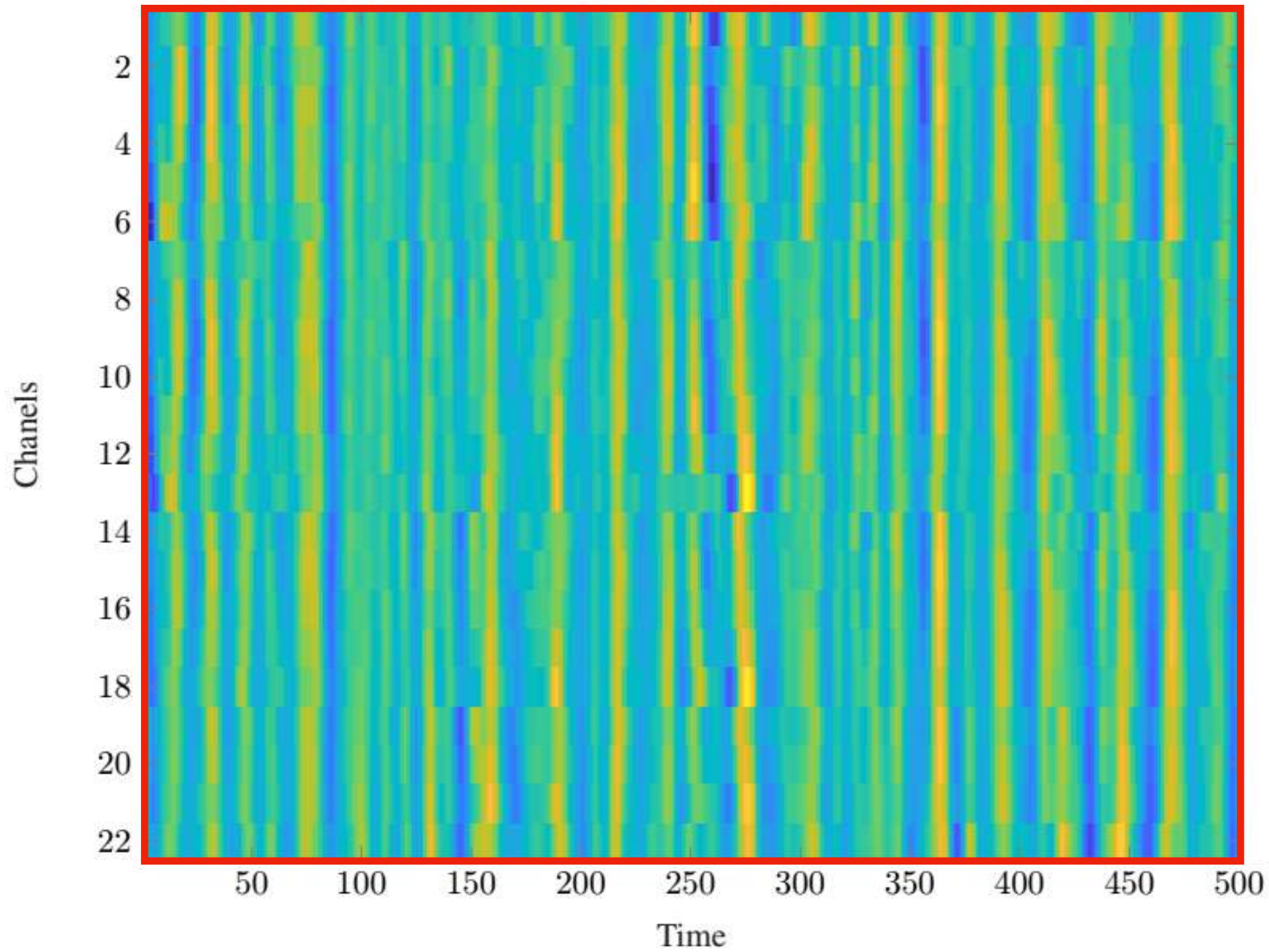


How do we measure stationarity?



Σ

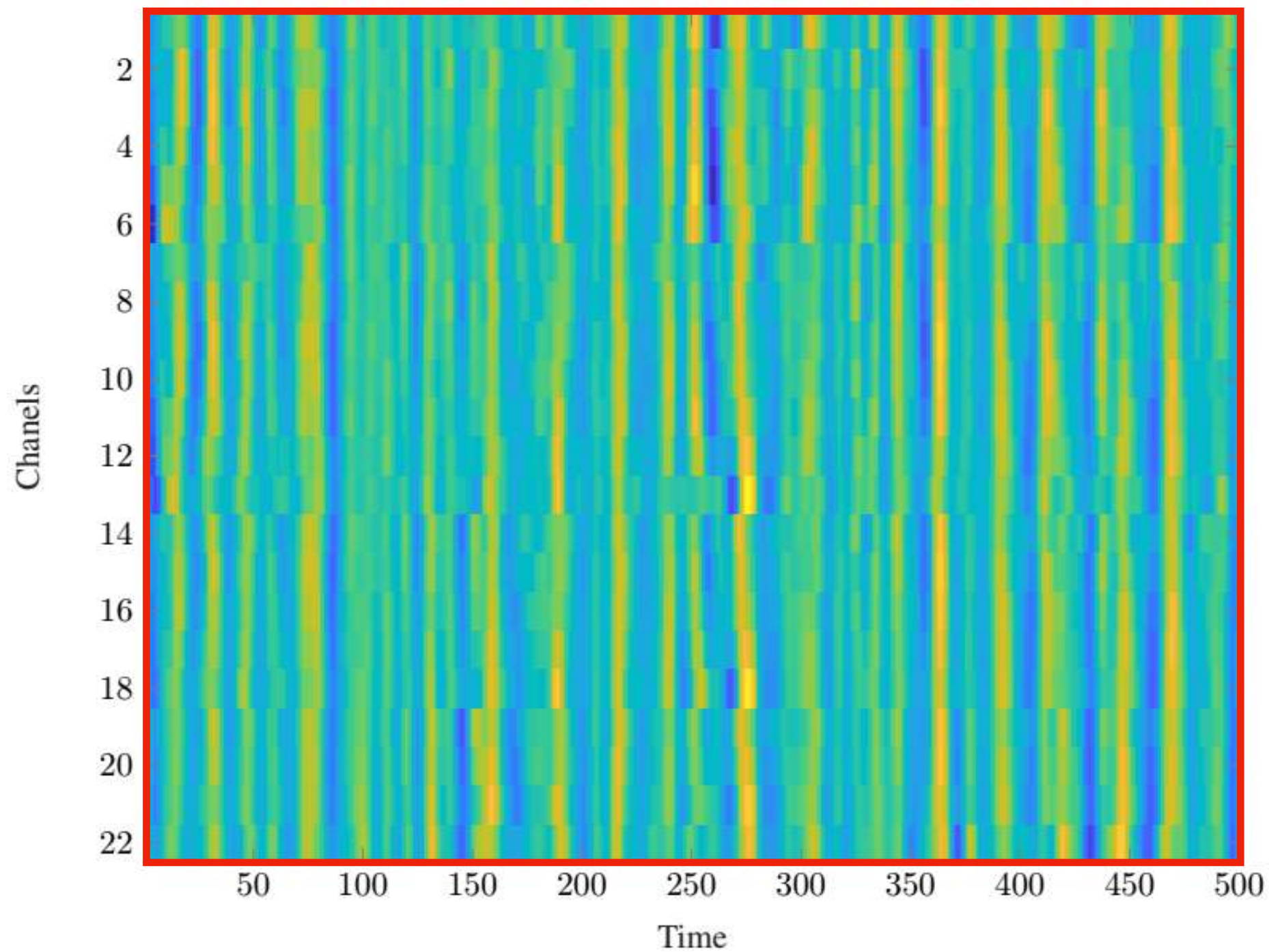
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Σ

Σ_1

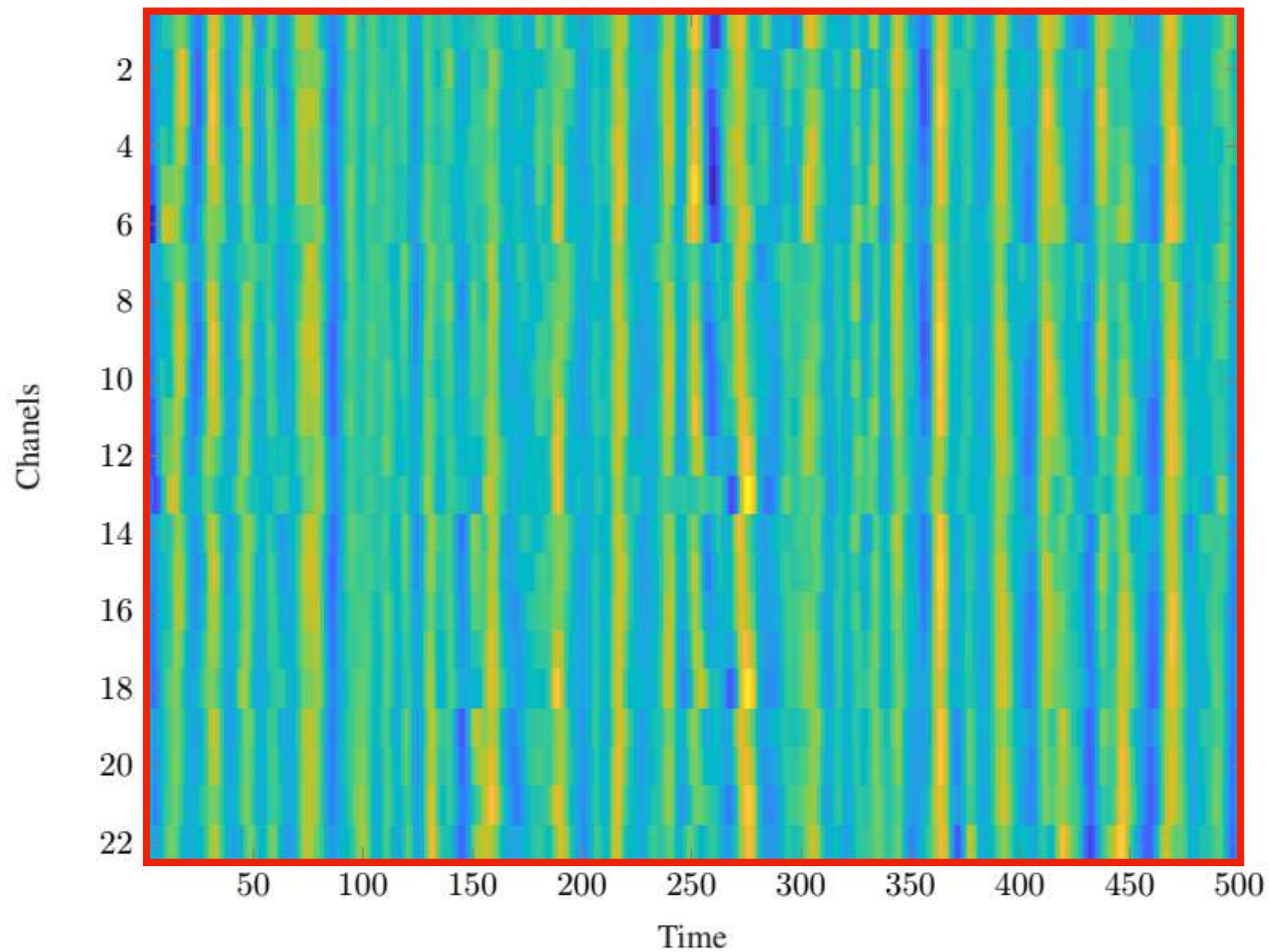
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Σ

$\Sigma_1 \quad \Sigma_2$

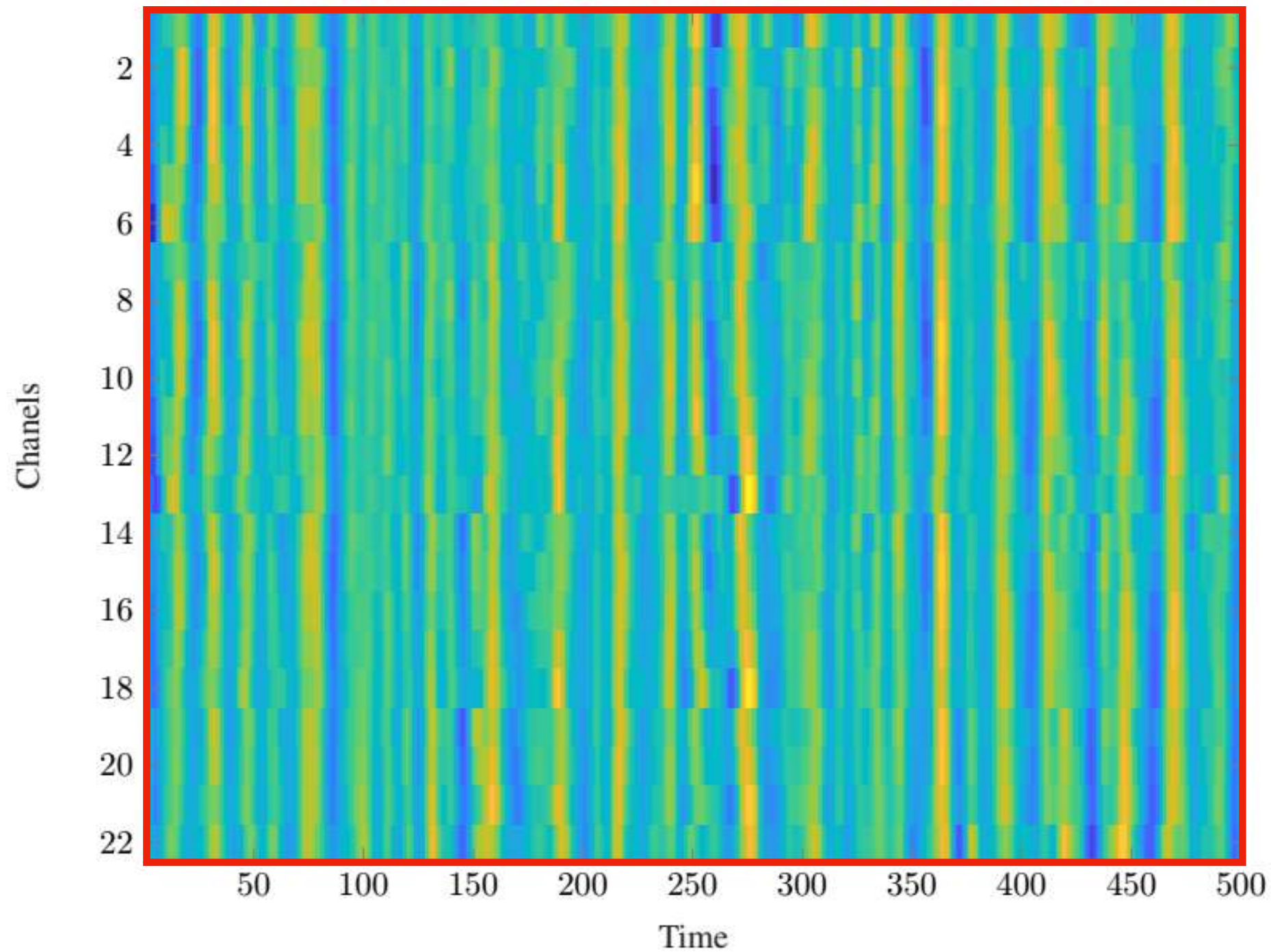
How do we measure stationarity?



Σ

$\Sigma_1 \quad \Sigma_2 \quad \Sigma_3$

How do we measure stationarity?



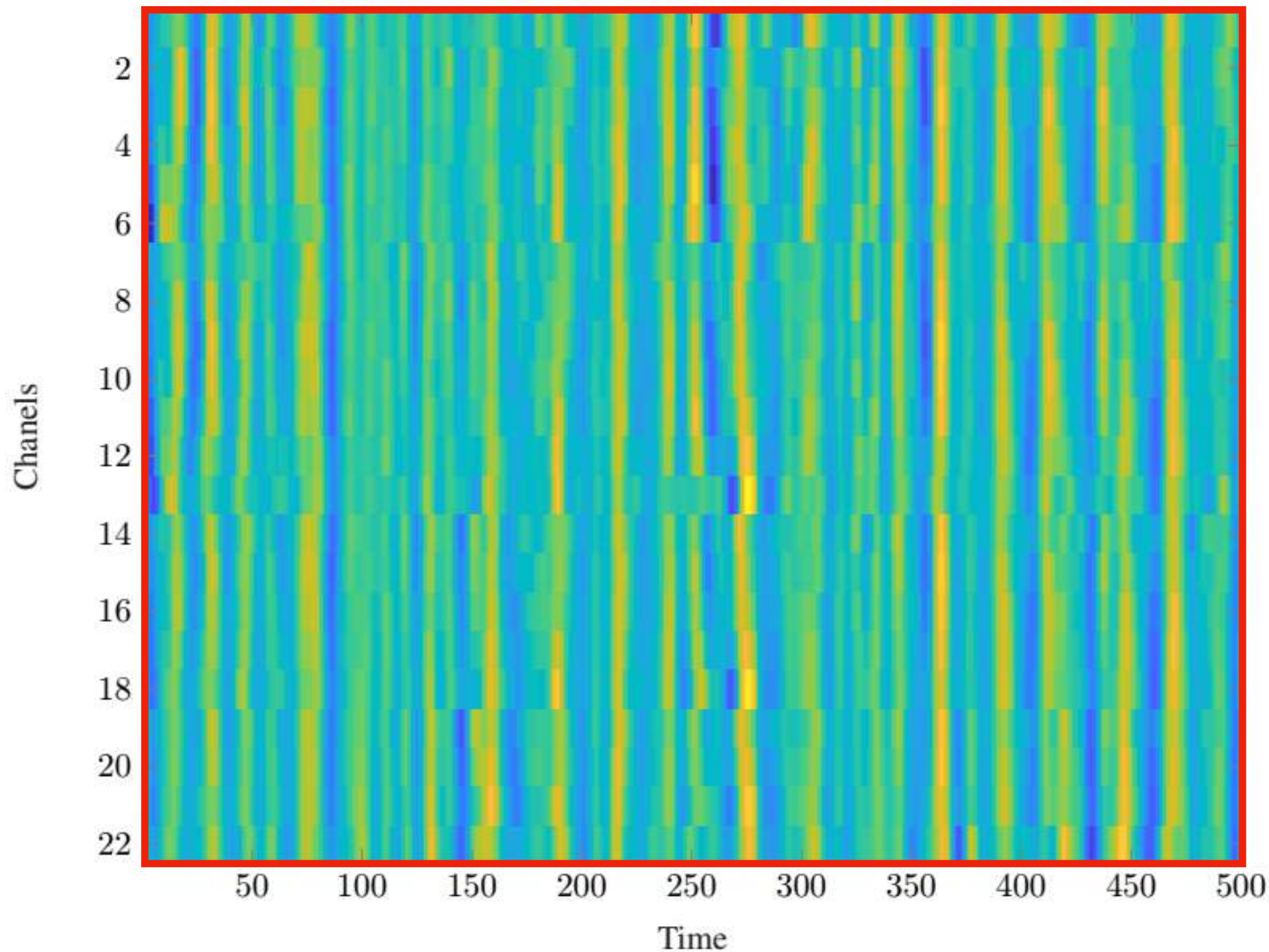
Σ

$\Sigma_1 \quad \Sigma_2 \quad \Sigma_3$

...

Σ_M

How do we measure stationarity?



$$\Sigma = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{\top}$$

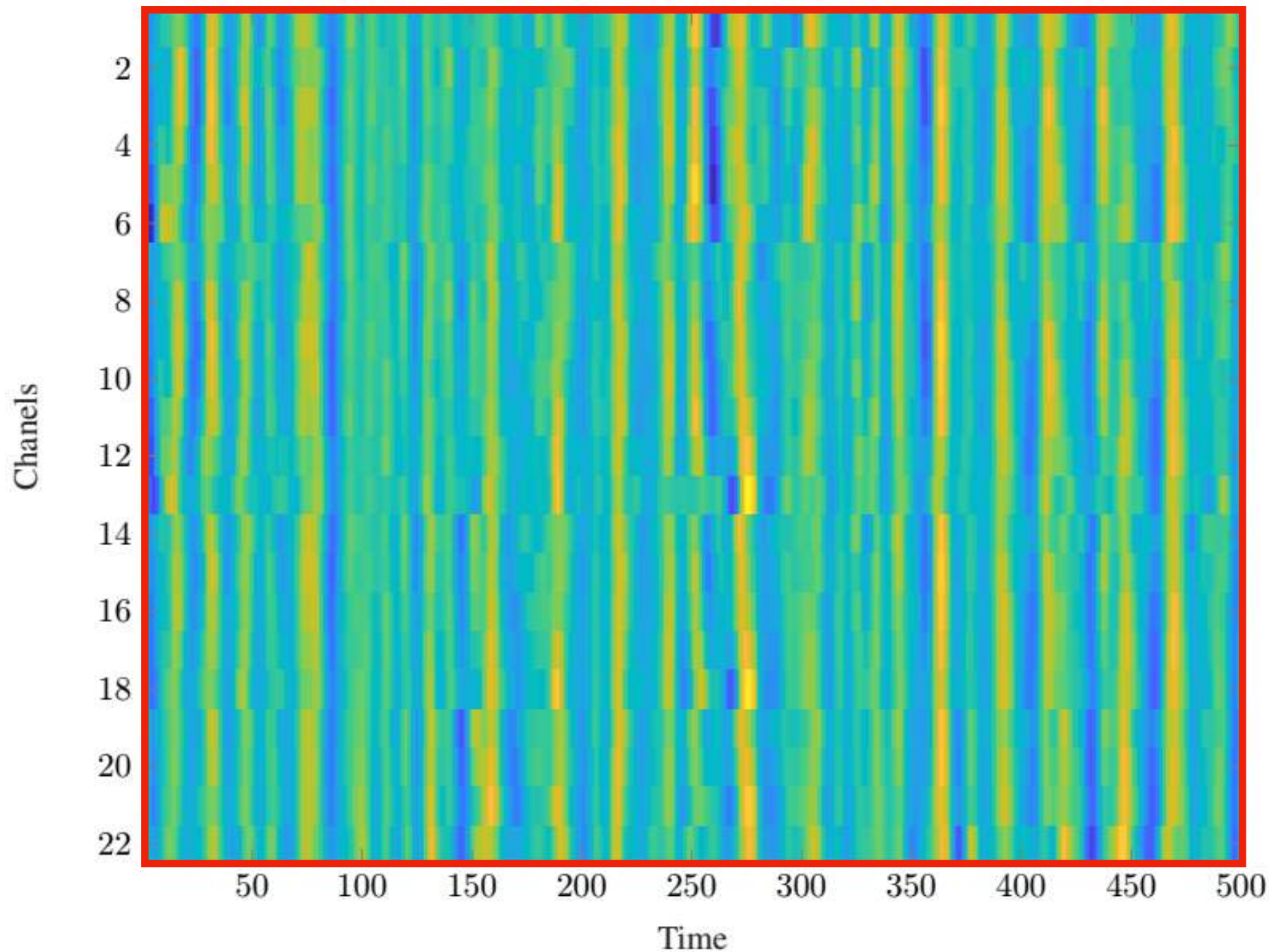
$$\Sigma_1 \quad \Sigma_2 \quad \Sigma_3$$

...

$$\Sigma_M$$

$$= \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^{\top}$$

How do we measure stationarity?

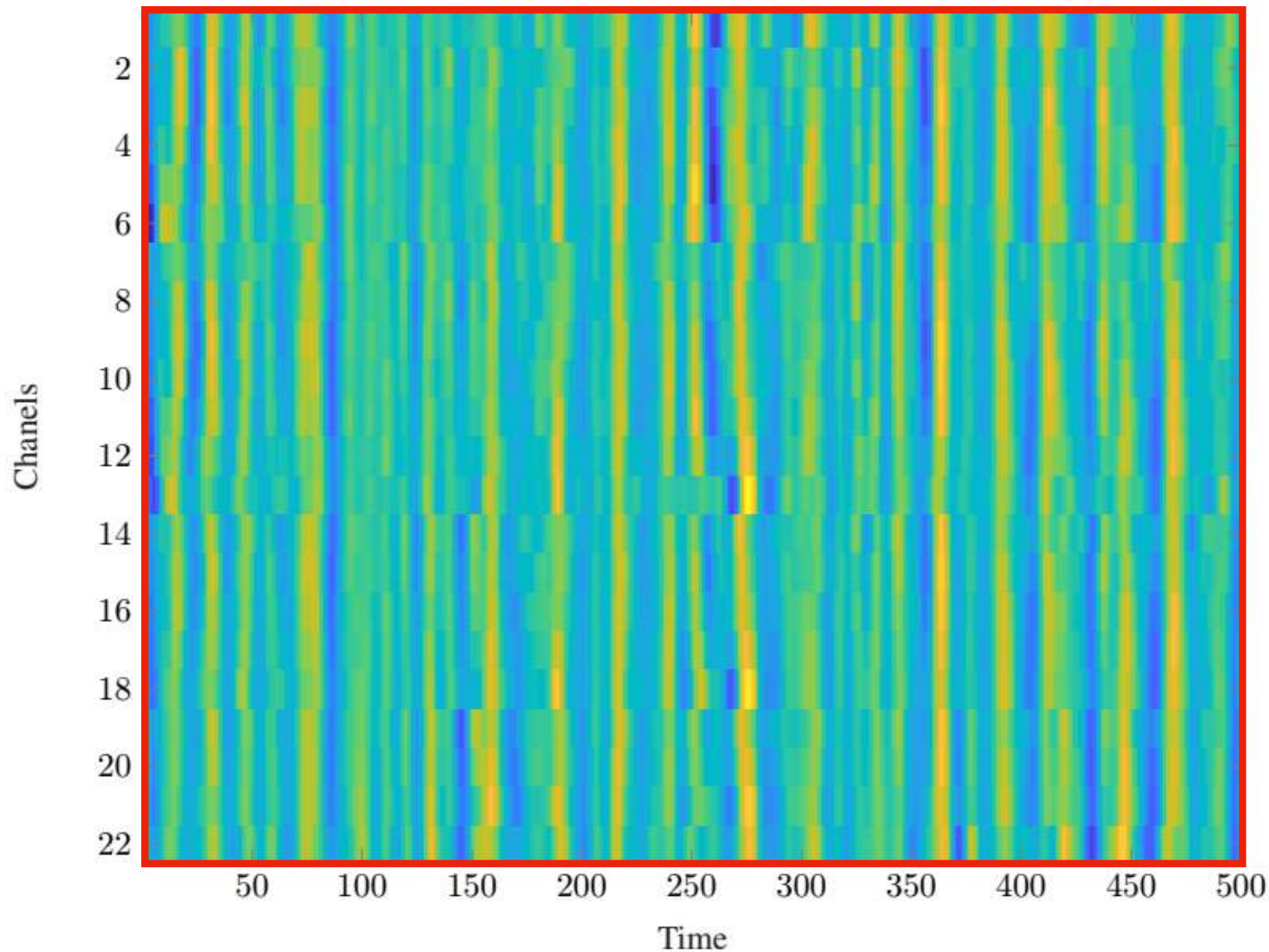


$$\Sigma = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$$

$$\Sigma_1 \quad \Sigma_2 \quad \Sigma_3 \quad \dots \quad \Sigma_M$$

$$= \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^T = \mathbf{V} \mathbf{V}^T \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^T \mathbf{V} \mathbf{V}^T$$

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$\mathbf{A}$ is diagonal iff
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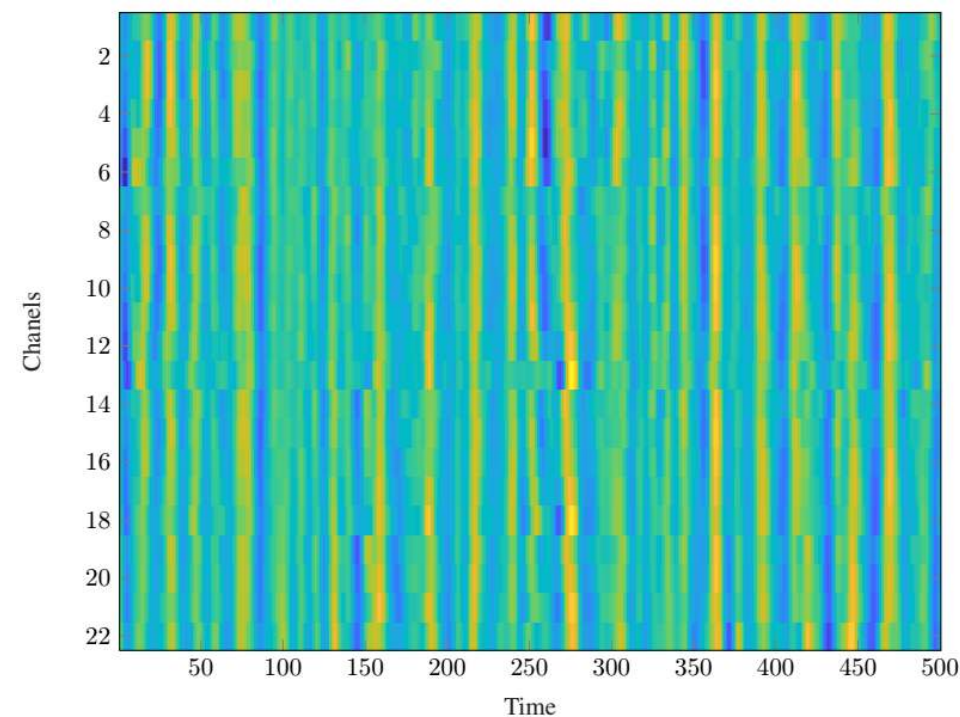
$$\Gamma(\Sigma_i) = \|\mathbf{A}_i \text{diag}(\mathbf{A}_i)^{-1}\|_F - \sqrt{N}$$

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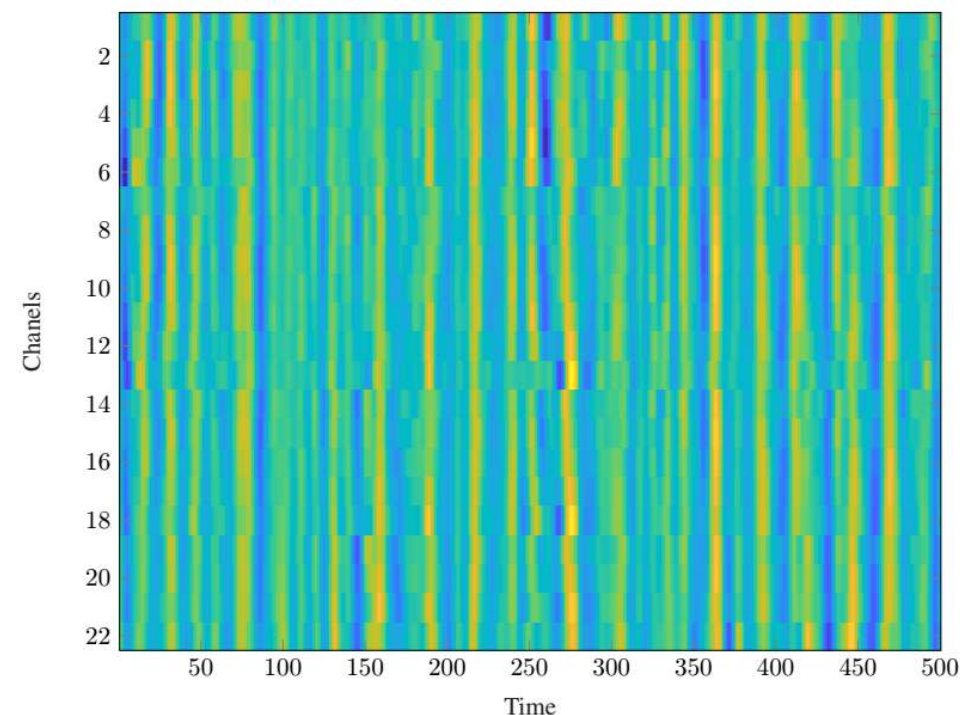


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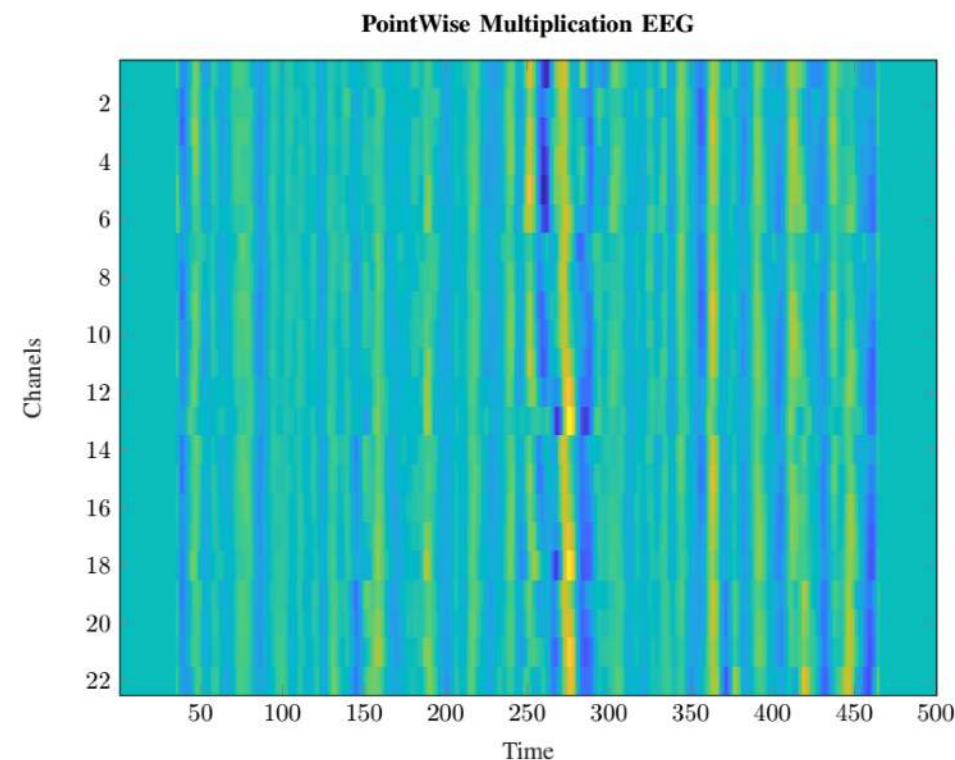


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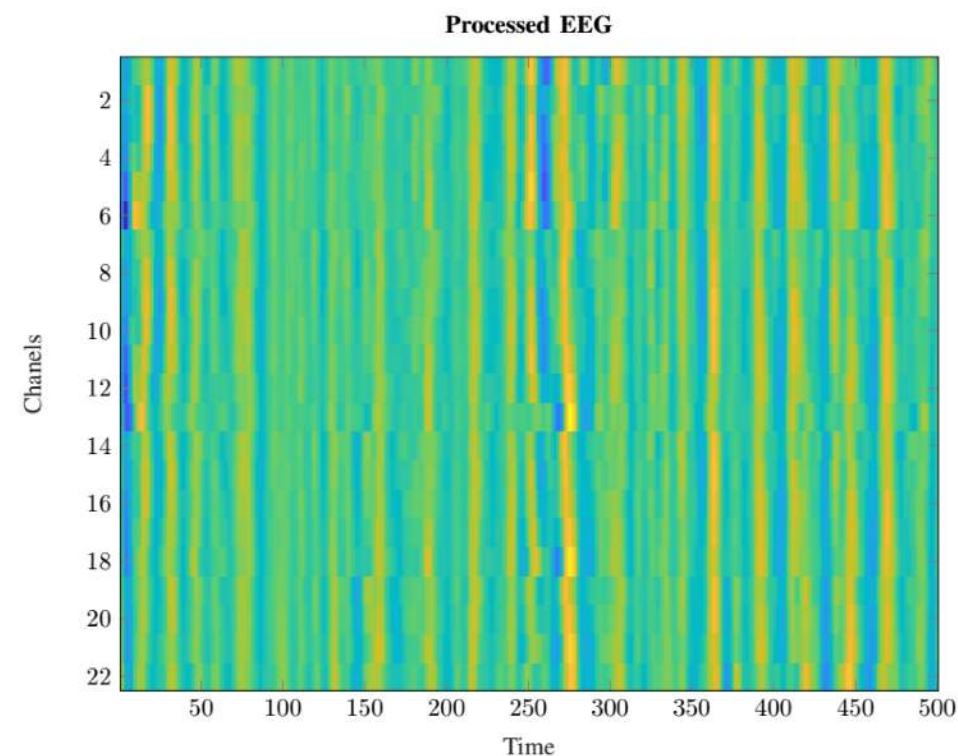


How do we measure stationarity?

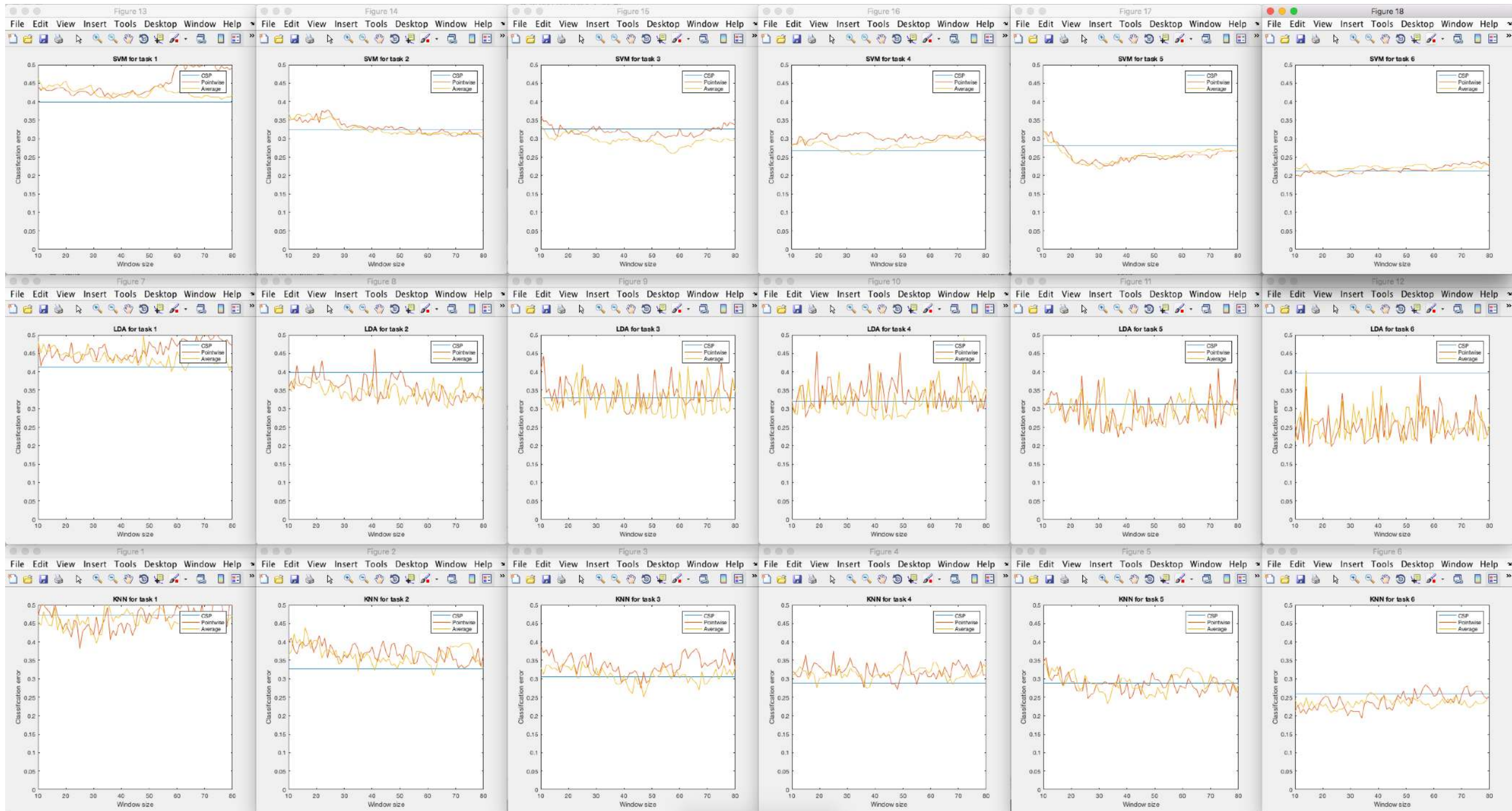
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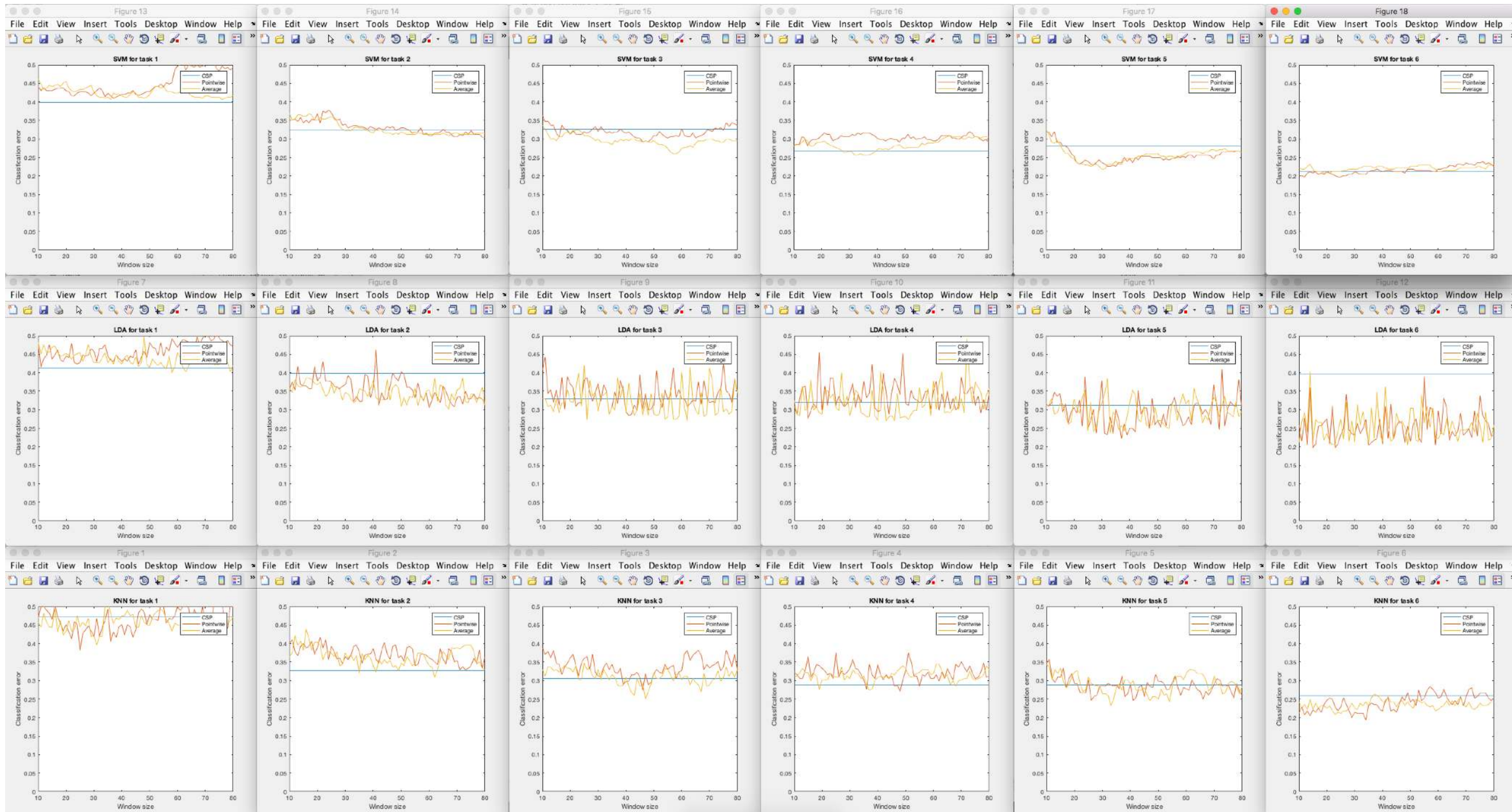
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Preliminary results (classification)



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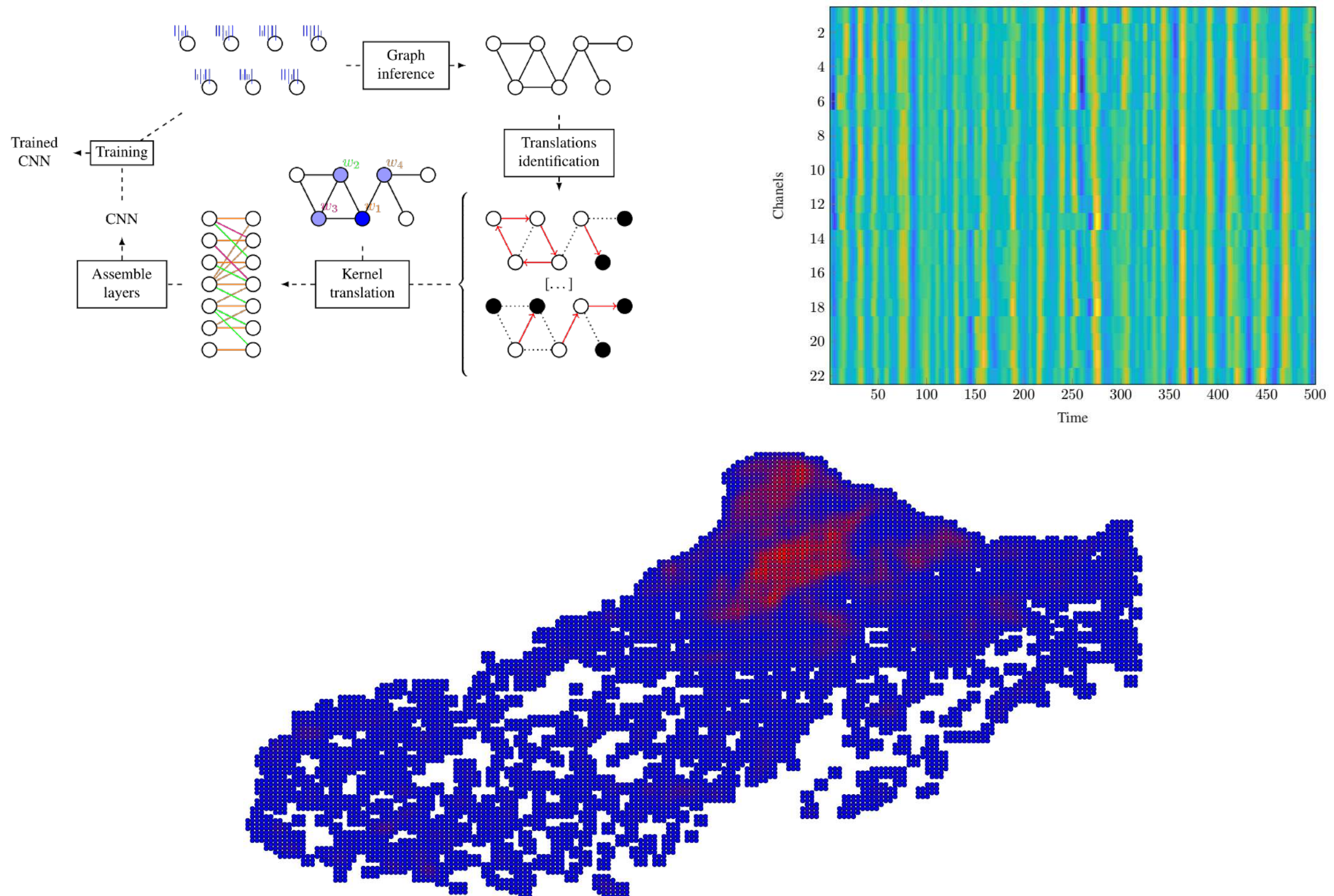


Room for improvement (cov. estimation, resting state, one window per trial...)

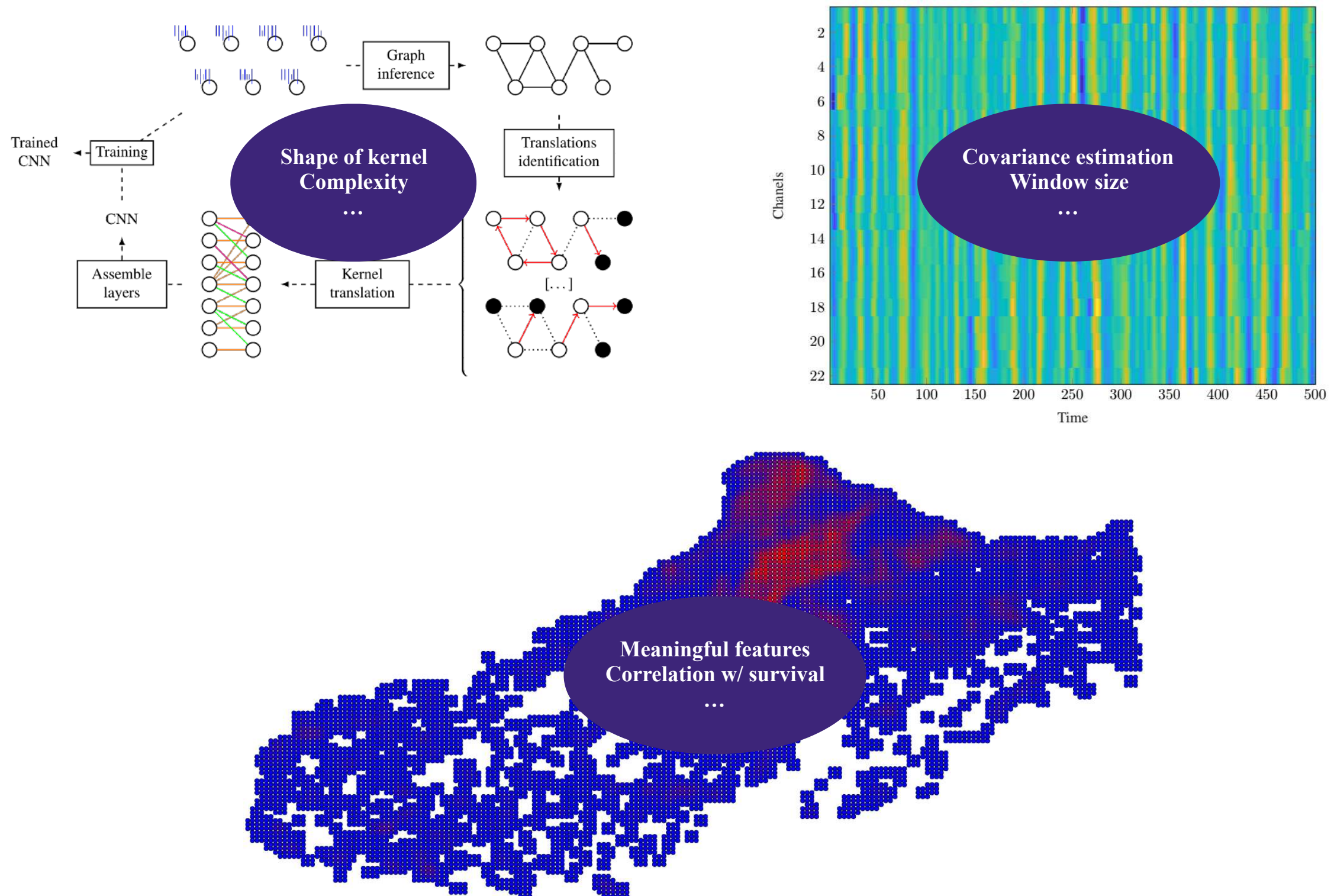
Outline

Conclusions

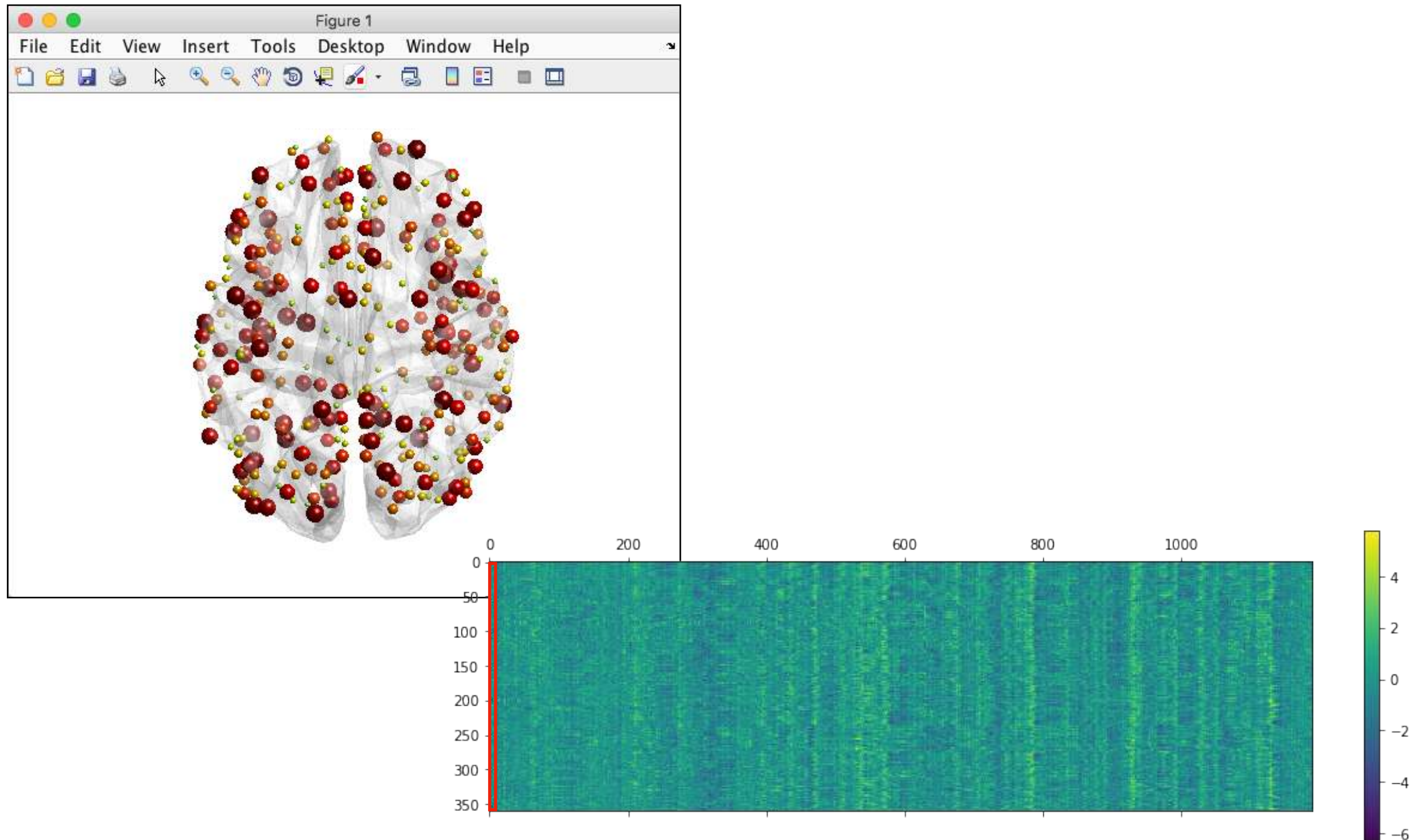
Those are only a few examples



A glimpse on future work

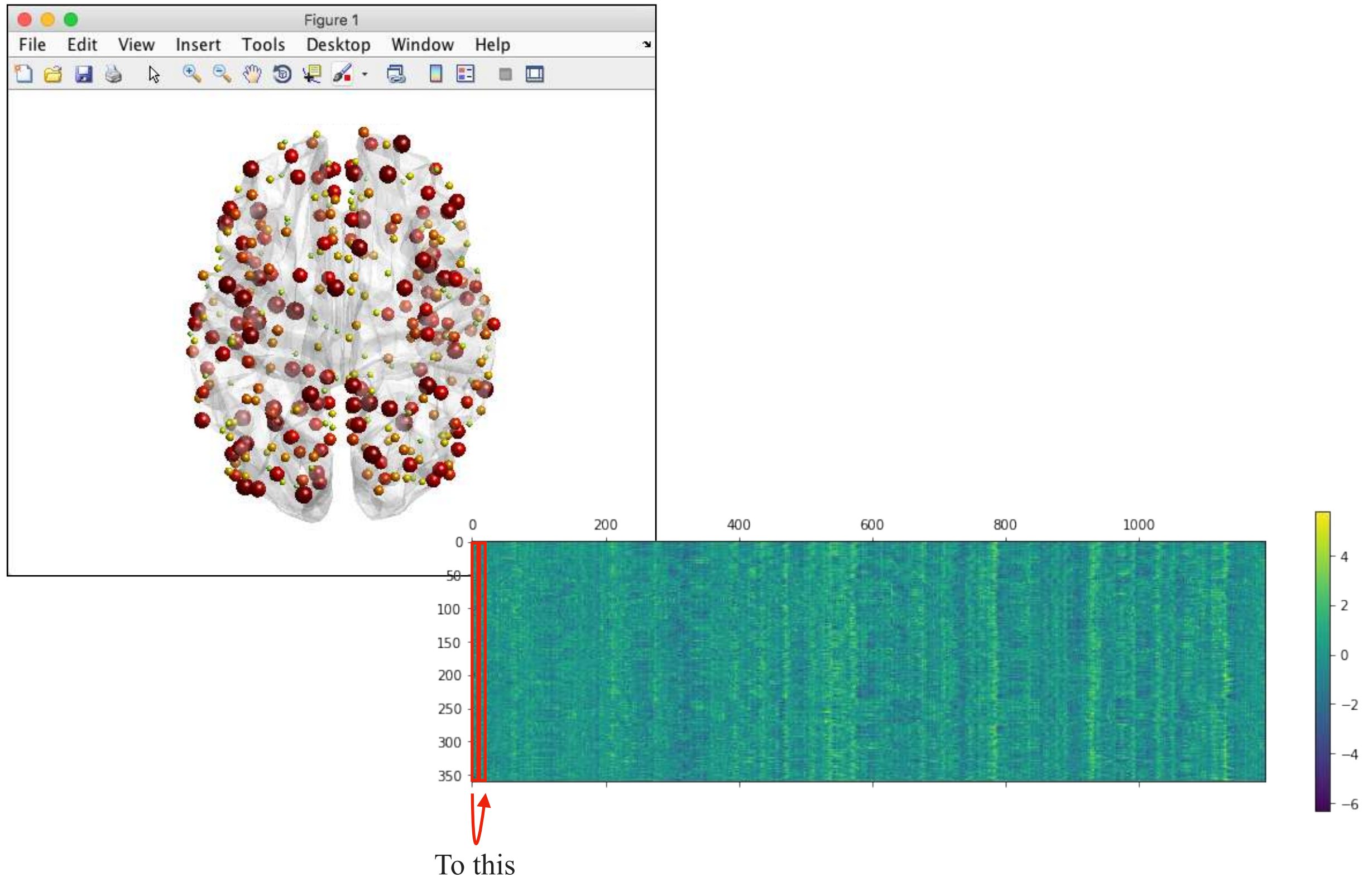


A glimpse on future work

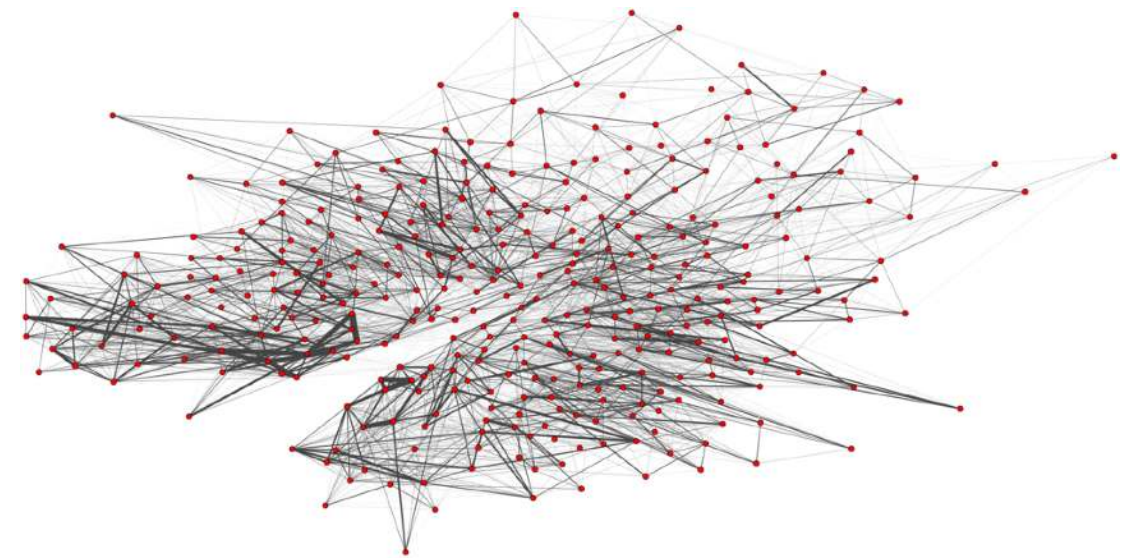
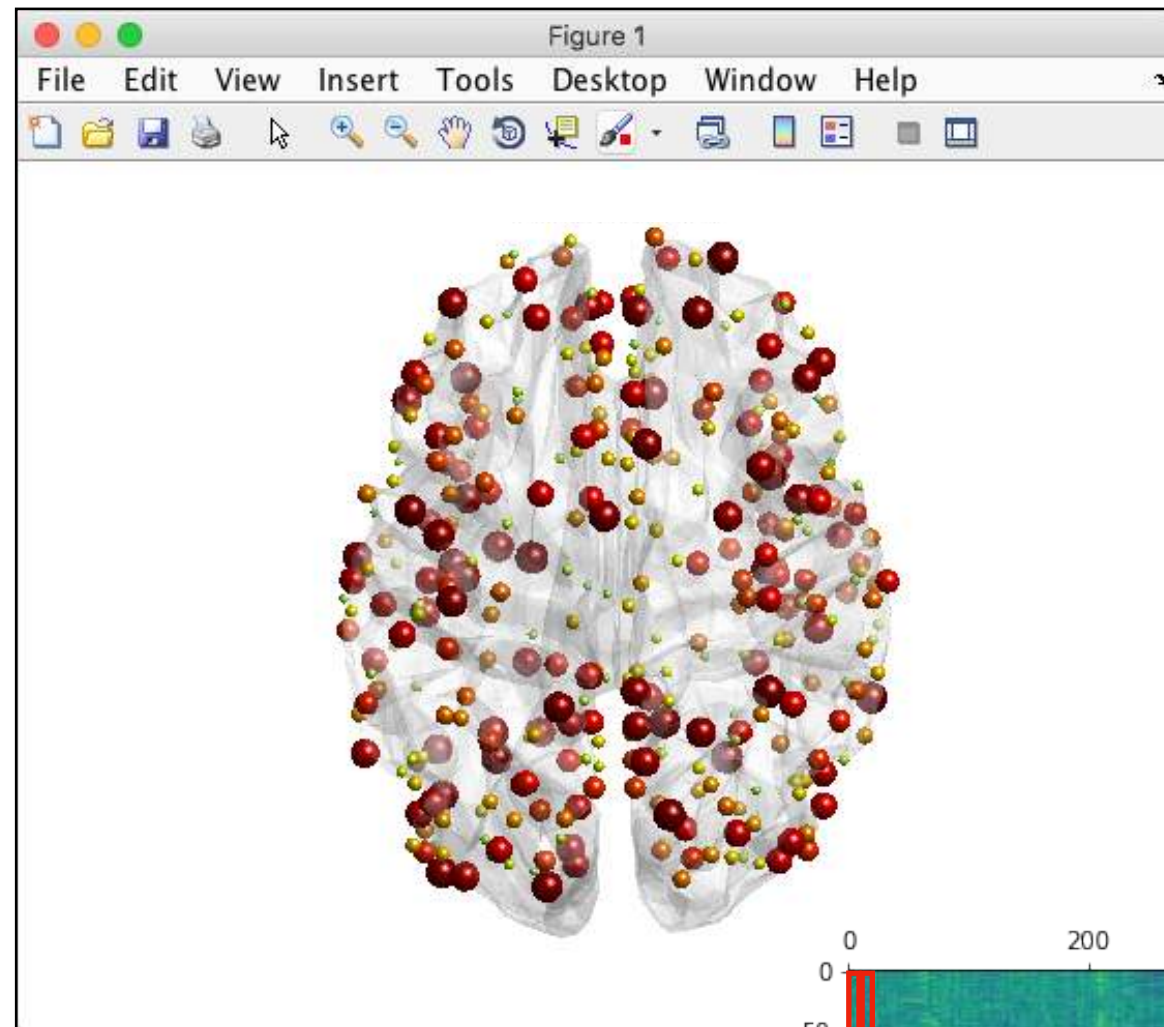


How to go
from this

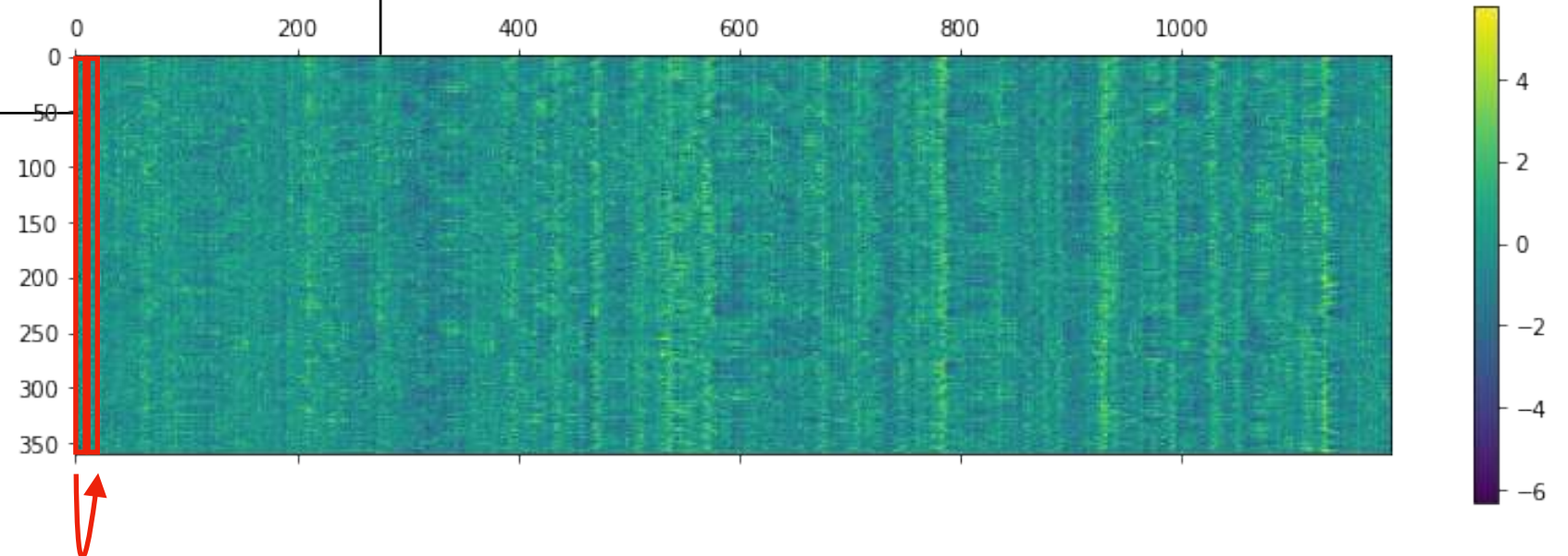
A glimpse on future work



A glimpse on future work



Using graph-constrained
optimal transport on the connectome



To this

A few applications of GSP to machine learning and medicine

Bastien Pasdeloup
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